

|                                                       |           |
|-------------------------------------------------------|-----------|
| <b>Terminology .....</b>                              | <b>1</b>  |
| <b>Unit: 1 Research &amp; Research Process .....</b>  | <b>16</b> |
| Definition of Research .....                          | 16        |
| Nursing Research .....                                | 16        |
| Need & Purpose of Nursing Research .....              | 19        |
| Importance & Significant of Nursing Research.....     | 19        |
| Scope & Areas of Nursing Research .....               | 20        |
| Problems in Nursing Health & Research.....            | 21        |
| Problem Solving Process .....                         | 23        |
| Scientific Method.....                                | 24        |
| Types of Research .....                               | 25        |
| Research Process/Steps of Research Process .....      | 28        |
| Hypothesis.....                                       | 39        |
| Assumption .....                                      | 45        |
| Differences between Assumptions & Hypothesis.....     | 47        |
| Differences between Delimitations & Limitations ..... | 47        |
| Ethics in Nursing Research.....                       | 48        |
| <b>Unit: 2 Research Problem/Question.....</b>         | <b>51</b> |
| Definition of Research Problem .....                  | 51        |
| Identification of Research Problem.....               | 51        |
| Criteria for Selecting a Good Research Problem .....  | 53        |
| Formulation of the Research Problem.....              | 54        |
| Research Objectives .....                             | 56        |
| Variables.....                                        | 58        |
| <b>Unit: 3 Review of Literature .....</b>             | <b>61</b> |
| Definition of Review of Literature .....              | 61        |
| Importance of Review of Literature.....               | 61        |

|                                                         |            |
|---------------------------------------------------------|------------|
| Purposes of Literature Review .....                     | 62         |
| Types of Review of Literature .....                     | 62         |
| Sources of Review of Literature .....                   | 64         |
| Steps of Review of Literature .....                     | 67         |
| <b>Unit: 4 Research Approaches &amp; Designs.....</b>   | <b>70</b>  |
| Definition of Research Design .....                     | 70         |
| Elements of Research Design .....                       | 70         |
| Factors Affecting Selection of the Research Design..... | 71         |
| Classification of Research Design (Methodology) .....   | 72         |
| Quantitative Research Designs.....                      | 74         |
| Qualitative Research Designs.....                       | 88         |
| <b>Unit: 5 Sampling &amp; Data Collection .....</b>     | <b>93</b>  |
| Sampling .....                                          | 93         |
| Definition of Population .....                          | 93         |
| Definition of Sample .....                              | 93         |
| Sampling Criteria/Characteristics of Good Sample.....   | 93         |
| Sampling Process .....                                  | 94         |
| Factors Influencing Sampling Process.....               | 96         |
| Types of Sampling Techniques.....                       | 97         |
| Data Collection .....                                   | 116        |
| Introduction .....                                      | 116        |
| Concepts of Data Collection .....                       | 116        |
| Data Sources.....                                       | 117        |
| Methods & Tools (Instruments) of Data Collection.....   | 117        |
| Types of Methods & Tools of Data Collection.....        | 118        |
| Procedure of Data Collection.....                       | 146        |
| <b>Unit: 6 Introduction to Statistics.....</b>          | <b>149</b> |
| Introduction.....                                       | 149        |
| Definition .....                                        | 149        |
| Uses & Applications of Statistics.....                  | 149        |

|                                                                  |            |
|------------------------------------------------------------------|------------|
| Types of Data/Scales of Measurement .....                        | 150        |
| Classification of Statistics .....                               | 151        |
| Normal Probability Curve (Z Score).....                          | 174        |
| Inferential Statistics .....                                     | 176        |
| Statistical Packages & Applications.....                         | 181        |
| <b>Unit: 7 Analysis of Data.....</b>                             | <b>183</b> |
| Introduction.....                                                | 183        |
| Purposes of Analysis of Data .....                               | 183        |
| Definition of Data Analysis .....                                | 183        |
| Interpretation Data .....                                        | 184        |
| <b>Unit: 8 Communication &amp; Utilization of Research .....</b> | <b>186</b> |
| Introduction.....                                                | 186        |
| Criteria/Finding for Communication of Research .....             | 186        |
| Writing a Research Report .....                                  | 186        |
| Utilization of Research Findings.....                            | 191        |
| Developing & Presenting a Research Proposal.....                 | 194        |
| Computer in Nursing Research .....                               | 198        |



## **Terminology**

### **1) Research-**

→ Research is a scientific, critical, purposeful and systematic process of investigation or experimentation, which involves disciplined inquiry in order to discover new facts or to generate reliable, valid and new knowledge or to bring solution to a problem of a conflicting or critical situation with data based recommendations or conclusion.

### **2) Research Proposal-**

→ Research proposal is a document prepared by a researcher in which the researcher specifies the topic of research study which also includes research problem its significances the research design, duration and the cast of the study data collection, data analysis plan etc.

### **3) Research Problem-**

→ Research problem is a situation or a state of concern or conflict where there is needs to find out solution to the problem or there is need for modification, alteration or improvement of the situation.

### **4) Research Objective-**

→ Research objectives are the gross (broad) or specific goals or aims of the research study that are generated from the researcher problem which help the researcher stay focused and which the researcher intends to achieve through the research study.

### **5) Basic Research-**

→ Basic research is a type of research which is taken up for generating more knowledge, formulating fresh theories etc. that help extending base of the knowledge advancing or adding to the existing knowledge of a specific discipline.

## **6) Applied Research-**

→ Applied research is a type of research study finding of which are applied in the clinical field or in the field of community for solving any problem in the field of problem or to enhance and the technique of practice.

## **7) Historical Research-**

→ Historical research is a type of research study in which the researcher attempts to gather data on the past events in a systematic and logical manner.

## **8) Descriptive Research-**

→ Descriptive research is the type of research study in which the researcher narrated or describes the characteristics of person, objects, attributes, situation etc and also the phenomena that occur.

## **9) Exploratory Research-**

→ Exploratory research is a type of research in which the researcher attempts to explore new facts, new theories or new dimensions to existing facts and theories.

## **10) Observational Research-**

→ Observational research is the type of research in which the researcher collects data by observing and recordings the behavior, characteristics, attributes, activities etc of the subject under study.

## **11) Evaluation Research-**

→ Evaluation research is the type of research study in which the researcher investigates about a program, practice, policy etc. and how well it has been working or functioning.

## **12) Quantitative Research-**

- Quantitative research is the types of research are transformed into precise measurements and qualification of the data.
- These data are collected through strictly controlled research design, these types of conducted for testing of the hypothesis.

## **13) Qualitative Research-**

- Qualitative research is the type of research in which the researcher does in depth investigation of a specific phenomenon in a holistic manner by using flexible research studies the objectives to an answer the research questions.

## **14) Clinical Research-**

- Clinical research is a type of research study that is conducted into clinical setting and in which the research finding are applied to the field of clinical practice to improve or to enhance the present status of practice.

## **15) Field Research-**

- Field research is the type of research in which the researcher conducts the study and collects data in the fields such as the data is collected of the individual natural behavior practice beliefs etc. in a natural or normal environment or in the real life situation.

## **16) Co-relational Research-**

- Co-relational research is the types of research in which the researcher attempt to explore inter relationship between the variables with in research study.

### **17) Experimental Research-**

→ Experimental research is the type of research in which the researcher control or manipulated the independent variables to cause on effect on the dependent variables.

### **18) Quasi experimental Research-**

→ Quasi experimental research is a type of research in which the subjects are not randomly chosen for the experimental treatment in this type of study the research manipulates independent variables and exercise certain control, which in turn helps in increasing the internal validity of the results.

### **19) Survey Research-**

→ Survey research is a type of research in which the researcher focuses on gathering data on behavior, characteristic activities, beliefs, and attributes, attitudes through direct questioning or observation.

### **20) Concept-**

→ Concept is the abstraction which is based on researcher observation of certain behavior or characteristics of people, objects & situations.

### **21) Phenomenon-**

→ Phenomenon is the concept in a research study which is under investigation.

### **22) Theory-**

→ Theory is the abstract generalization which predicts systematic explanations about the relationship among concept.

### **23) Generalization-**

→ Generalization is the implication of the research findings that applicable and justifiable for application to a larger population or to a majority or bulk of people or to the entire population.

Or

→ Generalization is the degree to which research studies justify increase that the research findings from a sample obtained through data analysis can be applied to the entire population to which the sample belongs.

### **25) Prediction-**

→ Prediction is the use of empirical evidence in making forecasts about variations of variables in a new setting with different subjects.

### **26) Hypothesis-**

→ Hypothesis or hypotheses (plural) are the formal statement of the tentative prediction or description or explanation of the relationship between two or more variables in a specific situation people or population.

### **27) Variables-**

→ Variables are the qualities properties or characteristics of people, objects, situation, concept, activities, attributes, attitudes etc. that changes or vary according to the stimulus.

→ The variables are “manipulative” and “measureable”.

### **28) Independent Variables-**

→ Independent variables are qualities or properties that can be manipulated by the researcher to cause an effect on the dependent variables. It is called “treatment” or experimental variables.

### **29) Dependent Variables-**

- Dependent variables are the qualities or properties or outcome that the researcher wants to predict which may occur in response to the treatment of independent variables.

### **30) Operational Definitions-**

- Operational definitions are the definitions of the variables, acts, proceeding etc. that are formulated by the research for the specific research study that has been undertaken.
- The researcher gives definition of the concept or of the variables would be measured. These are also called conceptual definition.

### **31) Assumptions-**

- Assumptions are the statement of basic principles or facts that are established and are universally accepted as true on the basis of reasoning, logic etc. without verification or proof, though they may not have been scientifically tested.

### **32) Delimitations-**

- Delimitations are the restriction or limitations of a research study which reduce the credibility or generalizability of the research findings. These are also known as weakness or research study.

### **33) Review of Literature-**

- Review of literature is a systematic scholarly, critical, scanning of related and relevant literature such as bodies, generals, research articles, research critics, research find to find out how much studies have been done in the area chosen by the researcher to conduct research study.

### **34) Primary Source-**

- Primary source is the first hand report of facts, events findings etc.
- In a research study the primary source is the original research report as prepared by the researcher who conducted the research study.

### **35) Research Methodology-**

- Research methodology is also called “Research Design” “research plan” or “research strategy” of a research study that gives guideline which directs the research steps the study process and helps in systematic data collection logical data organization and accurate data analysis in a research investigation.

### **36) Case Study-**

- Case study is a research methodology in which researcher studies and analysis the in depth information presented by one particular individual or group or one specific institution or a particular social unit.

### **37) Conceptual Framework-**

- This is also known as theoretical framework. This is the infrastructure of the research study in which inter related concepts and theories are assembled and presented in a rational manner that reflects the all features plan and arrangement of various elements of the researcher study that has been under taken.

### **38) Sample-**

- Sample is the sub-set of population or of situations that has been selected to make them participate in the respective research study.

→ This sub-set of population or objects/situation which represents the entire population.

### **39) Population-**

→ Population is the entire set of individual or of objects which have common characteristics or qualities.

### **40) Subject-**

→ Subject is an individual, an object etc belonging to a sample who participates in a research study and provides data.

### **41) Randomization-**

→ Randomization is the process of sampling in which sample is obtained in a random manner where each member of population has an equal chance to be given to include in the sample.

### **42) Random Sampling-**

→ Random sampling is a technique in which each member of population has an equal probability of being included in the sample that is obtained for a research study.

### **43) Probability Sampling-**

→ Probability sampling is a technique in which random procedure are used to choose a sample, simple random sampling, cluster sampling, systematic sampling etc are examples of the probability sampling.

### **44) Cluster Sampling-**

→ Cluster sampling technique in which large groups (clusters) are selected first such as students, nurses, teachers, doctors etc. then sub sampling is done to obtain smaller units of sample.

#### **45) Systematic Sampling-**

→ Systematic sampling is a technique in which every 10<sup>th</sup> (such as every 5<sup>th</sup> person) is chosen to be included in the sample.

#### **46) Non-probability Sampling-**

→ Non-probability sampling is a sampling techniques in which are obtained from a population by using a non-random method.

→ Convenience sampling quota-sampling etc. are example of the non-probability sampling.

#### **47) Convenience Sampling-**

→ Convenience sampling technique is a sampling technique where the readily available subjects or objects are selected as participants in a research study.

#### **48) Quota-Sampling-**

→ Quota sampling is a sampling technique in which non random selection of the students is done by the researcher.

→ In this the researcher specifies the characteristics of the sample to increase its representative.

#### **49) Purposive Sampling-**

→ Purposive sampling is a type of non-probability sampling method in which the researcher selects participants for the study on basis of certain purpose about which one will be most representative.

#### **50) Snowball Sampling-**

→ Snowball sampling technique in which selections of the participants are done by means of referrals from earlier participants, it is also called network sampling.

### **51) Data-**

→ Datum or data are the information that is systematically collected, organized presented and analyzed during the course of the research study. This information helps the researcher to draw conclusions.

### **52) Data Collection-**

→ Data collection is a written device or technique that is either prepared or obtained by researcher which the researcher uses to gather data such as questionnaire observation schedule, test etc.

### **53) Qualitative data-**

→ Qualitative data are the information that is collected in a qualitative research study. The data is non-numeric and is obtain through non-structured interview and is presented in a narrative form.

### **54) Questionnaire-**

→ Questionnaire is the method of data collection from the respondents by administering pre-set questions in a format.

### **55) Close Ended Questionnaire-**

→ Close ended questionnaire is a type of questionnaire which is used to gather information from the respondents and in this type of questionnaire the researcher selects multiple choice answer to all the question and the respondent have to “tick mark” against his/her choice from the answer given.

### **56) Open Ended Questionnaire-**

→ Open ended questionnaire is a type of questionnaire which is used to gather information from the respondents and in this type of questionnaire the researcher does not restricts the answer of

respondents and free to express their personal opinion in their own language in response to the questions.

### **57) Validity-**

→ Validity is the degree to which a data collection instrument measures what it intends to measure.

### **58) Content Validity-**

→ Content validity specifies certain degree to which the items of a research instrument or tool represent the whole of context.

### **59) Reliability-**

→ Reliability is the degree of consistency or dependability with which a research tool measures the attributes that the tool is designed to measure.

### **60) Interview-**

→ Interview is the method of data collection in which the researcher asks the questions from the subjects in a face to face set-up.

### **61) Rating Scale-**

→ Rating scale is the method in which the respondents are to rate something such as qualities or characteristics along orderly bipolar continuum such as very good, good, average, poor etc.

### **62) Pre-test-**

→ Pre-test is a procedure by which data are collected in experimental research before the experimental treatment is administered.

### **63) Post-test-**

→ Post test is a procedure by which data are collected in experimental research after the experimental treatment is administered.

#### **64) Experimental Treatment-**

→ Experimental treatment is the experiment intervention which is implemented in an experimental research to manipulate conditions, activities situations etc.

#### **65) Control Group-**

→ Control group is a group which consists of subjects in an experimental research who do not receive the experimental treatment. The results of the control group can be measured or compared.

#### **66) Pilot Study-**

→ Pilot study is a small scale version of the main research study which is conducted as a trial run in preparation of the main or major research study.

#### **67) Data Analysis-**

→ Data analysis is the process of after the data and synthesis of data after the hypotheses have been tested or the research problem has been solved.

#### **68) Statistics-**

→ Statistics is an estimate of a parameter, calculated from a data obtained through a sample in a research study.

#### **69) Statistical Analysis-**

→ Statistical analysis is the organization and analysis of the quantitative data using statistical procedures which include both descriptive and inferential statistics.

## **70) Descriptive Statistics-**

→ Descriptive statistics are statistics that are used to describe and summarize data such as mean, median and mode etc.

## **71) Inferential Statistics-**

→ Inferential statistics are the statistics which state whether the relationship observed in a sample are likely to occur in a large population. In another word, the statistics used allow inferences.

## **72) Central Tendency-**

→ Central tendency is a statistical index of special characteristics of a set of scores or value which comes from the center of the distribution of scores the three most common indices of the central tendency are the means, medium and mode.

## **73) Mean-**

→ Mean is a descriptive statistics, that is computed by adding (summing) all the scores and then dividing the total score or value by the number of subject.

## **74) Median-**

→ Median is descriptive statistics that represents exactly the middle score or value in a distribution of scores. The median is the score, the above and below which 50% of the score lies.

## **75) Mode-**

→ Mode is a descriptive statistics that represents the score or value which occur most frequently in a distribution of scores.

### **76) Standard Deviation-**

→ Standard deviation is the most frequently used statistics in a research study measuring the degree of variability in a set of scores.

### **77) Correlation-**

→ Correlation is the tendency for variation in which the variations of one variable is related to the variation of another variable.

### **78) Chi-square Test-**

→ Chi-square test is a method of statistics computation that assesses the existing relationship between the variables in research study.

### **79) Frequency Distribution-**

→ Frequency distribution is a systematic arrangement of numerical values from the lowest to the highest along with a count of a specific number of times each value is obtained.

### **80) Findings-**

→ Findings are the result of data analysis of the qualitative research study in a quantitative research study results of the hypothesis test are referred as the findings of a research study.

### **81) Research Result-**

→ Research results are the answer to research questions or solutions to research problems which are obtained through data analysis in a quantitative study the information is obtained through statistical tests.

### **82) Research Report-**

→ Research report is documents which are prepared by the researcher after completion of the research study and which summarizes the main features of the study include the research problem, research

methodology, research finding and interpretation of the data and also the implications of the findings.

### **83) Research Abstract-**

→ Research abstract is a briefly written description of the completed research study published in a research journal.

### **84) Research Critique-**

→ Research critique is an objective and critical appraisal or evaluation of the various dimensions of a research report.

### **85) Bibliography-**

→ Bibliography is the compilation of references found in books and periodic reports.

## **Research & Research Process**

### **Definition of Research-**

- Research is defined as a systematic and scientific process to answer questions about facts and relationship between facts.
- Research is systematic method of exploring, describing, explaining or establishing the existence of phenomena the factor that cause changes in phenomena and how the phenomena influence other phenomena.
- Research is a scientific, systematic, controlled, orderly and objective investigation to develop refine and expand body of knowledge.
- Research is a problem solving process, a systematic, intensive study directed toward full scientific knowledge of subject study.

### **Nursing Research-**

- Nursing research refers to the use of systematic, controlled, empirical and critical investigation in attempt to discover and confirm facts that related to specific problem or question about the practice of nursing.
- Nursing research can be defined as systematic process of collection and analysis of data on the organization delivery and outcome of the nursing care.
- Nursing research is a way to identify new knowledge to improve professional education and practice and use of resources effectively.
- Nursing research is scientific, systematic & orderly process to find out solution for problem concern to nursing and refining the nursing knowledge to improve quality of nursing care, nursing education and nursing administration.

### **Characteristics of Research-**

#### **1) Orderly & Systematic Process-**

- A research only is good if it is conducted in an orderly and systematic way.

**2) Based On the Current Professional Issues-**

→ Resolving current issue of any discipline is one of them fundamental purpose of research. Good research must be based on the current topic.

**3) Good Research Clearly Define Purpose-**

→ A research can only be conducted in effective manner. It is start with clearly define objective.

**4) Emphasize To Developing, Refining and Expand Professional Knowledge-**

→ The main purpose of the research is to develop the knowledge or refine the existing knowledge.

→ A good research always directed toward expenses of professional body of knowledge.

**5) Directed Toward Development or Testing Theories-**

→ Development or test theories are a systematic process of developing in a discipline.

**6) Finding Solution of a Problem-**

→ Find answer and solution of professional problem is another important person of a research.

→ Good research is always going toward pending an answer or solution.

**7) Dedicated To Develop Empirical Evidence-**

→ Another important purpose of research to generate empiric evidenced which can be use to improve the professional practice.

**8) To Collect First an Information-**

→ A successfully research is connected by collected data by directly from subject by different method like questioning, interview or observation.

**9) An Objective and Logical Process-**

→ Research information is collected through subjective manner will never lead satisfactory result therefore a good research emphasizes on objective and logical research process.

**10. Generate Finding to Refine and To Improve Professional Practice-**

→ A research cannot be considered successful completed until unless is contribute to add refine and improvement in professional practice the need of time.

**11) Use of Appropriate Methodology-**

→ Selection of a methodology in any research depend on various factors because each valuable and research a unique feature so use appropriate and suitable methodology.

**12) Conducted On Representative Sample-**

→ Generalization of research finding is any possible if study is conducted on a sample which has conducted on a sample which has characteristics similar to the population of the study good research is conductive on a representative sample.

**13) Use Appropriate Method and Tool of Data Collection-**

→ Base of nature of phenomena and subject understudy method or tool of data collection will be different.

→ E.g. – Questionnaire cannot be use for illiterate subject for a performance can only be measured through.

**14) Use of Valid and Reliable Data Collection Tool-**

→ Good research is always depending on validity and reliability of research tool.

**15) Carefully Recorded and Reported-**

- A research can only be considered good if research information is carefully recorded and reported because in absence of careful recording.

**16) Adequately & Appropriately Analyze Research-**

- A research activity is incomplete and who were until gather data adequately or appropriately analyze data use standardized and accepted method of data analysis.

**17) Adequately Communicating-**

- A research cannot be considered good until the information generalized is adequately ability its uses. A good research activity strives to community finding as widely as possible.

**Need & Purpose of Nursing Research-**

- Develop, refine and extend the scientific base of knowledge which is required for quality nursing care, education and administration.
- Enhance the body of professional knowledge in nursing.
- Provide foundation for evidence based nursing practice.
- Help in expensive knowledge which is essential for continue growth of nursing profession.
- Define the parameters of nursing.
- Refine and eliminate old knowledge.
- Identify nursing care practice.
- Develop and refine nursing theories and principles.
- Solve the problem or answer question related to nursing practice, nursing education and nursing administration.

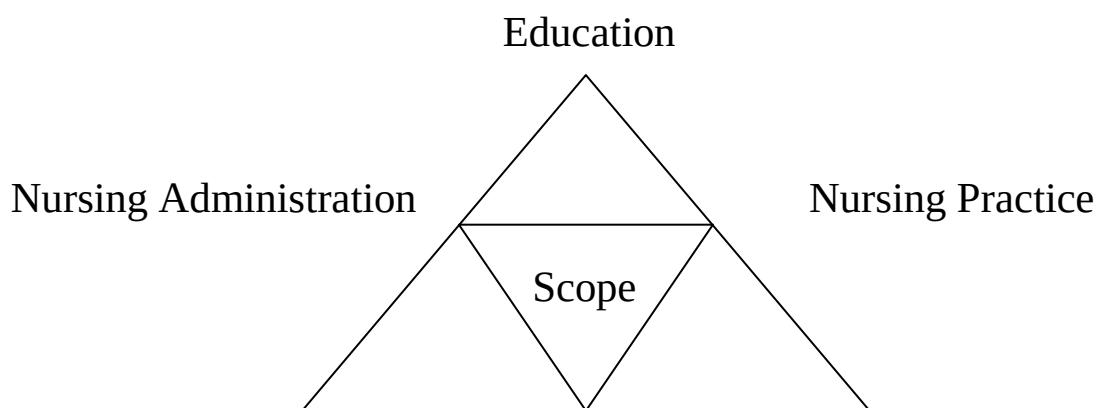
**Importance & Significant of Nursing Research-**

- 1) Development and utilization of nursing knowledge is essential for continue development in patient care nurses can develop or refine this knowledge by the help of nursing research.
- 2) Nursing research is essential for nurses to understand the various dimension of their profession.

- 3) Nurses need to get involved in research because the cost of care is increasing day by day; therefore, they must find solutions for cost-effective care through research.
- 4) Nursing research helps to:
  - Build and expand the body of nursing knowledge.
  - Validate and refine the adjusting nursing practice may help care efficient and cost-effective.

### **Scope & Areas of Nursing Research-**

→ They are the following areas:



### **1. Research in Nursing Practice-**

- 1) Health promotion, maintenance and disease prevention.
- 2) Patient aspect and quality of health care.
- 3) Promotion and risk reduction of health.
- 4) Patient centered care and care coordination.
- 5) Promotion of the health and well being of the older people.
- 6) Palliative and end of life care.
- 7) Development of evidence based practice.
- 8) Care implication of genetic testing and practice.
- 9) Nurses working environment.
- 10) Home care and community health nursing care.
- 11) Description of holistic nursing situation and social cultural religious and family practice which have health implication.

## **2. Research in Nursing Education-**

- 1) Testing the effectiveness and efficiency of the old teaching method.
- 2) Curriculum talk and learning experience of nursing student at undergraduate and post graduate level.
- 3) Extend of strict discipline required for the nursing student to improve the knowledge education.
- 4) Promoting clinical and classroom learning among nursing student.
- 5) Identify and managing problems of nursing student and motivate them.
- 6) Resolving any issue and phenomena related to teaching learning process of the nursing student.

## **3. Nursing Administration-**

- 1) Assessing existing organizational structure, control, communication, staffing pattern, benefits, performance evaluation practice and their effectiveness and to refine the old knowledge regarding nursing administrative phenomena.
- 2) Developing and testing different administrative model for the customer model for the customer satisfaction.
- 3) Retention and effective use of nursing personnel in providing the quality of nursing care.
- 4) Research can be conducted of any phenomena related to nursing administrative issue.

## **Problems in Nursing Health & Research-**

### **1) Fallibility of Discipline Research-**

- Research study in the field of nursing health and social science has some limitation each research question can be solving by the different approaches and various factor.
- It is always confusion to decide a best approach for solving research question.
- Proper discipline is use to solve a problem.

**2) Handling Multiple Variables-**

→ Most of the research study in the field of nursing and health and social science usually focus on the measurement of the multiple variables in a single attempt.

**3) Difficulty in Control of Exteriors Variables-**

→ Research in nursing health and social science is usually conducting in natural setting therefore it become very difficult to control over the exteriors variables while measuring the effect of independent variables on the depended variables.

**4) Minimal Possibility of Laboratory Research-**

→ Research in nursing health and social science usually deal with phenomena related to human. It becomes very difficult to conduct research study in laboratory.

**5) Lack of Standardized Tool-**

→ Research in nursing health and social science it deal with the natural phenomena where valid and reliable standardized tool are needed to generate empirical evidence, it is evident that is significant lack of valid and reliable tool to measure variables in nursing health and social science discipline.

**6) Measuring Qualitative Phenomena through Quantitative Means-**

→ It is usually observed that phenomena in the field of nursing qualitative in nature, it can be commonly seen that discipline usually measure. These qualitative phenomena usually by quantitative means where effectiveness of evidence get.

**7) Lack of Interest among Research-**

→ It is generally observe that people in the field of nursing health and social science conduct research study only for specific purpose as a

partially fulfillment of particular degree or due to specific compulsion and part of the job.

## **8) Ethical Consideration-**

→ Research study in the field of nursing deal with human being where safe guarding them right becomes an important issue.

## **Problem Solving Process-**

→ In problem solving process consist of a step that are following:

- 1) Problem definition
- 2) Problem analysis
- 3) Generating possible solution
- 4) Analysis the solution
- 5) Select the best solution
- 6) Implement the solution
- 7) Evaluation and revision

## **1) Problem Definition-**

→ The normal process for solving a problem initially involved definition the problem which is to be solve in first stage there is a need to write down, What exactly the problem which help to identify the real problem that is understanding and need an immediate solution.

## **2) Problem Analysis-**

→ The next step is to analysis how the problem affect the research and his or her current situation and the other people involve in the situation the analysis help in understanding the source of the problem and how it affect the current development and the researcher environment.

## **3) Generate Possible Solution-**

→ When the real problem is discover it contributing factor should be investigated focus must be identify and generating all possible solution for a problem.

**4) Analysis the Solution-**

→ In this action of the problem solving process various factor about each of them potential solution are investigated where in all the positive and negative aspect of each solution are analysis.

**5) Selecting the Best Solution-**

→ At this stage help attempt is made to compare the available solution an eventually the best solution is selected based on the careful judgment which is suppose smoothly.

**6) Implementing the Solution-**

→ The final step of the problem solving process is to practically solve the problem by implementing the separate solution.

**7) Evaluation and Revision-**

→ This is the final stage of the problem solving process where after implementing of the most potential solution and evaluation is made to judge the effectiveness of solution in solving the problem.

**Scientific Method-**

→ Scientific methods are defined as control systematic investigation that are rooted in objective reality and that aim to develop general knowledge about natural phenomena.

**Characteristics of Scientific Method-**

- They are orderly and systematic process.
- There are finding are based on empirical evidence.
- Scientific methods are based on assumption and hypothesis.
- They are basically conducted to develop or test theories.

## **Purpose of Scientific Method-**

→ The basic purposes of scientific method are description, exploration, explanation and identification of relationship of the fact.

## **Steps of Scientific Method-**

- 1) Selecting the topic and identify the research problem.
- 2) Defining the objective of the study.
- 3) Review of literature from theory and other related studies.
- 4) Define concept and variables.
- 5) Stating hypothesis about expected observation.
- 6) Identifying assumption and implication.
- 7) Determine the ethical implication.
- 8) Describing the research and method for data collection.
- 9) Defining study population and sample.
- 10) Planning the data analysis and collecting.
- 11) Collecting data from subject.
- 12) Analysis and interpretation of data.
- 13) Communication finding of the study.

## **Limitation of Scientific Method-**

- Moral and ethical problem.
- Human complexity.
- Measurement of problem.
- External variable control problem.

## **Types of Research-**

### **1. According to Purpose-**

#### **1) Basic Research-**

→ Basic research is performing without purpose. It is primarily concern with generation of new knowledge it is use to generate and expend

theories that describe explain to the discipline without record to its immediate use.

→ It is a formal and systematic process leading to the development of theories.

### ***Aims of the Basic Research-***

→ Gather and generate information.

→ Expend the body of knowledge to improve understanding about discipline.

→ Develop or refine theories.

### **2) Applied Research-**

→ Applied research refers to those studies which have functional purpose and practical and practical use or application.

→ The focus on finding on immediate solution to an existing problem researcher scientifically collect data to be used in clinical administrative or instructional area to find the solution to the problem and evaluate practice and identify need of the patient.

## **2. Based on Approach-**

### **Quantitative and Qualitative Research-**

| Parameter                              | Quantitative                                                                | Qualitative                                                          |
|----------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>1) General nature</b>               | ✓ Objective approach to seek measurement in numerical form.                 | ✓ Subjective approach to seek in depth discipline in narrative form. |
| <b>2) Knowledge of study variables</b> | ✓ Variables are clearly understood and define in advance by the researcher. | ✓ Researcher may have only rough ideas about variables in advance.   |

| Parameter                  | Quantitative                                                                                            | Qualitative                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3) Aim</b>              | ✓ To identify and count the variables to test or refine the theories and hypothesis.                    | ✓ To collect complete detail description of variables to develop theories and assumption.                                                     |
| <b>4) Research problem</b> | ✓ Problem does not change over the course of the study.                                                 | ✓ Problem is inductively and is reformulated during the course of study.                                                                      |
| <b>5) Planning</b>         | ✓ All aspect of study is carefully design before data is collected.                                     | ✓ The design emerges according to study.                                                                                                      |
| <b>6) Research design</b>  | ✓ Experimental, quasi experimental and non-experimental design may be used.                             | ✓ Phenomenology, ethnography case study and historical research.                                                                              |
| <b>7) Sample</b>           | ✓ Quantitative studies are conducted on large representative sample to generalize the research finding. | ✓ Qualitative research studies in depth study and therefore has small size, sample size; generalization of finding is generally not possible. |
| <b>8) Data</b>             | ✓ Quantitative data is easier to collect facilities the testing of the hypothesis.                      | ✓ Qualitative data is more time consuming process.                                                                                            |

| Parameter           | Quantitative                                                                       | Qualitative                                                                                                     |
|---------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>9) Tools</b>     | ✓ Structured or semi-structured tool to collect numerical data such as questioner. | ✓ Less formally structured instrument like participative observation in depth interview focus group discussion. |
| <b>10) Method</b>   | ✓ Researcher tends to remain objectively separated from the subject matter.        | ✓ Researcher tends to become subjectively immersed in the subject matter.                                       |
| <b>11) Analysis</b> | ✓ Analysis of numerical data through descriptive and inferential statistics.       | ✓ Analysis of data such as pictures, descriptive coding, diagram and thematic analysis.                         |

## Research Process/Steps of Research Process-

### 1) Quantitative Research Process-

- Quantitative research is a formal objective systematic process for generating information about the problem.
- Quantitative research is conducted to describe new situation event or concept.
- In the quantitative research variables are preselected and study are well defined in the first there are the following step of quantitative research process.

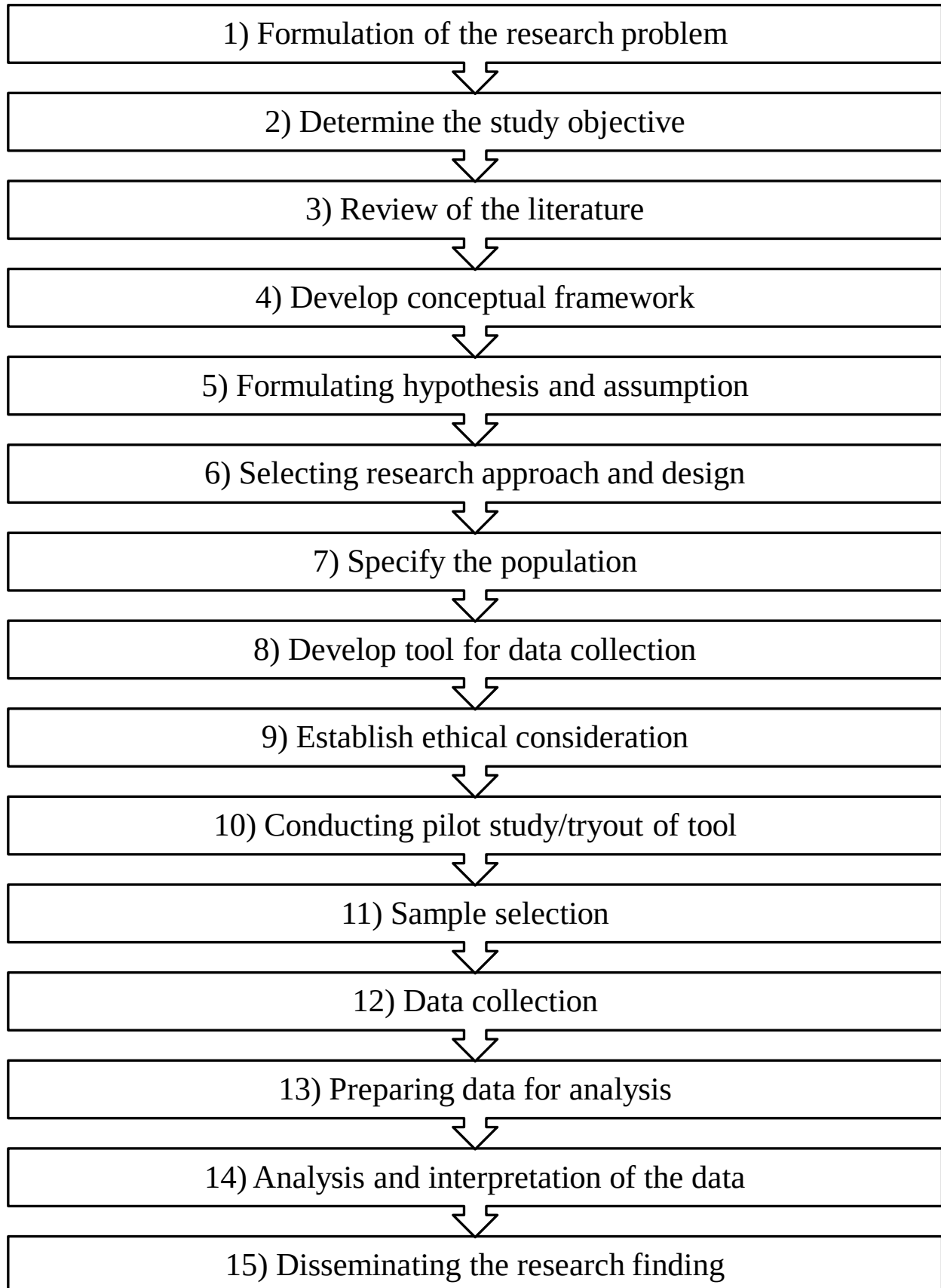


Fig. – Steps of Quantitative Research Process

## 1) Formulation of Research Problem-

- The foremost step of a research process is to discover and interesting and research problem.
- A good research needs a good research problem.
- It is an important step that the whole process can get waste of a research problem not clearly understood.
- The following dimension came under research problem step-

### i. Substantive Dimension-

- Research problem clinical or theoretical importance.

### ii. Methodological Dimension-

- What is the best possible way to study this problem?

### iii. Practical Dimension-

- Are adequate resources available to conduct the study?

### iv. Ethical Dimension-

- Would this problem respect the right of the study subject?

## 2) Determine Study Objective-

- There must be a clear direction to every research project and objective certainly this purpose researcher must have a list of objective for study which provides the guideline for research as to what exactly must be done during the course of a particular research study.
- This step of research process also include writing operational definition of the variables under study that also provide the define direction to research process.

## 3) Review of Literature-

- Literature of review is a summary of previous knowledge generated on the topic of study research can't be conducted in vacuum require a foundation of existing knowledge to plan on conduct a good research.

- Review of literature help the research to understand what is already known about a topic and what need to be further investigated effective literature review need certain basic skill in research such as critical thinking brain storming reading analysis and synthesis.
- Literature review play important role in the development of research project.

#### **4) Developing Conceptual Frame Work-**

- The basic aim of research is to develop refine or test theories nursing profession is considered to need of developing new theories or refining and testing to expend the professional knowledge most of the nursing research, plan to develop, conceptual framework based on the existing nursing or non-nursing theories.
- Conceptual framework not only provides meaning to the research problem but also help you to developing hypothesis or assumption for the research study.

#### **5) Formulating Hypothesis & Assumption-**

- Hypothesis is an assume statement suggesting in answer to a question which may or may not be true.
- E.g. – Co-relational study on alcohol intake and incidence of liver disease among people of an urban community at sikar in this sentence hypothesis will be alcoholic have higher incidence of disease.
- Assumption are the statement that are to be tested to be consider true before they have been scientifically tested.
- E.g. – Satisfaction is largely influence by pre-conserved expectation of customer.

#### **6) Research Approach & Design-**

- A research design is the systematic plan to obtain answer question research design is the blue print of research study which enables the researcher to know what when and how the study will be conducted.

- Who- specify the subject understudy.
- What- specify the variables which are to be focus upon and measured.
- When- specifies the time of study duration and frequency of the observation.
- Where- specifies the setting of study.
- How- specify how the data will be collected.

***In the quantitative research basically use design-***

**i. Experimental design-**

→ In experimental research there is random assignment of subject and availability of control group to compare with the experimental group and manipulation of independent variable to observe effect on dependent variable in an experimental group.

**ii. Quasi-experimental design-**

→ Quasi-experimental study manipulation of independent variable to observe effect on dependent variable but usually they do not control over exterior variable in manipulation.

**iii. Non-experimental design-**

→ In this design research variables are studied without manipulating them in natural setting for the purpose of description exploration explanation or identification of correlation between two variables.

**7) Specify the Population-**

- In quantitative research study is essential to lead about the characteristics and clearly describe the group of subject.
- Research population is an aggregated all the subject or object with specific characteristics.

→ E.g. – Study on prevalence of hospital acquired infection among patient admitted in intensive care unit (ICU) in this sentence population is patient admitted in ICU.

## **8) Developing Tool for Data Collection-**

→ This is the most important and crucial step of development. It is essential to device method and tool to measure research variable in this step researcher collect data through the structured and semi-structured tool.

## **9) Establishing Ethical Consideration-**

→ In the nursing science most of the research is conduct one human where it become more important to ethical consideration during research study can address the ethical issues.

- i. Taking inform consent from participant
- ii. Avoid error in data collection
- iii. Obtaining the permission from competent authority to conduct the study
- iv. Doing justice to participant in analysis data.

## **10) Conduct Pilot Study (Tryout of tool)-**

→ Pilot study is a kind of small skill researcher on the subject but these subjects are not a part of the actual study pilot study is conducted to ensure the flexibility of study and revise methodology.

## **11) Sample Selection-**

→ If it is not practically possible to conduct study on entire population researcher must select a representative part of population a sample can be selected by using either probability of non-probability sampling technique depend on various factor like nature of the study availability of time, money, resources and researcher knowledge about population.

**12) Data Collection-**

→ It is most time consuming step of researcher process which involve direct or indirect interaction with respondent to gather information pertaining to the topic under study. It must be carefully plan and implemented to collect relevant information by using preplan method technique and tool of the data collection.

**13) Preparing Data for Analysis-**

→ In qualitative studies careful checking of every tool for completeness and coding is the main activity during this step of the research process.

**14) Analysis & Interpretation of Data-**

→ In the quantitative research study numerical data must be organize in an orderly and sequence manner and processing is require because research question can be answered in numerical tool in the quantitative research data may be analyze by using the description and inferential statistics.

**15) Disseminating the Research Finding-**

- Research may fail in achievement of its objective if finding are not disseminated to its user.
- Research finding may be communicated through writing of research theories article or presenting a oral research report and scientific professional conference.

**2) Qualitative Research Process-**

- Qualitative research is a systematic and subjective approach to describe event of life experience and to give intended meaning the process of qualitative research is relatively less formally plan where planning of research move.
- There are the following step of qualitative research-

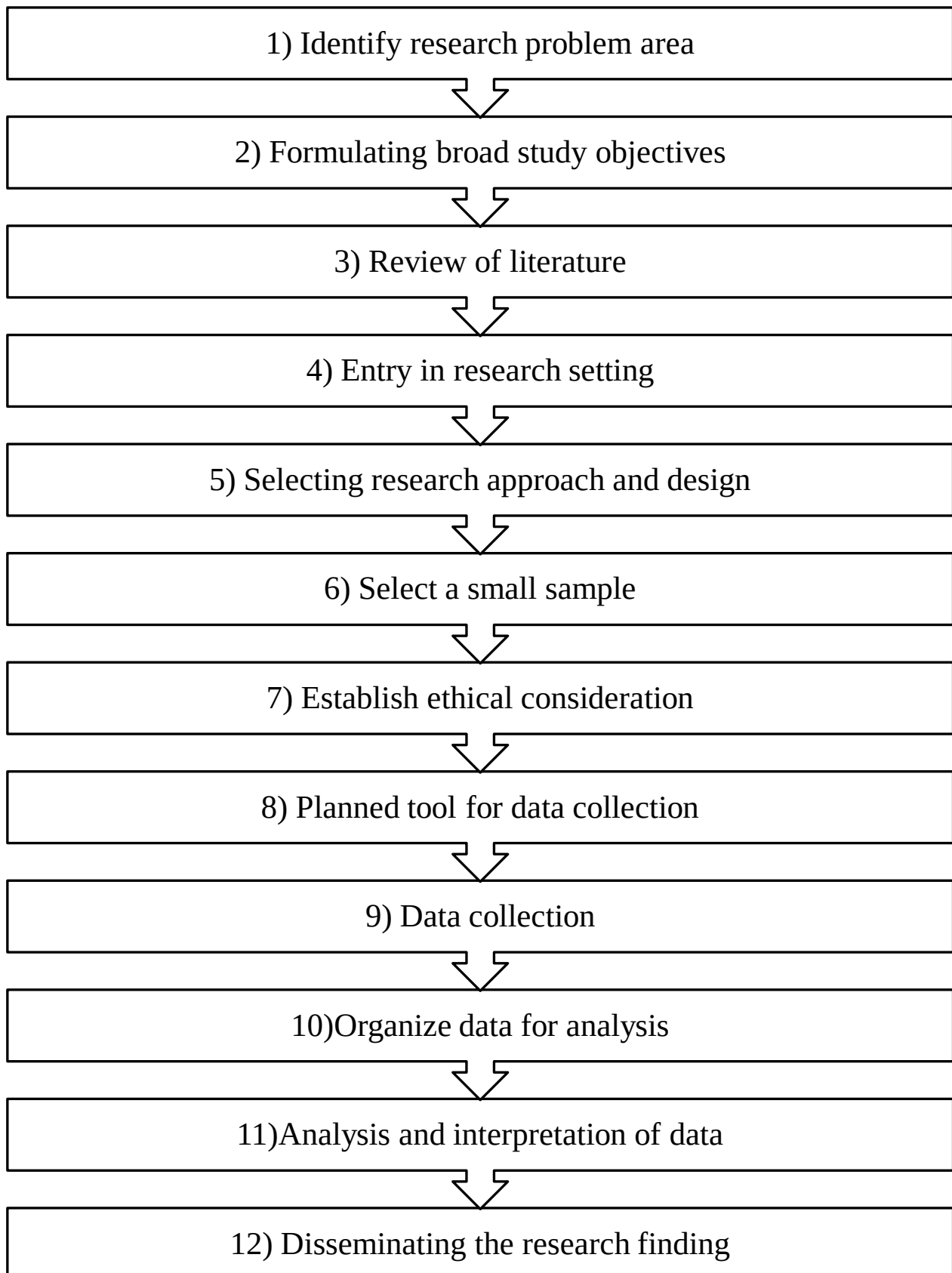


Fig. – Steps of Qualitative Research Process

### **1) Identify Research Problem Area-**

→ In qualitative research initially a broad research area is identified and then focus may be narrow down as the research get more information and experience in research setting qualitative research begin with in general topic of research problem and the group and population expected to study in qualitative study in initiate statement of problem could be a study on experience of tsunami survival in coastal region of the state of Tamilnadu.

### **2) Formulating Broad Study Objectives-**

→ Objective as guideline to conduct a research study in qualitative studies only broad objective may be plan which could later modified base on the need of the research.

### **3) Review of Literature-**

- There are different thought about literature review in qualitative study, qualitative research is a systematic approach of study a phenomena where researcher may learn about the work that has been already done on the particular phenomena under study.
- In qualitative studies understanding the phenomena and population understanding that help the research to gain some amount of prior information to have on effective future planning.

### **4) Entry in Research Setting-**

→ In qualitative study the researcher is not much aware about the population and phenomena. This is an important step for further planning and education of the further planning and education of the research process entry in research setting require prior permission from compliant authorities as well as contact with the people in the area of interest.

## **5) Selecting Research Approach & Design-**

→ There are a few common design use in the qualitative studies there are following design-

### **a. Phenomenological design-**

→ It may be use if the researcher is interested in a description of experience of people.

→ E.g. – A study or experience of people who survive in the disaster in north India.

### **b. Ethnographic design-**

→ It involves the collection and analysis of data about cultural group in ethnographic research, researcher like and become part of the culture.

### **c. Case study-**

→ Case study is in a depth examination of people object or institute.

### **d. Historical research-**

→ This approach of qualitative research is concerned with identification location evaluation and synthesizes data from the past.

## **6) Select a small sample-**

→ Qualitative studies are in depth inquiries there for generally a quit small sample (10-15). A selected to study there is no defined set of tool for exact sample size for a qualitative study.

## **7) Establish Ethical Consideration-**

→ Same as quantitative study there is more important concerned in qualitative study because of them more intimate nature of the

relationship that typically develops between the researcher and the study participant.

### **8) Planned Tool for Data Collection-**

→ Tool used for data collection in qualitative research are generally semi-structured and unstructured most of the item are open ended the answer of which should be taken in the form of sequence data may be collected through videotaping, tap recording, focus, group interview, photograph etc.

### **9) Collecting Data-**

→ In qualitative research data collection is the crustier and time consuming step of the research process interview and observation is two most common time of data collection method used in the qualitative studies.

### **10) Organizing Data for Analysis-**

→ Before analysis data is organized for data analysis and interpretation by the following technique-

- a) Listing
- b) Categorizing
- c) Comparing
- d) Laddering

### **11) Analysis & Interpretation of Data-**

→ In qualitative studies analysis of data actually begin with data collection and some of the researcher argue that there is no separate step of analysis of data in qualitative research usually involve and examination of word rather than number in the case of qualitative large amount of data is collected in form of word which could be 1000 pages of notes.

**Note-**

- They are no universally accepted tool for qualitative data. Qualitative data is analysis by descriptive coding, diagrams and arithmetic analysis.

**12) Disseminating Research Finding-**

- Research finding may be communicated by writing research thesis by articles or presenting an oral research report at scientific professional conference. Research finding must be communicated in a standardized format according to international, national or institutional guideline.

**Hypothesis-****Introduction-**

- A hypothesis is a formal tentative statement of the expected relationship between two or more variables under study.
- Hypothesis helps to translate the research problem and objectives into a clear explanation or prediction of the expected results or outcomes of the research study.
- Hypothesis also influences study design, sampling methods, data collection process, and interpretation of the research findings.

**Definition-**

- Polite and Beck define hypothesis as a statement of researcher's expectations about the relationship between the variables under investigation.
- A hypothesis is an assumption statement about the relationship between two or more variables that suggest an answer to the research question.

## **Importance of Hypothesis in Research-**

- Hypothesis enables the researcher to objectively investigate new areas of discovery.
- It provides objectivity to the research activity.
- The formulation of hypothesis provides a study with focus and guides the researcher about the specific aspects of research problems to investigate.
- It provides the insight to the researcher about what data to collect and what not to collect, thereby providing focus to the study.
- It also provides directions to conduct research such as defining the sources and relevance of data.
- It provides clear and specific goals to the researchers.
- It provides a bridge between theory and reality.
- It suggests which type of research is likely to be most appropriate.

## **Characteristics of a Good Hypothesis-**

- A good hypothesis must be written in declarative statements, using present tense.
- It must contain variables, population under study, and should be relevant to the research problem and objectives, which can be easily tested.
- The main characteristics of a good hypothesis are as follows:

### ***1) Conceptual clarity-***

- Hypothesis should consist of clearly defined and understandable concepts.
- It should be stated in very clear terms, the meaning and implication of which cannot be doubted.

### ***2) Empirical referents-***

- Research must have an ultimate empirical referent.
- No usable hypothesis can embody moral judgements.

**3) Objectivity-**

→ Hypothesis must be objective, which facilitates objectivity in data collection and keeps the research activity free from researcher value-judgement or bias.

**4) Specificity-**

→ A good hypothesis should be specific, not general, and should explain the expected relations between variables.

**5) Consistency-**

→ A hypothesis should be consistent with an existing body of theories, research findings, and other hypotheses.

→ It should correspond with existing knowledge.

**6) Simplicity-**

→ A hypothesis should be formulated in simple and understandable terms. It should require fewer conditions and assumptions.

**Sources of Hypotheses-**

→ These are following sources of hypothesis:

- ✓ Magazines & newspapers
- ✓ Research reports
- ✓ Encyclopedia & dictionary
- ✓ Theses
- ✓ Electronic database
- ✓ Books
- ✓ Journals
- ✓ Conference papers

## Types of Hypothesis-

→ These are following types of hypothesis:

### 1) Simple & Complex Hypotheses-

#### *a) Simple hypothesis:*

→ It is a statement that reflects the relationship between two variables.

→ E.g. – The lower the level of haemoglobin, the higher the risk of infection among postpartum women.

#### *b) Complex hypothesis:*

→ It is a statement that reflects the relationship between more than two variables.

→ E.g. – Satisfaction is higher among patients who are older and dwelling in rural areas than those who are younger and dwelling in urban areas.

### 2) Associative & Causal Hypotheses-

#### *a) Associative hypothesis:*

→ It reflects a relationship between variables that occurs or exists in natural settings without manipulation.

→ E.g. – Communication skills of health care providers and cost of care are related to the satisfaction of patients.

#### *b) Causal hypothesis:*

→ It predicts the cause-and-effect relationship between two or more dependent and independent variables in experimental or interventional settings, where the independent variable is manipulated by the researcher to examine the effect on the dependent variable.

→ E.g. – Prevalence of pin site infection is lower in patients who receive pin site care with hydrogen peroxide as compared to patients who receive pin site care with povidone-iodine solution.

### 3) Directional & Non-directional Hypotheses-

#### ***a) Directional hypothesis:***

- It specifies not only the existence, but also the expected direction of the relationship between variables.
- Directional hypothesis states the nature of the relationship between two or more variables such as positive, negative, or no relationship.
- E.g. – There is a positive relationship between years of nursing experience and job satisfaction among nurses.

#### ***b) Non-directional hypothesis:***

- It reflects the relationship between two or more variables, but it does not specify the anticipated direction and nature of relationship such as positive or negative.
- E.g. – There is relationship between years of nursing experience and job satisfaction among nurses.

### 4) Null & Research Hypotheses-

#### ***a) Null hypothesis ( $H_0$ ):***

- It is also known as statistical hypothesis and is used for statistical testing and interpretation of statistical outcomes.
- E.g. – There is no relationship between smoking and the incidence of coronary artery disease.

#### ***b) Research hypothesis ( $H_1$ ):***

- It is also known as alternative hypothesis. It states the existence of a relationship between two or more variables.
- E.g. – There is a relationship between smoking and incidence of lung cancer.

## Formulating Hypotheses-

- Hypotheses are usually formulated in quantitative research studies, which investigate the interplay of dependent and independent variables such as experimental and correlation research studies, while in other studies use of hypothesis is inappropriate.
- There is no single way to develop an excellent hypothesis.
- These are following point to follow formulation hypothesis:
  - Hypotheses are always written before starting the study and should not be changed after the study results are examined.
  - They are primarily derived from the statement of research problem, study variables, operational definitions, and relevant theories.
  - Researcher uses null form as well as alternative form for formulating research hypothesis.
  - Hypothesis should be stated using declarative sentence and present tense.
  - Hypothesis must be stated in empirically testable form. For example, nurse practitioners are better health care providers than physicians.
  - It is better that one hypothesis contains only single predication about relationships of study variables.
  - Ideally, the number of hypothesis should be restricted to six or less because more number of hypotheses leads to confusion and increased work in data collection, analysis, and interpretations of data.
- E.g. – A well-stated research hypothesis is written in declarative directional– alternative form, which predicts the nature of relationship between dependent and independent variables and also including population under study.
  - There is a significant positive correlation between years of experience and level of nurse's job satisfaction.
  - Male patients have higher satisfaction with nursing care as compared to their female counterparts.

## **Assumption- Introduction-**

- An assumption is a realistic expectation which is something that we believe to be true.
- In other words, an assumption is an act of faith which does not have empirical evidences to support it.
- Assumptions provide a basis to develop theories and research instruments and, therefore, influence the development and implementation of research process.
- Assumptions are essential and an integral part of the research.

## **Definition-**

- Assumptions are statements that are taken for granted or are considered true, even though they have not been scientifically tested.

## **Uses of Assumption in Research-**

- Research is built upon assumptions, as a foundation is needed to move forward. One must assume something to discover something.
- Assumptions guide the research argumentation, evidence investigation, generalization and conclusion of the study.
- Assumptions listed in research paper may be good sources of the research questions.
- Assumption about the phenomenon under study may help researcher to declare statements of the hypotheses.
- The assumptions found to be true through research studies help in expansion of professional body of knowledge.
- The power of a research study is only as strong as the clarity of the basic assumptions of the particular study.

## **Types of Assumptions-**

→ These are following types of assumptions:

### **1) Universal Assumptions-**

→ They are also known as descriptive assumptions.

→ Universal assumptions are beliefs that are assumed to be true by a large part of society, but testing such assumptions is not always possible.

→ E.g. – There is a supernatural power that governs this universe.

### **2) Theoretical Assumptions-**

→ Assumptions may also be drawn from existing theories.

→ If a research study is based on an existing theory, the assumption of the particular theory may become an assumption of that particular research study.

→ E.g. – A study based on Roy's Adaptation Model will use assumption of this particular theoretical model.

### **3) Prescriptive Assumptions-**

→ They are based on the implicit preferences of the particular person's values.

→ E.g. – Assumptions on sexuality among adolescents made by Indian and western researchers could be different.

### **4) Research Study Assumptions-**

→ Some common sense assumptions may be developed to conduct a particular study, which could be based on the observations mentioned in the findings of a previous research or the personal experiences of the researcher and so on.

→ E.g. – Prevalence of coronary artery disease is more common among urban people as compared to rural people.

### 5) Warranted Assumptions-

- These are stated along with evidence to support.
- E.g. – Regular prayers bring success because they boost morale. It is always better to use warranted assumptions in research studies.

### 6) Unwarranted Assumptions-

- These are stated without any supportive evidence.
- E.g. – Almighty God exists everywhere in this universe.

### Differences between Assumptions & Hypothesis-

| S.N. | Assumption                                                                                                              | Hypothesis                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1)   | Assumptions are basically beliefs and ideas that we hold to be true.                                                    | Hypothesis is a prediction.                                                              |
| 2)   | Often with little or no evidence and are not statistically tested in research.                                          | Can be statistically tested and may be accepted or rejected.                             |
| 3)   | Beliefs about the variables.                                                                                            | Beliefs about the variables.                                                             |
| 4)   | Based on the beliefs, researchers attempt to discover the correlation only through testing hypothesis driven from them. | Predicts a relation between variables and is statistically tested to conclude the study. |
| 5)   | Assumptions are generally formulated through qualitative.                                                               | Used in quantitative research studies.                                                   |

### Differences between Delimitations & Limitations-

| S.N. | Delimitations                                                         | Limitations                                                                                                                                                        |
|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1)   | The delimitations of study are the explanation of the scope of study. | The limitations of the study are those characteristics of design or methodology that impacted or influenced the interpretation of the findings from your research. |
| 2)   | Delimitation study allows the writer to explain why certain           | They are the constraints on generalizability, application to                                                                                                       |

| S.N. | Delimitations                                                                                                                  | Limitations                                                                                                                                                 |
|------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | aspects of a subject were chosen & why others were excluded.                                                                   | practice and utility of findings.                                                                                                                           |
| 3)   | Delimitations are set so that your goals do not become impossibly large to complete.                                           | Limitations that are the result of the ways in which you initially chose to design the study and the method used to establish internal & external validity. |
| 4)   | Delimitations are boundaries that are set by the researcher in order to control the range of a study.                          | Limitation point out strengths and weakness of the research.                                                                                                |
| 5)   | E.g. – Research questions, variables, theoretical objectives that you have adopted and populations chosen as targets to study. | E.g. – If you conducted a survey to obtain your research results, your sample (participants) were asked to respond to the survey questions.                 |

## Ethics in Nursing Research- Definition-

→ Ethics in nursing research can be defined as the act of moral principles that the researcher has to follow while conducting nursing research to ensure the rights and welfare of individuals, groups, or community under study.

## Importance of Ethics in Nursing Research-

→ Following are some of the important reasons to support importance of ethics in nursing research:

- 1) Protect the vulnerable group and other study participants from harmful effects of the experimental interventions.
- 2) Safeguard the participants from exploitation by researchers.
- 3) Establish the risk– benefit ratio for the study subjects.

- 4) Ensure the fullest respect, dignity, privacy, disclosure of information, and fair treatment for study subjects.
- 5) Build the capability of subjects to accept or reject participation in study and to have access to informed written consent for participation in research study.

## **Code of Ethics in Nursing Research-**

→ These are following code of ethics in nursing research:

### ***1) Informed Consent Document-***

- In nursing research involving human participants, the investigator must obtain the informed consent of the prospective participant.
- It is the responsibility of the investigator to prepare an informed consent document (ICD), which includes (a) participant information sheet (PIS) and (b) informed consent form (ICF), and get it approved by ethical committee (EC) and let each participant read the document carefully, clarify all the doubts and obtain an informed written voluntary consent from each study participant before the data collection begins in the study.

### ***2) Institutional Ethical Committee-***

- It has been observed that large number of nursing studies conducted by nurse researchers in India especially as postgraduate thesis are not submitted for review by the ethical committee and the research is conducted only based on permission obtained from the authority of selected study setting.
- The scientific and ethical review of a research proposal is a primary responsibility of the EC (Ethical Committee).

**3) ICMR Ethical Guidelines for Biomedical and Health Research-**

- The Indian Council of Medical Research (ICMR) brought out the 'Policy Statement on Ethical Considerations Involved in Research on Human Subjects' in 1980 and revised these guidelines in 2000 as the 'Ethical Guidelines for Biomedical Research on Human Subjects'.
- In 2006, a revised Ethical Guidelines for Biomedical Research on Human Participants was released.

**4) Code of Ethics for Nurses in India-**

- Indian Nursing Council (INC) has published the Code of Ethics for Nurses in India in the year 2006.
- The code of ethics for nurses is critical for building professionalism and accountability.
- Ethical considerations are vital in any area dealing with human beings including nursing research because they represent values, rights, and relationships.
- These are following code of ethics:
  - The nurse respects the uniqueness of an individual in provision of care.
  - The nurse respects the rights of individuals as partners in care and helps in making informed choices.
  - The nurse respects individuals' rights to privacy, maintains confidentiality and shares information judiciously.
  - The nurse maintains competence in order to render quality nursing care.
  - The nurse is obliged to practice within the framework of ethical, professional and legal boundaries.
  - The nurse is obliged to work harmoniously with members of the health team.
  - The nurse commits to reciprocate the trust invested in nursing profession by the society.

## **Research Problem/Question**

### **Definition of Research Problem-**

- Research problem is a troubling area or topic which needs an inquiry to get better understanding or solution.  
E.g. – Female feticide in India could be a troubling topic of the present time.
- Problem is a statement that asks what relationship between two or more variables.
- Research problem is an area of concern where there is a gap in the knowledge base needed for professional practice.

### **Identification of Research Problem-**

- Identification of research problem is the first and most important step in the research process.
- Generally, a broad area is selected and then a board topic is delimited or narrowed down to a specific one–sentence statement of the problem.

### **Sources of the Research Problem-**

#### **1) Personal experience-**

- Day to day personnel experience of researcher may serve as good sources of ideas to formulate a research problem.  
E.g. – A researcher observe domestic violence suffered by wife of alcoholic husband.

#### **2) Practical experience-**

- Nurses get plenty of ideas to formulate research problem from their clinical experience.  
E.g. – Nurse Find the unrestricted visiting hour a surgical ward reduce the analgesic demand among post operative patient.

**3) Critical appraisal of literature-**

→ When he critically studies books and articles related to the subject of over interest including research report articles etc.

**4) Previous research-**

→ A body of knowledge should be developing on a sound foundation of research finding at the end of research further research problem are suggested base on the previous research.

**5) Existing theories-**

→ Research is a process of theory development and theory testing nurses use many theories from other discipline in their practice.

**6) Social issues-**

→ Some topics are suggested by more social or political issues of relevance to the health care community.

E.g. – HIV female criticize, sexual harassment, domestic violence and gender equality in health care and in research are some of the current social and political issues.

**7) Brain storming-**

→ Brain storming sessions are good technique to find new question where an intensified discussion among interested people of the profession is conducted to find more ideas to formulate a good research problem.

**8) Folklores-**

→ Common belief could be right or wrong.

E.g. – It is generally believed that study just before the test decrease the score.

**9) Exposure to the field situation-**

→ During field exposure researcher get variety of experience which may provide, plenty of ideas to formulate research problem.

E.g. – Working in field of researcher observe specific traditional practice for cure of a disease condition.

**10) Consultation with expert-**

→ Experts are believed to a sound experience of their respective field which may suggest a significant problem to be studied.

**Criteria for Selecting a Good Research Problem-****1) Significant to nursing profession-**

→ Research problem is significant for nursing profession when it is directed to develop or refine the body of professional knowledge that fulfill the following criteria-

- a) Nursing profession and patient nurse and health care fraternity will benefit from the study.
- b) The result will improve clinical nursing practice.
- c) Promote nursing theories.

**2) Original-**

→ Every research problem should be new and unique; it is the key responsibility of a researcher that an innovative knowledge is use for selecting a research problem.

**3) Feasible-**

→ Feasibility is an essential consideration of any research project, feasibility of research problem reference to time availability of subject facilities equipment money and ethical consideration should be checked.

→ These are feasible:

- Time
- Cost
- Equipments & supplies
- Administrative support
- Peer support
- Availability of subject
- Researcher competence
- Ethical consideration

#### **4) Solvable-**

→ Problem selected is considered good only if it is solvable. E.g. – A researcher select a research problem to know the existing of god in this universe.

#### **5) Current-**

→ A good research problem must be based on the current problem and need of a profession.

#### **6) Interesting-**

→ A research problem can only be considered good if it is in accordance with researcher field of interest.

### **Formulation of the Research Problem-**

→ Formulation of research problem is a complex and long process which include following steps-

- i. Selection of a research area.
- ii. Review of literature and related theories.
- iii. Delimiting the research topic.
- iv. Evaluating the research problem.
- v. Formulating final statement of research problem.

**i. Selection of Research Area-**

→ Formulation with the research begin with selection of a broad research topic from personnel experience literature, previous research and theories in which researcher is interested and has significance for nursing profession.

E.g. – A researcher get an idea to conduct a study on the female feticide. In this sentence researcher initially begin with such broad research topic.

**ii. Review of Literature & Theories-**

→ After getting a broad idea for research, researcher need to review the nursing literature is review to know what has already be done in this selected area of research.

→ Researcher can plan a research topic to further expand the existing body of knowledge in respective area of research.

**iii. Delimiting the Research Topic-**

→ In this step researcher processing as a general research area of interest to more specific topic of research to conduct a study.

E.g. – Researcher decides to conduct study on female feticide later in the stage researcher limit it to specific research topic.

→ A study on perception of women about causes and prevention of female feticide in selected rural commonly of Sikar district.

**iv. Evaluating the Research Problem-**

→ Researcher is clear about the specific research problem. Research problem must be carefully evaluated for its significant and feasibility.

**v. Formulating Final Statement of Research Problem-**

→ After establishing the significance researcher finally formulate a final statement of research problem.

→ Statement of research problem could be *declarative or interrogative format*.

***i) Declarative format-***

→ E.g. – A descriptive study of prevalence of bed sore among unconscious patient admitted in intensive care unit of S.K. hospital, sikar.

***ii) Interrogative format-***

→ An interrogative format a research problem is in the question form.

→ E.g. – What is the effect of morning sickness on pregnant women in selected rural community of sikar district?

**Research Objectives-**

→ Research objective is a clear, conscious, declarative statement which provides direction to investigate the variables. Research objective focus on the way to measure the variables.

→ Objective are directed toward identify the relationship or difference between variables.

**Characteristics of Research Objectives-**

→ Research objective will be SMART (S-specific, M-measurable, A-attainment, R-realistic, T-time bound).

→ These are following specific characteristics:

✓ Measurable

✓ Attainable

✓ Realistic

✓ Time-bound

✓ Relevant

✓ Feasible

✓ Logical

✓ Observable

## **Need of Research Objectives-**

→ Formulation of research objective will help the researcher-

### **1) Focus-**

→ With clearly define objective researcher can focus on study in this the researcher will concentrate on the way to achievement.

### **2) Avoid-**

→ Formulation of research objective help the researcher avoids the collection of data which are not restrict necessary for understanding and solving problem.

### **3) Direction-**

→ Specific objective will facilitate the development of research methodology and will help to orient the collection analysis interpretation and utilization of data.

## **Types of the Research Objectives-**

→ The research objective are two types-

### **1) General Objective-**

→ General objective are broad goal to be achieve.

### **2) Specific Objective-**

→ Specific objective are short term and narrowing focus general objective can be broken into small logical, small part to the form of specific objective.

## **Variables-**

### **Introduction-**

- Variables are anything that has quantity and quality that varies.
- In other words, variables are qualities, properties, or characteristics of person, things, or situations that change or vary.

### **Meaning of Variables-**

- Variable is central idea in Research. Simply defined as Variable is a concept that varies. It can be in quality, intensity, amount & types.
- Variable is a measureable characteristics that varies. It may change from group to group, person to person or within person over time
- E.g. – Gender is a variable it can take two values: Male & Female.

### **Types of Variables-**

- These are following types of variables:
  - 1) Independent variables
  - 2) Dependent variables
  - 3) Research variables
  - 4) Demographic variables
  - 5) Extraneous variables

#### **1) Independent Variable-**

- It is a presumed cause/stimulus or activity that is manipulated or varied by the researcher to create the effect on the dependent variable.
- In non-experimental research, it is more appropriate to use term ‘predictor variable’ instead of independent variable, because ‘presumed cause’ is not manipulated by the researcher.
- E.g. – IQ varies with age.

**2) Dependent Variable-**

- It is a presumed effect of the independent variable, which researcher wants to predict or explain.
- E.g. – A researcher is observing the effect of ginger intake (independent variable) on morning sickness symptoms (dependent variable) among pregnant women.

**3) Research Variables-**

- In descriptive, exploratory, comparative, and qualitative research studies, variables are observed or measured in natural setting as they exist, without manipulating or imposing the effect of intervention or treatment.
- E.g. – An exploratory study on factors contributing to sleep disturbance among patients admitted in selected intensive care units of AIIMS, New Delhi. In this research study, factor contributing to sleep disturbance is a research variable, which is observed in natural setting without manipulating it.

**4) Demographic Variables-**

- In most research studies, researchers make the attempt to study the sample characteristics and present them in research findings.
- In addition, sometimes researchers even try to establish relations of the demographic variables with the research variables.
- These characteristics and attributes of the study subjects are considered as demographic variables.
- Common demographic variables are age, gender, educational status, religion, social class, marital status, habitat, occupation, income, medical diagnosis, etc.

### 5) Extraneous Variables-

- Extraneous variables are the factors that are not part of the study but may affect the measurement of the study dependent or outcome variables.
- They are commonly known as ‘confounding variables’ or confounders.
- These variables are usually recognized and controlled by researchers in quasi-experimental and experimental studies.
- E.g. – A study is conducted to assess the effect of two different pin site care protocols on prevention of pin site infection among patients with external skeletal fixation. In this study, pin site care protocols are independent variables and pin site infection is the dependent variable.

## **Review of Literature**

### **Definition of Review of Literature-**

- Literature review is a body of text that aims to review the critical point of knowledge on a particular topic of research.
- Review literatures are defined as broad comprehensive in depth systematic and critical review of scholarly publication and publish printed or audio visual material and personal communication.
- Literature review is an evaluative report of information found in the literature related to selected area of study, the review describe, summarize, evaluate and clarify topic and give a frame work for study.

### **Importance of Review of Literature-**

- Identification of research problem and development or refinement of research question.
- Generation of useful research question or project activities.
- Orientation to what is known and not known an area of inquiry to a certain what research can best contribute to knowledge.
- Discovery of unanswered question.
- Identification of relevant theoretical or conceptual framework for research problem.
- Identification or development of new or refine clinical intervention to test through empirical research.
- Help in planning the methodology of present research study.
- It also helps a development of research instrument.
- Identification of suitable design and data collection method for research study.

**Purposes of Literature Review-**

- Place each in the context of its contribution to the understanding of subject under review.
- Describe the relationship of each study to other research study under consideration.
- Identify areas of prior scholarship to prevent duplication of efforts.
- Point a way forward for further research.
- See what has and has not been investigated.
- Develop general explanation for observe variation in a behavior.
- Identify potential relationship between concepts and to identify research hypothesis.
- Identify data sources that other researchers have use.
- Develop alternative research project.

**Types of Review of Literature-**

→ These are following types-

- 1) Traditional or narrative literature review
- 2) Systematic literature review
- 3) Meta-analysis
- 4) Meta-synthesis

**1) Traditional or Narrative Literature Review-**

- This type of literature review presents the summary of literature and draws conclusions about the topic in question.
- In this, the body of literature is composed of a summary of facts on the subject area from the primary and secondary sources.
- The main aim of this type of literature review is to gather a volume of literature in a specific subject area, summarizing, and synthesizing it.

## **2) Systematic Literature Review-**

- The systematic literature review is significantly different from the traditional or narrative literature review, which includes a more systematic and rigorous approach to literature review in a specific subject area.
- The main aim of systematic literature review is to find the answer for well-focused questions of clinical practice.
- It is suggested that a systematic literature review should include the methods used for searching, evaluating and synthesizing the literature.

## **3) Meta-analysis-**

- Meta-analysis is an advanced form of systemic literature review, which includes a large body of findings from quantitative studies and computes the statistical analysis in order to draw the integrated and cumulative inferences and conclusion.
- The main aim of meta-analysis is to involve findings of several quantitative studies on single subject area and carry out statistical computations on them using standardized statistical techniques and procedures.

## **4) Meta-synthesis-**

- Contrary to meta-analysis, meta-synthesis is a non-statistical technique used to integrate, evaluate, and interpret the findings of multiple qualitative research studies.
- Findings from phenomenological, grounded theory or ethnographic studies may be integrated and used.
- Such studies may be combined to identify their common core elements and themes.

**Sources of Review of Literature-**

→ Literature can be reviewed from two main sources:

- 1) Primary sources
- 2) Secondary sources

**1) Primary Sources-**

- Primary sources are original, peer-reviewed, and published research journal articles reported by original researchers.
- In other words, primary sources are the research reports, which are description of studies written by researchers who conducted them.
- A primary source is written by a person who developed the theory or conducted the research, or is the description of an investigation written by the person who conducted it.
- E.g. – An original qualitative study on patient experiences in the ICU: Hupcey, J. E. (2000). Feeling safe: the psychosocial needs of ICU patients. *Journal of Nursing Scholarship*, 32, 361–367.

**2) Secondary Sources-**

- Secondary source research documents are description of studies prepared by someone other than the original researcher.
- They are written by people other than the individuals who developed the theory or conducted the research.
- The secondary sources include the comments and summaries of multiple research studies on one topic, for example, systemic reviews, meta-analysis, meta-synthesis, etc.
- E.g. – A literature review on patient experiences in the ICU: Stein-Parbury, J. & Mckinley, S. (2000). Patient experiences of being in an intensive care unit: a select literature review. *American Journal of Critical Care*, 9, 20–27.

## ***The Main Sources of Review of Literature-***

### **1. Electronic Sources-**

- Computer-assisted literature search has revolutionized the review of literature.
- These searches, however, for a variety of reasons may not provide the desired references.
- General literature search can be conducted through search engines, like Yahoo ([www.yahoosearch.com](http://www.yahoosearch.com)), Google ([www.google.com](http://www.google.com)).
- A search engine is a collection of software programmes that collect information from the Web, index it and put it in a format that is accessible visually on screen at an on-site library or in downloadable, readable written format.
- The most useful and relevant nursing databases are as follows:

#### ***1) CINAHL (Cumulative Index to Nursing and Allied Health Literature)-***

- Accessible at <http://www.cinahl.com>, it contains citations of nursing literature published after 1988.
- Even full-text articles are available on CINAHL Plus, a paid web page.

#### ***2) MEDLINE (Medical Literature Analysis and Retrieved System Online)-***

- It is another electronic source of literature review commonly used by nurses.
- The National Library of Medicine provides free access to MEDLINE through PubMed, available at [www.pubmed.com](http://www.pubmed.com) or <http://www.ncbi.nih.gov/entrez/query.fcgi>.

**3) PubMed-**

→ PubMed can be used to search research abstracts, available at <http://www.pubmed.com>.

**4) British Nursing Index-**

→ The British Nursing Index (BNI) is a leading UK nursing database providing bibliographic references to journal articles from all the major British nursing and midwifery journals, as well as a selection of English-language international journals.

→ It is available from Royal College of Nursing Library e-library database: <http://www.proquest.com/products-services/bni.html>.

**5) MedlinePlus-**

→ It is the National Library of Medicine's websites for consumer health information.

→ It is available at <http://www.medlineplus.gov>.

**6) Nursing & Health Science-**

→ It is a full-text collection of 24 journals published by Sage Publication.

**7) Cochrane Database of System Reviews-**

→ Health care-related literature can be searched from this source, available at <http://www.cochrane.org>.

**8) PsycINFO-**

→ The PsycINFO database belongs to American Psychological Association, and covers literature from psychological or related disciplines.

→ It may be searched at <http://www.psychinfo.com>.

## **2. Printed Sources-**

- Printed sources are also used for literature review. Printed research summary may be located from published abstracts, such as Nursing Research Abstract, Psychological Abstracts, Dissertation Abstract International, Masters Abstract International, etc.
- Following are the main printed sources that can be used to review the relevant literature:
  - ✓ Journals
  - ✓ Research reports
  - ✓ Unpublished dissertations and theses
  - ✓ Magazines and newspapers
  - ✓ Conference papers and proceedings
  - ✓ Encyclopaedias and dictionaries
  - ✓ Books

## **Steps of Review of Literature-**

- These are following steps:

### **Step I: Understanding the concepts of research problem and identifying the key search terms-**

- The journey of literature search begins with understanding and examining the concepts of research problem so that essential key search terms may be identified.
- The key search terms serve as foundation for the overall search of literature.

**Step II: Identifying the relevant sources-**

- After understanding the concepts of research problem and identification of key search terms, we need to locate the relevant sources for literature review.
- The journals are considered as superior and up-to-date source compared to books and other resources, because books are considered outdated by the time they are published after a long process.

**Step III: Searching the literature-**

- Having selected relevant sources of literature review, the next step is to identify in a structured way the appropriate and related information.
- The traditional method of literature search is manual search from the hard copy of journals that are specially related to the topic of interest or those that are likely to cover the topic.

**Step IV: Analyzing and synthesizing the literature-**

- After collecting the relevant literature, it is the stage to analyze and synthesize the collected literature.
- There are several sub stages or systems to achieve this purpose.
- The most popular and simple method is referred to as the preview, question, read, summarize (PQRS) system.

**Step V: Writing the literature review-**

- This is a final step of literature review after completing the adequate appraisal of literature.
- It is time to give a serious thought about how to write a draft of literature review.
- It is essential to avoid long and confusing words and jargon and keep sentences short with one clear message.

**Step VI: Referencing and reviewing final draft of literature-**

- It is the last step of the literature review where the final draft of literature is reviewed for the completeness, accuracy and relevance of the content.
- Furthermore, the reference citations and list of references should be checked with author's name and year of publication to ensure the accuracy of in-text citation and the reference list.
- The creation of list of references with complete detail such as full names of authors, year of publication, title of article, name of journal, volume, issues of journal and page number of the particular article must be archived carefully from the beginning of the literature, so that towards the end, reference listing becomes easy.

## **Research Approaches & Designs**

### **Definition of Research Design-**

- Research Design is a master plan specifies the method and procedure for collecting and analyzing the needed information in a research study.
- Research design is a plan of how, when, where data are to be collected analyzed.
- Research design is the researchers overall plan for answering the research questions or testing the hypothesis.

### **Elements of Research Design-**

- Research design is also known as blue print.
- Research design is also interchangeable with the term methodology there are the following:

#### **1) Approach-**

- It involves the description of plan to investigations the phenomenon under study in a structured, unstructured or combination of method.

#### **2) Population Sample & Sampling Technique-**

- Research design also provides the researcher with direction about populous sample and sampling technique which will be used for the research study.

#### **3) Time, Place & Source of Data Collection-**

- Time (Specified date, month and year of the study), Location (Study setting and source of the data are the other implement essential total in ensure effective planning to conduct a research study).

#### **4) Total & Method of Data Collection-**

- This element of research design involve the description of depend total and method of data collection, questionnaires, direction observation, interview, other method.

**5) Method of the Data Analysis-**

→ A research design must also include the description of the method of data analysis either quantitative or qualitative data analyzed.

**Factors Affecting Selection of the Research Design-**

→ The selection of the research design may be influenced by the following factors:

**1) Nature of the research problem-**

→ This is the key factor which helps the researchers to decide about the selection of research design through the identification of problem. The researcher decides whether it should be investigated through an experimental, quasi-experimental, and non-experimental approach.

**2) Purpose of the study-**

→ Study may be conducted for the purpose of prediction, description, and exploration and correlation research variables.

**3) Research knowledge and experience-**

→ Selection of the research design is largely influenced by researcher knowledge and experience because they avoid using those designs where there is a lack of confidence, relevant knowledge, and experience.

**4) Research interest and motivation-**

→ Interest and motivation level help the researcher decide about the particular research design.

**5) Research ethics & principles-**

→ The incorporation and application of ethical and legal principles in the research design are essential.

→ This includes moral obligation such as respect for participants and their rights.

**6) Subject/Participate-**

→ The number and availability of study may influence the selection of the research design.

**7) Resources-**

→ Research can't be conducted research without resources such as many equipment, facilitates and support from colleagues.

**8) Time-**

→ Time is also a major deciding factor for the selection of research design.

→ E.g. – A researcher need more time to conducted longitudinal studies.

**9) Possible control on extraneous variables-**

→ Design can maximize the results decreased errors and control preexisting or impair condition that may affect the outcome of study.

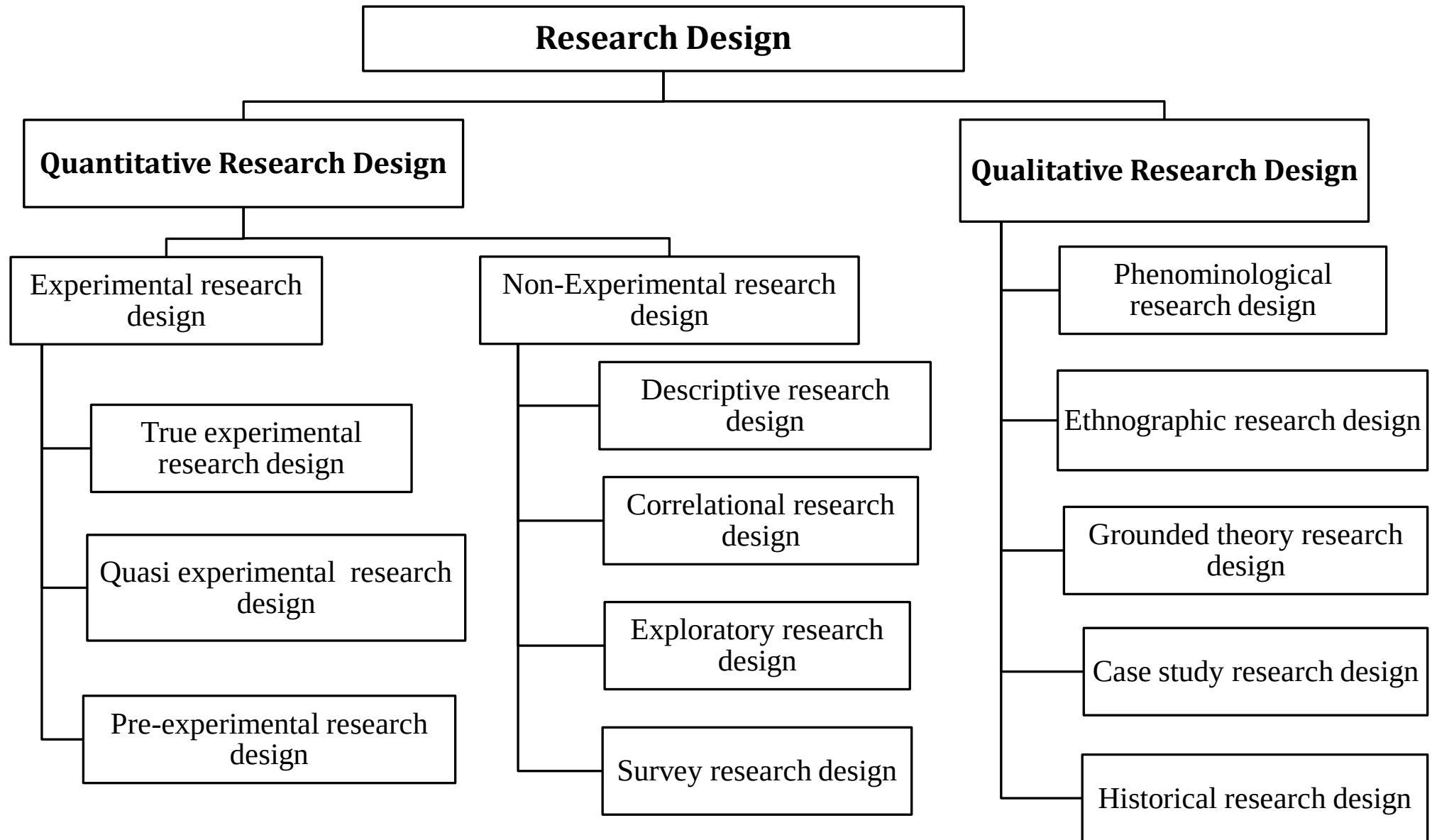
**10) User of the findings-**

→ A research design also involves various methods of data collection and data analysis while choosing a research design researcher must ensure that research design is appropriate for the user study findings.

**Classification of Research Design (Methodology)-**

→ Research methodology is broadly classified as the following:

1. Quantitative Research Design
2. Qualitative Research Design



## **1. Quantitative Research Designs-**

### **I. Experimental Research Design-**

- Experimental is the most scientifically research method. It is defined as observation under control conditions.
- Experimental research differs from non-experimental design in one important aspect.
- Experimental research designs are concerned with examination of the effect of the independent variable on the dependent variable where the independent variable is manipulated to treatment and intervention.

### **Characteristics of the Experimental Research Design-**

#### **1) Manipulation-**

- Manipulation referred to conscious control of the independent variable by the researcher through treatment and intervention to observe its effects on dependent variable.

#### **2) Control-**

- Control is another essential element of true experimental design. Control referred to the use of control group and controlling the effect of the exterior variable on the dependent variable.

#### **3) Randomization-**

- Randomization means that every subject has an equal chance of being assigned to experimental and control group. This is also known as random assignment of subject.
- According to the characteristics experimental research design may be classified into following types:
  - 1) True experimental research design
  - 2) Quasi experimental research design
  - 3) Pre-experimental research design

### 1) True Experimental Research Design-

- True experimental research design are where researchers have complete control over the extraneous variable and can predict confidently that the observed effect on the dependent variable is only due to the manipulation of independent variables.
- Mostly three characteristics of experimental design: Randomization, control and manipulation design.

### Types of the True Experimental Research Design-

- I. Post test control design
- II. Pre and post test design
- III. Solomon four group
- IV. Factorial design

#### I. Post Test Control Design-

- Composed of two randomly assigned groups: experimental and control, but neither of which is pre-tested before the implementation of the treatment. On the experimental group, treatment is implemented, and on the control group, no treatment is implemented. A post-test is carried out on both groups to assess the effect of manipulation.
- This design can be useful where it is not possible to pretest the subject.

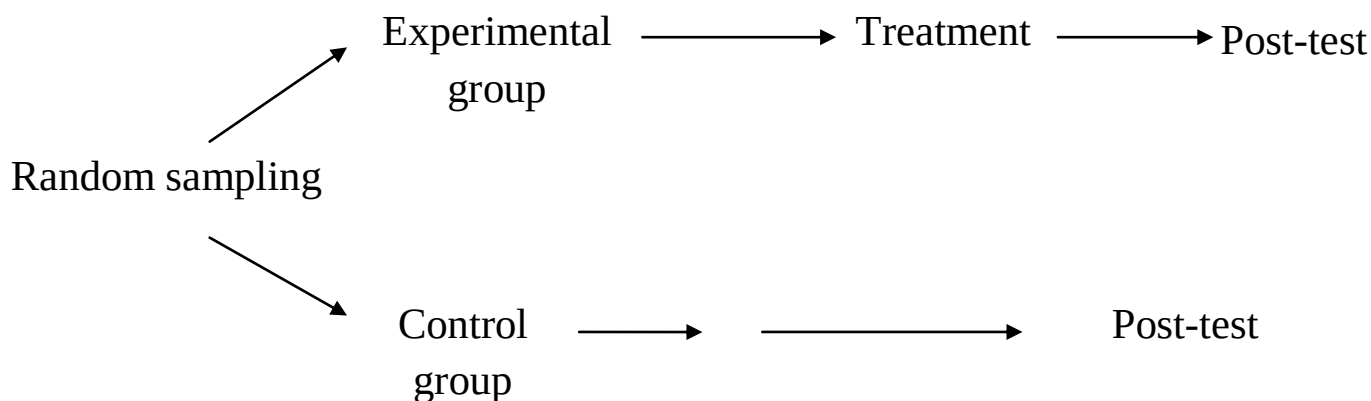


Fig. – Post-test control design

## II. Pre-test & Post-test Design-

→ In this design subject are randomly assign to either the experimental or control group effect of the dependent variable on both the group is seen before treatment (Pretest) than treatment is carried of experimental group only and after treatment observation dependent variable is mode on both the group to examine the effect of the manipulation of independent variable on the dependent variable.

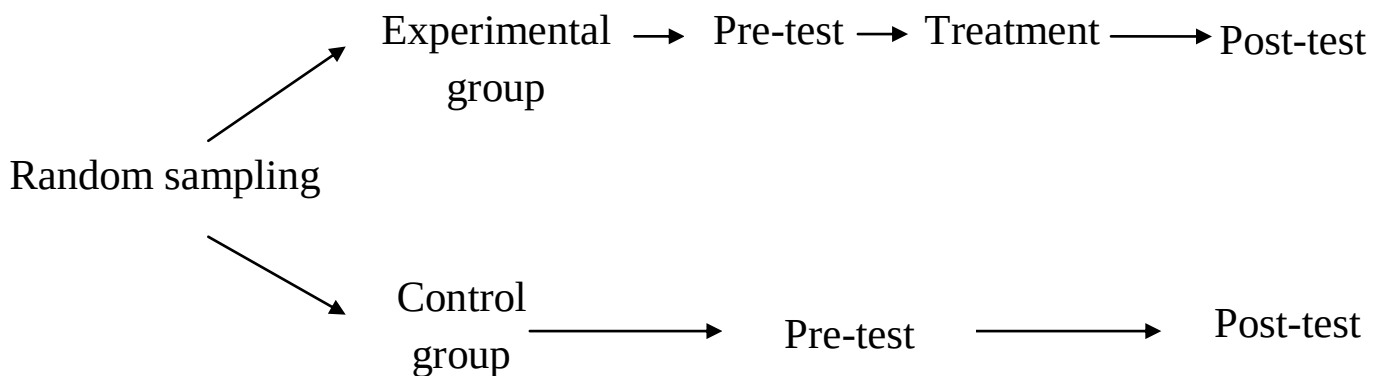


Fig. – Pre-test & Post-test control design

## III. Solomon Four Group's Experimental Research-

- This research method consists of two experimental groups and two control groups.
- In this method, the sample which is randomly selected is divided into 2 main groups that is one experimental group and one control group.
- Further the experimental group is sub-divided into 2 sub-groups and the control group also is sub-divided into 2 sub-groups.
- In this research method pre-test is administered to one experimental sub-group and one control sub-group and pre-test data is collected. Then the experimental treatment is administered only to both the experimental sub-groups.
- Than post test administrated both experimental sub groups and both control sub groups. Then observation is made on all the 4 sub groups and post test data is collected.

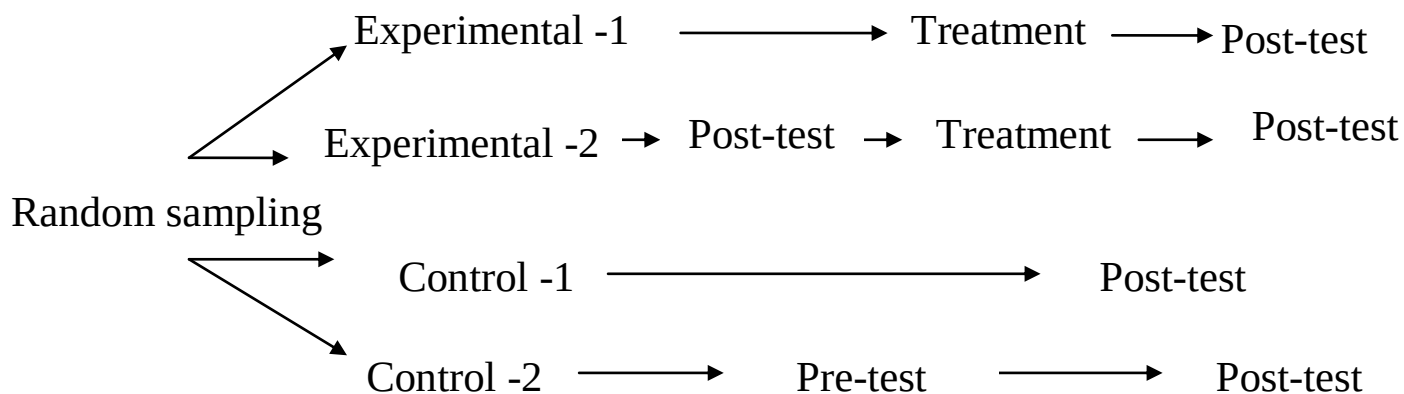


Fig. – Solomon four group's experimental research

#### IV. Factorial Experimental Research Design-

- In factorial design researcher manipulated two or more independent variables simultaneously to observe their effect on the dependent variables. This design is useful when there are more than independent variables called factors to be tested.
- E.g. – A researcher wants to observe the effect of two different protocols of mouth care on prevention of ventilator associated pneumonia when performed at different frequencies in a day.

| Frequency in hours | Protocol of mouth care       |                       |
|--------------------|------------------------------|-----------------------|
|                    | Chlorhexidine ( $\alpha_1$ ) | Saline ( $\alpha_2$ ) |
| 4 hrs $\beta_1$    | $\alpha_1\beta_1$            | $\alpha_2\beta_1$     |
| 6 hrs $\beta_2$    | $\alpha_1\beta_2$            | $\alpha_2\beta_2$     |
| 8 hrs $\beta_3$    | $\alpha_1\beta_3$            | $\alpha_2\beta_3$     |

#### Advantages of True Experimental Research Design-

- ✓ Experimental research designs are considered the most powerful design to establish the causal relationship between independent and dependent variables.
- ✓ In this study the control environment is to a greater degree of purity in observations.

- ✓ Condition not found in natural setting can be created in an experimental settings where the independent variables manipulated by investigator.
- ✓ Experimental approach can be condition in a short period of time that may take near to occur naturally.

### **Disadvantages of True Experimental Research Design-**

- ✓ Most of the time the results of experimental research design can't be replicated in study conducted on human being due to ethical problem.
- ✓ Experimental study conduct in the naturally in like hospital and community of composed of control of exteriors variables.
- ✓ Experimental study takes prolonged time.

### **2) Quasi Experimental Research Design-**

- Quasi experimental research design involved the manipulation of independent variables to observe the effect on dependent variable but it lack at least one of the two characteristics of true experimental design & randomization of control group.

### **Types of the Quasi Experimental Design-**

- There are the following:

#### **I. Non Randomization Control Group Design-**

- It is also known as non equipment control group design. This design is identical to the pretest, post-test control group except there is no as random assign of the subject in experimental and control group.
- In this design experimental and control group are selected randomization variables are observed before intervention later experimental group receive treatment after the post test observation of the variables.

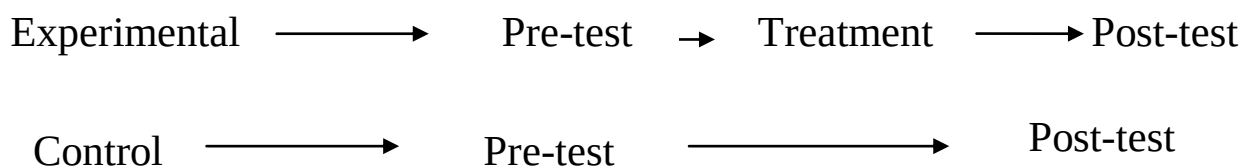
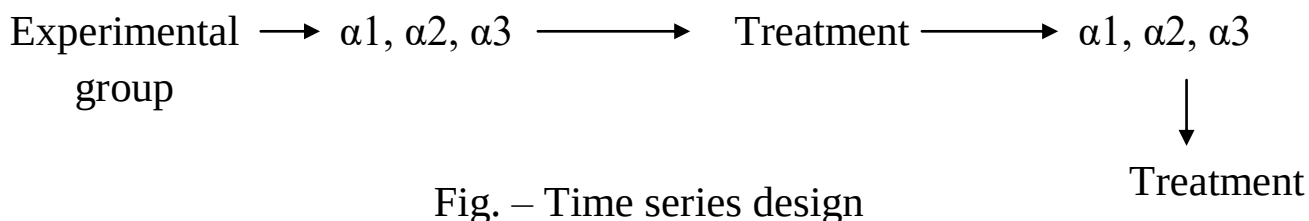


Fig. – Non-randomization control group

## II. Time Series Design-

- This design is useful when the experimental wants to measure the effect of a treatment over a long period of time.
- The experimental would continuous to administer the treatment and measure the effect time of course of experiment.



## Advantages of Quasi Experimental Research Design-

- ✓ Quasi experimental designs are more frequently use because they are more practical and feasible to conduct research study in nursing.
- ✓ This design is more suitable for real, natural setting.
- ✓ It allow researcher to evaluate the impact of independent variables under naturally occur condition.
- ✓ It may be able to establish causal relationship of variables.

## Disadvantages of Quasi Experimental Research Design-

- ✓ There is no control over extraneous variables influencing the dependent variable.
- ✓ The observation of control group over the variables results of the design less reliable and being of established casual effect of relationship.

## 3) Pre-Experimental Research Design-

- This research design is considered very weak because the researcher has very little control over the experiment.

## Types of the Pre-Experimental Research Design-

### I. One Shot Case Design-

- In this research design a single experimental group is exposed to a treatment an observation are made after the implementation of the treatment.
- There is no random assignment to the subject to the experimental and no control group.

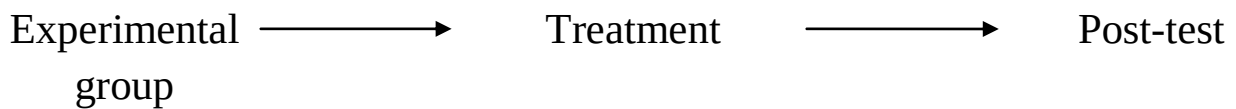


Fig. – One shot case design

### II. One Group Pretest & Post-test Design-

- It is the simplest type of Pre-experimental design where only the experimental group is subjected as the study subjects a pretest observation of the dependent variables is made people implementation to the selected group.
- The treatment is administered and finally a post test observation of the dependent variables is the carried out to assess the efforts of treatment on the group.

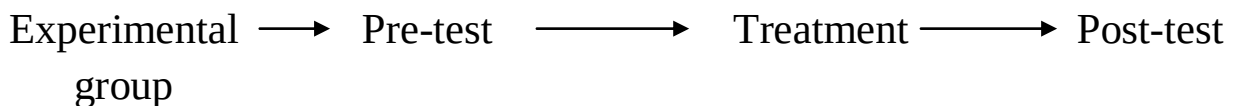


Fig. – One group pre-test & post-test design

**Advantages of Pre-Experimental Research Design-**

- ✓ Very simple and convenient to conduct a study in natural setting.
- ✓ Most suitable free design for the beginner field of the research experimental.

**Disadvantages of Pre-Experimental Research Design-**

- ✓ It is a very weak experimental design to established casual relationship between independent and dependent variables because it controls no threats to internal validity.
- ✓ It has very little control over the research.

**II. Non-Experimental Research Design-**

→ Non experimental research design is one of the broad category of research design in which the researcher observe the phenomenon as they occur naturally and no external variables are introduced in this design variables nor is the setting control.

**Need of the Non-experimental design-**

- ✓ Study in which the independent variable can't be manipulated.
- ✓ Studies or research situation where it is not practically possible to conduct experiment.
- ✓ Descriptive type study that does not required any experimental approach.

**Types of the Non-Experimental Research Design-**

- 1) Descriptive research design
- 2) Correlation research design
- 3) Exploratory research design
- 4) Survey research design

***1) Descriptive Research Design-***

→ The purpose of descriptive study is to observe describe and document aspects of a situation it naturally occur and sometimes to survey and starting point of hypothesis generation of theory development.

## **Main Feature of Descriptive Design-**

- ✓ Descriptive design are use to observe document and Descriptive a phenomena occurring in natural setting without any manipulation or control.
- ✓ Descriptive studies are design to gain more information about characteristics within a particular field of inquire.
- ✓ Descriptive study provides impression of a situation it occurs in natural setting.
- ✓ Descriptive studies don't involve the manipulation of variables and variables are study as they exist in the real word.

## **Types of Descriptive Research Design-**

→ These are the following types:

### **I) Univariate Descriptive Design-**

- Univariate Descriptive designs are under taken to describe the frequency of occurrence of phenomenon. This design doesn't focus on the study of a single variable there may be one or more variables involved in the study.
- E.g. – A researcher is interested in assessing the experience of patient suffering from rheumatoid arthritis in this study the researcher may described symptoms experience by patient under type of treatment they receive during course of disease.

### **II. Exploratory Descriptive Design-**

- Exploratory design is use to identify explore describe the existing phenomenon and it related factor it is not only a simple description of the frequency occurrence of phenomena but it in depth exploration and a study of related factor to improve further understanding about a less understood phenomenon.

- E.g. – An exploratory study to assess the multi factorial dimensions of the fall and home safety measures for alternately people living in selected community in sikar.
- E.g. – An Exploratory study on factor affecting bowel movement in arthritic patient admitted in S.k. hospital sikar.

### **III. Comparative Design-**

- It involves comparing and contrasting two or more sample in study subject on one or more variable obtain at single point of time.
- This design is used to compare to group on the basis of selected attribute such as knowledge level perception and attitudes.
- E.g. – A Comparative study in health among rural and urban old people in Sikar district.

### **2) Correlational Research Design-**

- This is a non-experimental design where researcher examines the relationship between two or more variables in a natural setting without manipulation of independent variable.
- In other words, it is a research design where researchers study the relationship of two or more variables without any intervention.

### **Main Features of Correlational Design-**

- In correlational studies, the researchers examine the strength of relationships between variables by determining how change in one variable is correlated with change in the other variable.
- Generally, correlational studies have independent and dependent variables, but the effect of independent variable is observed on dependent variable without manipulating the independent variable.
- In some correlational studies, identification of the independent and dependent variables is difficult; however, in most correlational studies,

the independent variable is identified, which, without any intervention, influences the dependent variable.

### **Types of Correlational Research Design-**

→ These are following types-

- i. Cohort research design
- ii. Case control research design
- iii. Analytical cross sectional research design

#### **i) Cohort Research Design-**

- A cohort is a group of people who have something in common and who remain a part of a group over an extended time.
- In health care, the subjects in cohort studies are selected by some defining characteristic suspected of being a precursor to or risk factor for a disease or health effect.

### **Types of Cohort Research Design-**

→ These are following three types:

- a) Prospective cohort design
- b) Historical cohort design
- c) Ambispective cohort design

#### **a) Prospective Cohort Design-**

- A prospective cohort study is an observational longitudinal study, where a cohort (a group of subjects with similar characteristics) is followed over a time period, but cohorts differ in certain factors under study to determine how these factors affect the rate of certain dependent outcomes.
- E.g. – A researcher may follow a cohort of middle-aged nurses who vary in terms of body mass index, to the hypothesis that the 15-year

incidence rate of coronary artery disease will be highest among the obese nurses, followed by overweight and the healthy weight.

### ***b) Historical Cohort Design-***

- The historical cohort design is also known as retrospective cohort design, where one can also undertake a cohort study by using information collected in the past and then kept in records or file.
- E.g. – One researcher wanted to assess study outcomes in men with prostate cancer treated with a specific type of radiation therapy. He plans to evaluate existing records to look at survival and tumour recurrence in 1607 men who were treated between 2009 and 2016 and had at least four prostate-specific antigen measurements after radiation.

### ***c) Ambispective Cohort Design-***

- The ambispective cohort study design moves both forward and backward in time.
- With this design, the exposure is measured twice historical and in real time during the study period.
- The design allows a more precise measure of exposure in terms of time.
- E.g. – A study about hip replacement (outcome in the future) can start with adults aged 20–40 years, and their current and future physical activity (contemporary exposure), as well as their school and other historical records can be measured to determine their childhood experience with organized sports-type, intensity and duration.

### ***ii) Case Control Research Design-***

- A design in which the researcher studies the current phenomenon by seeking information from past is also known as a retrospective research design.
- In this design, the researcher links the present phenomenon with the past events. In other words, the researcher has a backward approach to

study a phenomenon, where he/she moves from effect to identify the cause.

- In case-control studies, causes of a disease are investigated after the occurrence of a disease.
- E.g. – A researcher investigates the history of smoking in patients diagnosed with lung cancer.

### **iii) Analytical Cross-Sectional Design-**

- Descriptive cross-sectional study involves single group of subjects with purpose to identify, assesses and describes a phenomenon by collecting data at single point of time.
- E.g. – A descriptive cross-sectional study to assess the prevalence of anxiety at first day of clinical posting among nursing students studying in selected nursing institutes of Kerala and Tamil Nadu.

### **3) Exploratory Research Design-**

- Exploratory research design is most primitive research design, which is used to study a phenomenon, which is not well understood to operationally define problem, generate hypothesis and improve a research design.
- It is used when topic of research is new and helps to operationally define the problem and generate hypothesis.
- E.g. – An exploratory study to assess the multifactorial dimensions of falls and home safety measures for elderly people living in selected communities in the city of Ludhiana.

### **4) Survey Research Design-**

- A survey is a research design used to collect information from different subjects within a given population having same characteristics of interest.

- If a survey is conducted on a sample of population, it is called sample survey, and if the entire population is involved, it is called a population survey, such as census, and so on.
- It provides superficial information on what people do, eat, and seek health care and so on, which is collected through face-to-face interview, questionnaire telephonic or electronic interviews.
- It provides extensive rather than intensive results.
- Survey is used to obtain information about prevalence, distribution and interrelations of phenomenon in a population such as political opinion polls, customer survey and health survey.
- These are following types of survey:
  - ✓ Descriptive survey
  - ✓ Comparative survey
  - ✓ Written survey
  - ✓ Oral survey
  - ✓ Electronic survey

### **Advantages of Non-Experimental Research Design-**

- Non-experimental research designs tend to be closest to real-life situations.
- Non-experimental research designs are most suitable for the nursing research studies.
- There are many research situations in which it is simply not practical to conduct a true experiment.

### **Disadvantages of Non-Experimental Research Design-**

- The major disadvantage of non-experimental researches is that the results obtained and the relationship between the dependent and independent variables can never be absolutely clear and error-free.

- Non-experimental studies are conducted for comparative purposes using non-randomly selected groups, which may not be homogeneous and tend to be dissimilar in different traits or characteristics that may affect the authenticity and generalizability of the study results.

## **2. Qualitative Research Designs-**

- Qualitative research approaches have somewhat short and less well-defined plans.
- In qualitative research study design, elements typically evolve over the course of the project.

### **Types of Qualitative Research Design-**

- These are following types-

- 1) Phenomenological research design
- 2) Ethnographic research design
- 3) Grounded theory
- 4) Case study research design
- 5) Historical research design

#### ***1) Phenomenological Research Design-***

- Phenomenology is a movement in philosophy that has been adapted by certain sociologists to raise an understanding of the relationship between states of individual consciousness and social life.
- As an approach within sociology, phenomenology attempts to uncover how human awareness is implicated in the production of social action, social situation and social world.

### **Types of Phenomenological Research Design-**

- These are following types-

**i) Realistic Phenomenological Research-**

→ It focuses on gathering the universal abstract of various types of information, including human actions, motives and results.

**ii) Constitutive Phenomenological Research-**

→ This includes the philosophy of natural sciences.

→ This procedure entangles suspending acceptance of the pre-given position of conscious life as something that exists in the world, and is carried out to obtain an ultimate inter subjective grounding for the world and the positive sciences of it.

**iii) Existential Phenomenological Research-**

→ This is concerned with topics, such as actions, conflicts, desires, finitude, oppression and death.

**iv) Hermeneutical Phenomenological Research-**

→ It utilizes vital experiences as a device for better consideration of the political, social, cultural or historical aspect in which those experiences happen.

→ E.g. – A phenomenological study on lived experiences of tsunami victims in selected villages of Tamil Nadu.

**2) Ethnographic Research Design-**

→ Ethnographic research is a method of conducting enquiry of a life process by studying individuals, artifacts or documents in their natural setting.

→ It includes both anthropology and historical forms of research. In healthcare research, ethnography provides access to health beliefs and healthcare practices in particular cultural or sub-cultural group.

**Characteristics of the Ethnographic Research-**

→ Ethnographers learn about cultural groups in which they are interested through the extensive fieldwork.

- Ethnographic research is a labour-intensive and time-consuming endeavour, where even months or years of fieldwork can be involved.
- A certain level of intimacy with cultural group members is required to study culture. Intimacy can develop over time and by working together directly with those cultural group members who are active participants.
- E.g. – Ethnographic study on socio-cultural beliefs of the people about antenatal, natal and post-natal care of women in selected tribal communities of Rajasthan.

### ***3) Grounded Theory-***

- The term grounded theory means that the theory developed from the research is ‘grounded’ or has its roots in the data from which it was derived.
- Grounded theory has become an important research method for the study of nursing theories of phenomena relevant to nurses.
- It is an approach to study social process and social structures.

### **Characteristics of the Grounded Theory-**

#### ***1) Overview and phases of grounded theory-***

- Grounded theory begins with a research situation. Within that situation, the task of researcher is to understand what is happening there (core variable), and how the players manage their role.
- After each hour of data collection, the researcher notes down key issues; this is known as note taking.

#### ***2) Methodology-***

- The steps of the grounded theory research occur simultaneously.
- An important methodological technique in grounded theory is constant comparison. Although this method is frequently considered to be an inductive means of developing theory.

**3) Sources of data collection-**

- For data collection, the researcher will use all his/her senses.
- There is a lot to be learned just by observing; some of it is evident within minutes of entering a situation.
- Sources of data collection vary with the focus of enquiry, the purpose of investigation and guidelines suggested by research approach being utilized.

**Types of Grounded Theory-**

- These are following two types-

**i) Substantive Theory-**

- It is grounded in data on a specific substantive area, such as postpartum depression.

**ii) Formal Theory-**

- Substantive theory serves as a springboard for developing a higher, more abstract level of theory from a complicated substantive grounded theory study regarding a particular phenomenon.
- E.g. – A study on the response and adaptation process of patients diagnosed with cancer at Nehru Hospital, PGIMER, and Chandigarh.

**4) Historical Research Design-**

- The systematic collection and critical evaluation of data relating to past occurrences of a particular phenomenon is also a tradition that relies primarily on qualitative data.
- Historical research is undertaken to answer questions concerning causes, effects or trends relating to past events that may shed light on present behaviour or practices.

**Characteristics of Historical Research Design-**

- It is important not to confuse historical research with a review of literature about historical events.
- Historical research involves the careful study and analysis of data about past events.
- Historical research is a critical investigation of events, their development, experiences of the past, the careful weighing of evidence of the validity of courses of information from the past, and the interpretation of the weighed evidence.
- The purpose is to gain a clearer understanding of the impact of the past on present and future events related to the life process.
- E.g. – A historical study on inception and development of Punjab Nurses Registration Council.

**5) Case Study Research Design-**

- Case studies are in-depth examination of people, places or institutions.
- Robson defines case study as the development of detailed, intensive knowledge about a single ‘case’ or a small number of related cases.
- Case study is often used when the selected case might offer insight into a unique situation or when researcher wants to know more about a particular phenomenon within a real-life context.
- In nursing sciences, case study methodology is used since a long time for in-depth study of a single patient or a group of patients to generate knowledge for solving nursing problems of patients suffering with specific disease conditions.
- E.g. – A case study on the availability and utilization of the emergency services at Civil Hospital, Ludhiana, Punjab.

## **Sampling & Data Collection**

### **Sampling-**

#### **Definition of Population-**

- Population is the aggregation of all the units in which a researcher is interested.
- In other words, population is the set of people or entities to which the results of a research are to be generalized.
- E.g. – A researcher needs to study the problems faced by postgraduate nurses of India; in this the ‘population’ will be all the postgraduate nurses who are Indian citizens.

#### **Definition of Sample-**

- Sample may be defined as representative unit of a target population, which is to be worked upon by researchers during their study.
- In other words, sample consists of a subset of units which comprise the population selected by investigators or researchers to participate in their research project.

#### **Sampling Criteria/Characteristics of Good Sample-**

- These are following criteria-

##### **1) Representative-**

- A representative sample is one that the key characteristics of which are closely related to those of the population.

##### **2) Free from bias and errors-**

- A good sample is one which is free from deliberate selection of the subjects for study.

→ Sample should be free from simple random sampling errors or sampling bias.

### **3) No substitution and incompleteness-**

→ A sample is said to be good if once a subject is selected for the study, it is neither replaced nor is it incomplete in any aspect of researchers' interest.

### **4) Appropriate sample size-**

→ Generally, it is believed that in quantitative studies the larger the sample size better is the probability of the goodness of the sample.

### **Sampling Process-**

→ Sampling is a process of selecting a part of the assigned population to represent the entire population.

→ The procedure of sampling should be systematically organized.

→ These are following stapes-

### **1) Identifying & Defining the Target Population-**

→ The first step of the sampling process is the identification and defining the target population.

→ Target population consists of the total group of people or objects that meet the designated set of criteria of interest of the researcher.

### **2) Describing the Accessible Population & Ensuring Sampling Frame-**

→ It is not always possible to have access to each subject included in the target population.

→ Therefore, researcher must establish a description about the accessible population, which is readily available for research.

### **3) Specifying the Sampling Unit-**

→ The researcher must establish the specific inclusion and exclusion criteria to select a particular sampling unit.

### **4) Specifying Sample Selection Methods-**

→ It is one of the important stages of the sampling process, where the researcher decides whether sample will be drawn from the population by using probability or non-probability sampling techniques.

### **5) Determining the Sample Size-**

→ It is very essential to determine the size of sample, so that the researcher can plan the implementation of the sampling process accordingly, details of sample size.

### **6) Specifying Sampling Plan-**

→ Before the selection of a particular sample, the researcher must have a final sampling plan, so that sampling process can be implemented without any undue problems.

### **7) Selecting a Desired Sample-**

→ Finally, a researcher draws a representative sample from the accessible population, which requires the implementation of the plan of the sampling process, and ultimately selects a representative part of the population, which is used by the researcher for data collection in research study.

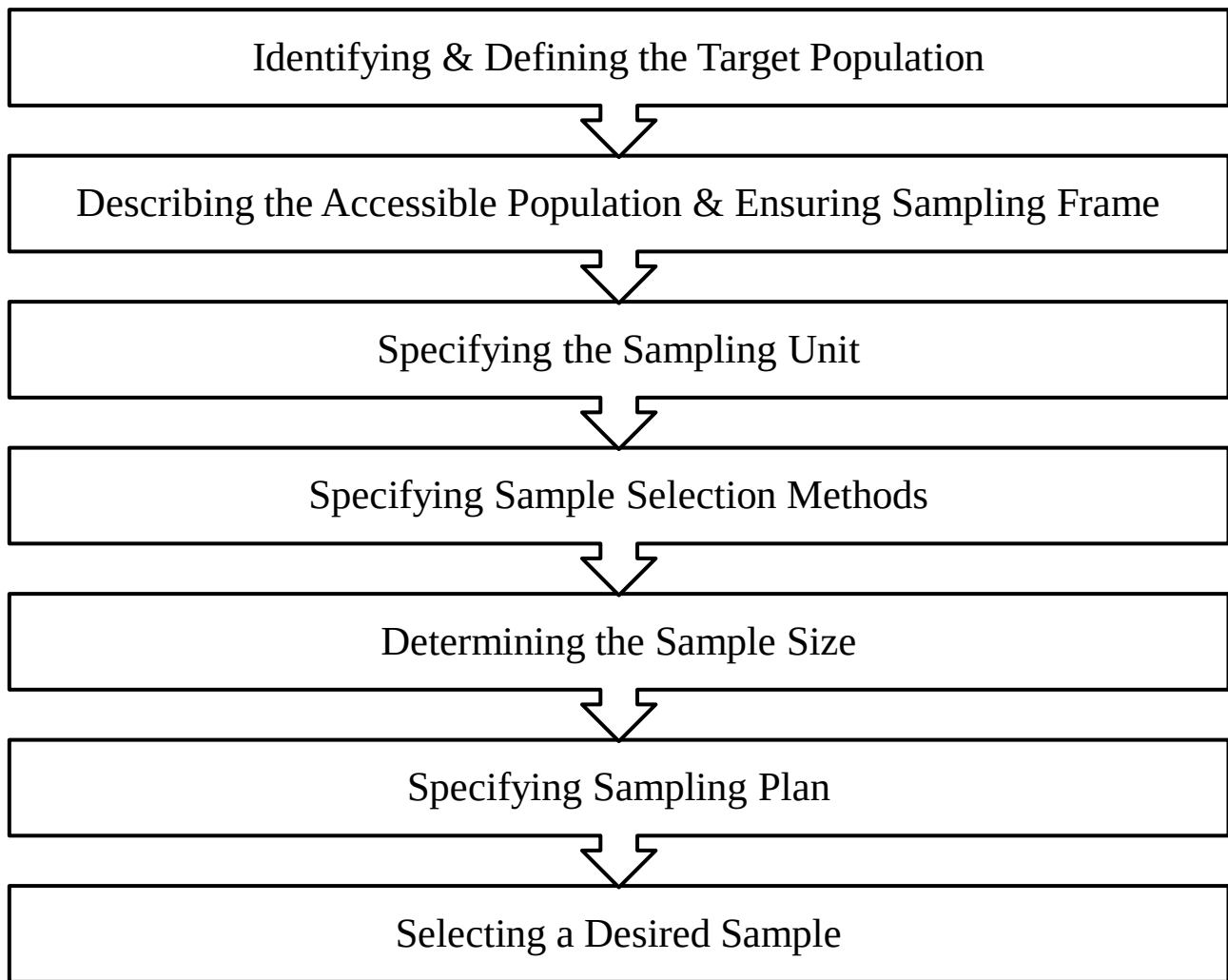
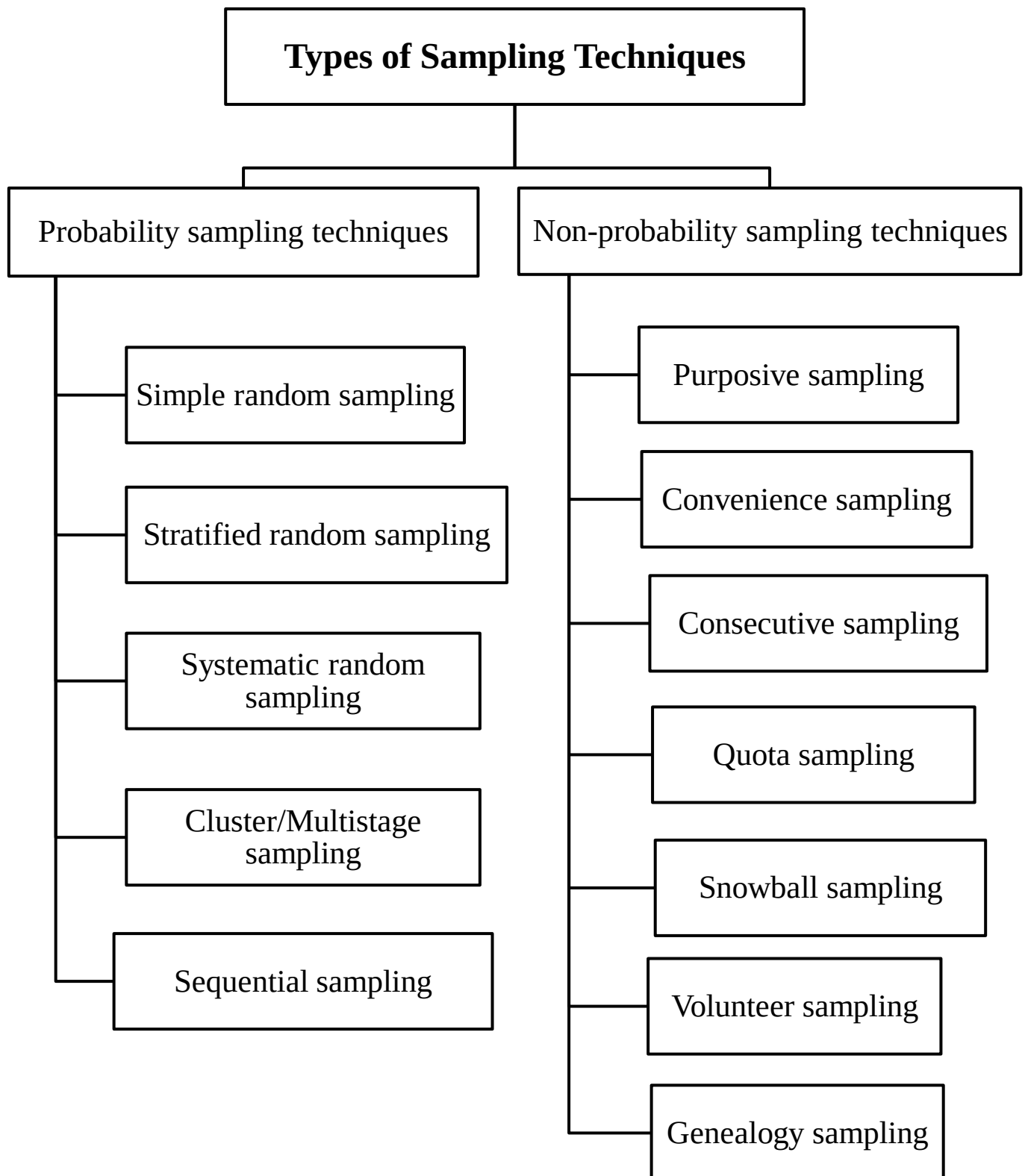


Fig. – Stages of the Sampling Process.

### Factors Influencing Sampling Process-

→ These are following factors-

- ✓ Inexperienced investigator
- ✓ Lack of honesty
- ✓ Lack of interest
- ✓ Intensive workload
- ✓ Inadequate supervision
- ✓ Inappropriate sampling techniques
- ✓ Sample size
- ✓ Large geographic area
- ✓ Lack of time
- ✓ Lack of cooperation

**Types of Sampling Techniques-**

## **1. Probability Sampling Techniques-**

- It is based on the theory of probability.
- It involves random selection of the elements/members of the population.
- In this, every subject in a population has equal chance to be selected as study sample.

### **Features of Probability Sampling-**

- The technique probability sampling involves the samples to be gathered in a process that provides equal chances to all the individuals in the population of getting selected.
- In this sampling technique, the researcher surely has to guarantee that each individual has an equal opportunity for getting selected.
- This is feasible only if the researcher uses randomization.
- The absence of both sampling and systematic bias is the advantage of utilizing a random sample. If random selection is made in a proper manner, the sample is representative of the whole population.
- The effect of this is an absent or minimal systematic bias and is a difference between the results from the sample and those from the population.

### **Types of Probability Sampling Techniques-**

- Probability sampling techniques are classified in five subtypes:
  - 1) Simple random sampling
  - 2) Stratified random sampling
  - 3) Systematic random sampling

4) Cluster or multistage sampling

5) Sequential sampling

### 1) Simple Random Sampling Technique-

- This is the most pure and basic probability sampling design.
- In this type of sampling design, every population member has a similar chance of being picked as the subject.
- The whole process of sampling is carried out in a single step, with each subject chosen independently of the other members of the population.
- The first step of the simple random sampling technique is to identify the accessible population and prepare a list of all the elements/members of the population.
- The list of the subjects in population is known as sampling frame and a sample can be drawn from sampling frame by using following methods:

**a) The lottery method-** There are a variety of methods to carry out simple random sampling. The oldest and mechanical is the lottery method. Each member of the population is attributed a unique number. Each number is placed inside a hat or bowl and mixed in a thorough manner.

**b) The use of table of random numbers method-** This is most commonly and accurately used method in simple random sampling. Random tables present several numbers in rows and columns. Researcher initially prepares a numbered list of the elements/members of the population, and then with a blindfold chooses a number from the random table.

**c) The use of computer method-** Nowadays random tables may be generated from the computer, and subjects may be selected as described in the use of random table. For populations with a

small number of members, it is advisable to use the first method, but if the population has many members, a computer-aided random selection is preferred.

### **Merits-**

- One of the best things about simple random sampling is the ease of assembling the sample.
- It is also considered a fair way of selecting a sample from a given population since every member is given equal opportunity of being selected.

### **Demerits-**

- This method does not make use of knowledge about a population which researcher may already have.
- Lots of procedures need to be done before sampling is accomplished.

| <b>No.</b> | <b>Name of student</b> | <b>No.</b> | <b>Name of student</b> |
|------------|------------------------|------------|------------------------|
| 1          | Amit pareek            | 2          | Manik Malhotra         |
| 3          | Bajrang lal            | 4          | Nandini Murthy         |
| 5          | Bhagwan Singh          | 6          | Deepak Verma           |
| 7          | Dwarka Prasad          | 8          | Dhanesh                |
| 9          | Mohm. Khalid           | 10         | Rajveer Dangi          |
| 11         | Jitendra Kumar         | 12         | Onkar mal              |
| 13         | Hansraj Jakhar         | 14         | Pankaj Singh           |
| 15         | Pradeep Sood           | 16         | Manoj Kajala           |
| 17         | Rohit Verma            | 18         | Rajkumar Verma         |
| 19         | Rajendra Saini         | 20         | Rajesh Chenchi         |
| 21         | Satish Goswami         | 22         | Indraaj Meena          |
| 23         | Ranjeet Puri           | 24         | Sunny Singodiya        |
| 25         | Pramod Prajapat        | 26         | Rajesh Sangwan         |
| 27         | Sandeep Jakhar         | 28         | Vikas Kumawat          |
| 29         | Ravi Verma             | 30         | Suresh Kajla           |

Fig. – Sampling Frame of the Study Conducted on College Students

|    |    |    |   |    |    |    |    |    |    |
|----|----|----|---|----|----|----|----|----|----|
| 4  | 2  | 5  | 7 | 2  | 10 | 15 | 17 | 20 | 30 |
| 26 | 22 | 18 | 2 | 5  | 1  | 7  | 12 | 21 | 16 |
| 8  | 19 | 14 | 7 | 24 | 3  | 6  | 9  | 2  | 4  |

Fig. – Extract from a Table of Random Numbers

## 2) Stratified Random Sampling-

- This method is used for heterogeneous population.
- Stratified sampling is a probability sampling technique wherein the researcher divides the entire population into different homogeneous subgroups or strata, and then randomly selects the final subjects proportionally from the different strata.
- The strata are divided according to selected traits of the population, such as age, gender, religion, socio-economic status, diagnosis, education, geographical region, type of institution, type of care, type of registered nurses, nursing area specialization, site of care, etc.

## Types of Stratified Random Sampling-

### a) *Proportionate Stratified Random Sampling-*

- In this, the sample chosen from each stratum is in proportion to the size of total population. When viewed against the entire population, the sample size of each stratum in this technique seems proportionate to the population size of the stratum.
- E.g. – Researcher has three strata containing 100, 200, and 300 population sizes, respectively. Moreover, the researcher selects a sampling fraction that is  $\frac{1}{2}$ . Then, the researcher should randomly sample 50, 100, and 150 subjects from each stratum, respectively.

***b) Disproportionate Stratified Random Sampling-***

- In this subtype, the sample chosen from each stratum is not in proportion to size of total population in that stratum.
- Sampling fractions is the only differences between proportionate and disproportionate stratified random sampling. The different strata have dissimilar sampling fractions with disproportionate sampling.
- E.g. – If a researcher wants to study biophysical profiles of nursing students in a college of nursing, which contains 100 students from Himachal Pradesh, 200 students from Haryana, and 300 students from Punjab. The researcher chose different sampling fractions and then randomly selects a sample of 50 subjects from each stratum.

***Merits-***

- It ensures representation of all groups in a population.
- Researchers also employ stratified random sampling when they want to observe existing relationships between two or more subgroups.
- The researcher can representatively sample the most inaccessible subgroups and even the smallest in the population with stratified sampling. This permits the researcher to sample the rare extremes of a provided population.

***Demerits-***

- Proportionate stratification requires accurate information on the proportion of population in each stratum.
- Large population must be available from which to select subjects.
- There is always a possibility of faulty classification and hence increase in variability.

### 3) Systematic Random Sampling-

- Systematic random sampling can be likened to an arithmetic progression, wherein the difference between any two consecutive numbers is the same.
- It involves the selection of every Kth case from a list of groups such as every 10th person on a patient list or every 100th person from a phone directory.
- The desired sample size is established at some number (n) and the size of population must be known or estimated (N).

$$K = N/n$$

- E.g. – A researcher wants to choose about 100 subjects from a total target population of 500 people. Therefore,  $500/100 = 5$ . Therefore, every 5th person will be selected.

#### ***Merits-***

- Researchers find this technique convenient and simple to carry out.
- Distribution of sample is spread evenly over the entire given population.
- Less cumbersome, time-consuming, and is cheaper than simple random sampling technique.

#### ***Demerits-***

- If first subject is not randomly selected, then it becomes a nonrandom sampling technique.
- Sometimes this may result in biased sample.
- If sampling frame has non-randomly distributed subjects, this sampling technique may not be appropriate to select a representative sample.

#### **4) Cluster or Multistage Sampling-**

- Cluster sampling means random selection of sampling unit consisting of population elements. Then from each selected sampling unit, a sample of population elements is drawn by either simple random selection or stratified random sampling.
- This method is used in cases where the population elements are scattered over a wide area, and it is impossible to obtain a list of all the elements.
- E.g. – A researcher requiring to survey the academic performance of Indian high school students.
  - He/she can divide the entire Indian population into different clusters (e.g. cities).
  - Depending on his/her research through simple or systematic random sampling, the researcher then selects a number of clusters.
  - Then the researcher can either include all the high school students as subjects or he/she can chose a number of subjects from each cluster through simple or systematic random sampling from the selected clusters (randomly selected cities).

#### **Types of Cluster Samples-**

##### ***a) One-stage Cluster Sample-***

- Recall the example given above; one-stage cluster sample occurs when the researcher includes all the high school students from all the randomly selected clusters as sample.

##### ***b) Two-stage Cluster Sample-***

- In this, initially the researcher lists all the clusters appearing in the population. After this, the clusters are selected normally by simple random sampling (SRS). Then usually by simple random sampling or

often by systematic sampling, the elements or units in the selected clusters of the first stage are then sampled in the second stage.

***c) Multistage Cluster Sample-***

→ In multistage cluster sample, the sampling is done at more than the two levels by initially identifying clusters as population at different levels and selecting them using simple random sampling technique and finally selecting units (elements) using simple random sampling or systematic sampling.

***d) Probability Proportion to Size Cluster Sampling-***

→ It is a variant of cluster sampling, when size of clusters is not same and there is risk of over-sampling from smaller size clusters and under sampling from the larger size clusters. Then it is most useful when the sampling units vary considerably in size because it assures that those in larger sites have the same probability of getting into the sample as those in the smaller sites, and vice versa.

***Merits-***

- This sampling technique is cheap, quick, and easy for a large population.
- Large populations can be studied, and require only list of the members.
- Enables investigators to use existing division, such as districts, villages/ towns, etc.

***Demerits-***

- This technique is the least representative of the population from the various kinds of probability sampling.

- The tendency of individuals within a cluster is to get similar characteristics.
- With a possibility of high sampling error, this is also a possibility of error in the probability sampling technique.

### 5) Sequential Sampling-

- This method of sample selection is slightly different from other methods. Here the sample size is not fixed.
- The investigator initially selects small sample and tries out to make inferences; if not able to draw results, he or she then adds more subjects until clear-cut inferences can be drawn.
- E.g. – A researcher is studying association between smoking and lung cancer. Initially researcher takes a smallest sample and tries to draw inferences.
- In this table given values, out of 50 subjects, 28 smokers had almost double incidence of lung cancer as compared to 22 smokers.

|                   |             |                 | Having lung cancer |     |
|-------------------|-------------|-----------------|--------------------|-----|
| Number of subject | Smokers (A) | Non smokers (B) | (A)                | (B) |
| 20                | 7           | 12              | 2                  | 1   |
| 30                | 18          | 22              | 5                  | 3   |
| 50                | 28          | 22              | 10                 | 4   |

#### **Merits-**

- Facilitates to conduct a study on best-possible smallest representative sample.
- Helping in ultimately finding the inferences of the study.

#### **Demerits-**

- With this sampling technique, it is not possible to study a phenomenon which needs to be studied at one point of time.
- Requires repeated entries into the field to collect the sample.

## 2. Non-Probability Sampling-

- Non-probability sampling is a technique wherein the samples are gathered in a process that does not give all the individuals in the population equal chances of being selected in the sample.
- In other words, in this type of sampling every subject does not have equal chance to be selected because elements are chosen by choice not by chance through nonrandom sampling methods.
- Generally, it is believed that nonrandom methods of sampling are more likely to produce a biased sample than random methods.

### Features of Non-probability Sampling-

- Non-probability sampling is a technique wherein the samples are gathered in a process that does not give all the individuals in the population equal chances of being selected.
- Most researchers are bound by time, money, and workforce, and because of these limitations, it is difficult to randomly sample the entire population and it is often necessary to employ another sampling technique, the non-probability sampling technique.
- In contrast with probability sampling, non-probability sample is not a product of a randomized selection processes. Subjects in a non-probability sample are usually selected on the basis of their accessibility or by the purposive personal judgement of the researcher.

### Types of Non-probability Sampling Techniques-

- Non-probability sampling techniques are classified in seven subtypes:
  - 1) Purposive sampling
  - 2) Convenience sampling
  - 3) Consecutive sampling
  - 4) Quota sampling
  - 5) Snowball sampling

- 6) Volunteer sampling
- 7) Genealogy sampling

## 1) Purposive Sampling-

- Purposive sampling is more commonly known as ‘judgmental’ or ‘authoritative sampling’.
- In this sampling technique, samples are chosen by choice not by chance, through a judgement made by the researcher based on his or her knowledge about the population.
- Researcher might decide purposely to select subjects who are judged or believed to be typical or representative of the accessible population.
- Particularly experts who have in-depth knowledgeable about the accessible population under study may be asked to provide help to select the sample in purposive sampling.
- E.g. – A researcher wants to study the lived experiences of post-disaster depression among people living in earthquake-affected areas of Gujarat. In this case, a purposive sampling technique is used to select the subjects who were the victims of the earthquake disaster and have suffered post-disaster depression living in earthquake-affected areas of Gujarat. For this study, the researcher selected only those people who fulfill the criteria as well as particular subjects that are the typical and representative part of population as per the knowledge of the researcher.

### ***Merits-***

- Simple to draw sample and useful in explorative studies.
- Saves resources, requires less fieldwork.

### ***Demerits-***

- Requires considerable knowledge about the population under study.
- It is not always reliable sample, as conscious biases may exist.

## 2) Convenience Sampling-

- Convenience sampling is probably the most common of all sampling techniques because it is fast, inexpensive, easy, and the subjects are readily available.
- In convenience sampling, which is a non-probability sampling technique, subjects are selected due to their convenient accessibility and proximity to the researcher.
- The subjects are chosen because of the fact that they are very easy to recruit for the study. Moreover, the researcher does not opt for choosing subjects that are representative of the entire population.
- Sometimes, it is also known as an accidental sampling. Subjects are chosen simply because they are easy to recruit.
- E.g. – If a researcher wants to conduct a study on the older people residing in Ludhiana, and the researcher observes that he can meet several older people coming for morning walk in a park located near his residence in Ludhiana, he can choose these people as his research subjects. These subjects are readily accessible for the researcher and may help him to save time, money, and resources.

### ***Merits-***

- This technique is considered easiest, cheapest, and least time consuming.
- This sampling technique may help in saving time, money and resources.

### ***Demerits-***

- The most obvious criticism about convenience sampling is sampling bias, and that the sample is not representative of the entire population.
- It does not provide the representative sample from the population of the study.

### 3) Consecutive Sampling-

- Consecutive sampling is very similar to convenience sampling except that it seeks to include all accessible subjects as part of the sample.
- This non-probability sampling technique can be considered as the best of all non-probability samples because it includes all the subjects that are available, which makes the sample a better representation of the entire population.
- It is also known as total enumerative sampling.
- E.g. – If a researcher wants to study the activity pattern of post-kidney-transplant patients, he can select all post-kidney transplant patients who meet the designed inclusion and exclusion criteria, and who are admitted in post-transplant ward during a specific time period.

#### ***Merits-***

- There is very little effort on the part of the researcher when performing this sampling technique.
- It is not expensive, not time consuming, and not workforce intensive.

#### ***Demerits-***

- The researcher has no set plans about the sample size and sampling schedule.
- It always does not guarantee the selection of representative sample.

### 4) Quota Sampling-

- Quota sampling is a non-probability sampling technique wherein the researcher ensures equal or proportionate representation of subjects, depending on which trait is considered as the basis of the quota.
- The bases of the quota are usually age, gender, education, race, religion, and socio-economic status.

- E.g. – If the basis of the quota is college level and the researcher needs equal representation, with a sample size of 100, he or she must select 25 first-year students, another 25 second-year students, 25 third-year, and 25 fourth-year students.
- The quota sampling technique appears like stratified random sampling technique; however, in this sampling technique, sample is selected without random process from different quotas.

***Merits-***

- Economically cheap, as there is no need to approach all the candidates.
- Suitable for studies where the fieldwork has to be carried out, like studies related to market and public opinion polls.

***Demerits-***

- Not possible to estimate errors.
- Bias is possible, as investigator/ interviewer can select persons known to him.

**5) Snowball Sampling-**

- Snowball sampling is a non-probability sampling technique that is used by researchers to identify potential subjects in studies where subjects are hard to locate, such as commercial sex workers, drug abusers, etc.
- E.g. – A researcher wants to conduct a study on the prevalence of HIV/ AIDS among commercial sex workers. In this situation, snowball sampling is the best choice for such studies to select a sample.
- Researchers use this sampling method if the sample for studies that are very rare or are limited to very small subgroups of the population. This type of sampling technique works like chain referral. Therefore, it is also known as chain referral sampling.

## Types of Snowball Sampling-

→ Snowball sampling are classified into three subtypes:

### a) *Linear Snowball Sampling-*

→ In this, each selected sample is asked to provide reference of only one similar subject; where a linear chain is created by the completion of desired sample

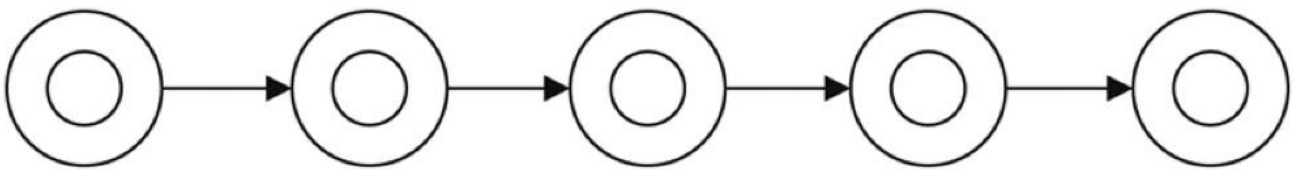


Fig. – Linear Snowball Sampling

### b) *Exponential Non-discriminative Snowball Sampling-*

→ In this, each sample member is asked to provide reference of at least two similar subjects, because of which the size of the sample grows exponentially and a large sample size can be achieved.

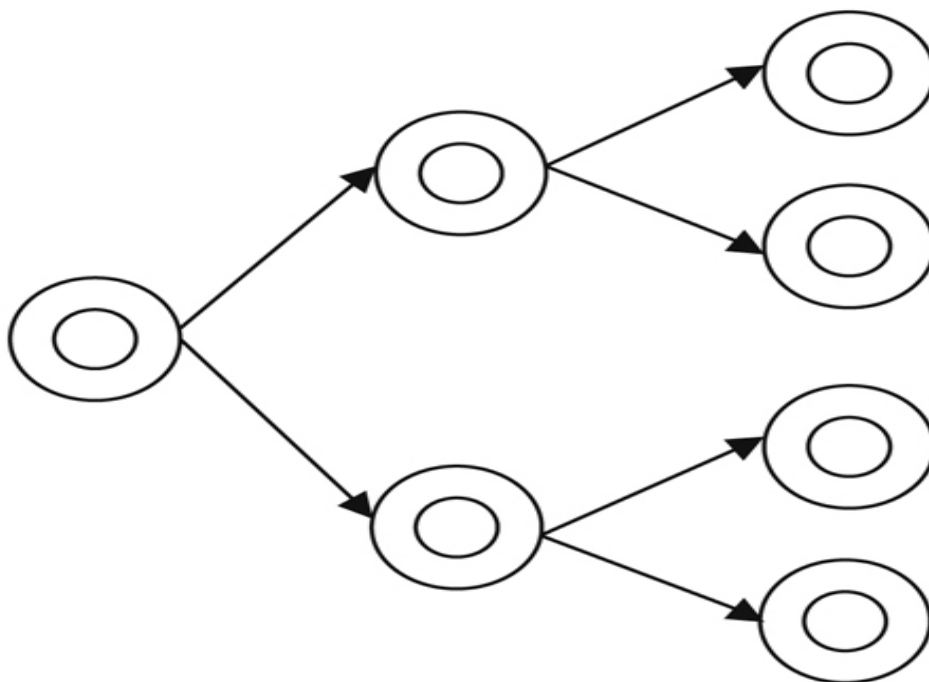


Fig. – Exponential Non-discriminative Snowball Sampling

### c) Exponential Discriminative Snowball Sampling-

→ In this, initially one sample is selected and asked for two references of similar subjects, out of which at least one subject must be active to provide further references and another could be non-active in providing references.

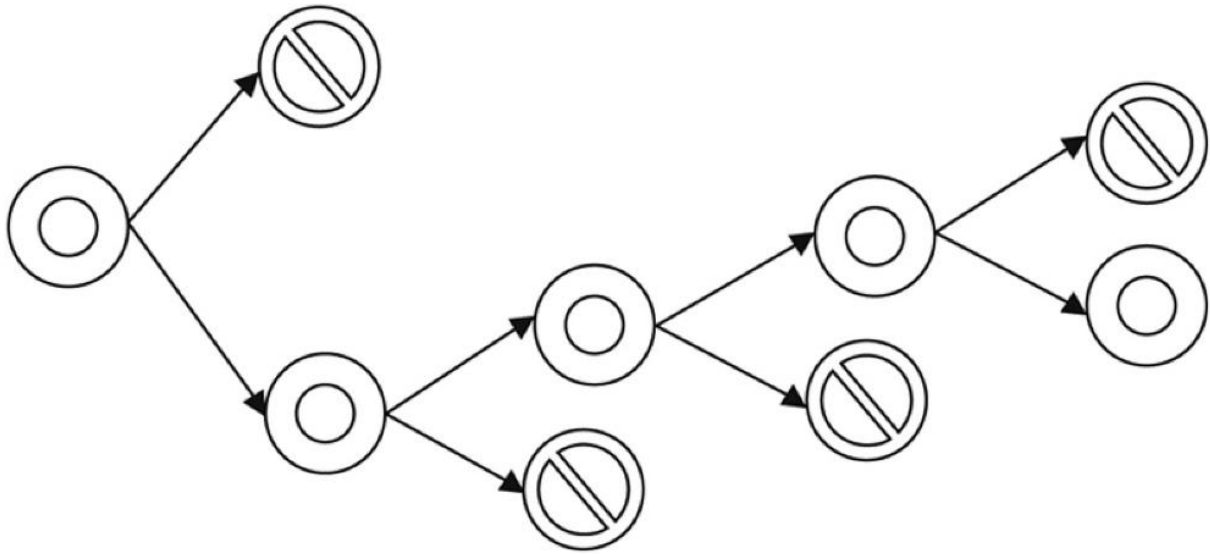


Fig. – Exponential Discriminative Snowball Sampling

***Merits-***

→ When utilizing other sampling methods, the chain referral process permits the researcher to reach populations that are difficult to sample.

→ The process is simple, cheap, and cost-efficient.

### ***Demerits-***

→ The researcher has less control over the sampling method. The subjects that the researcher can get depend chiefly on the previous subjects that were observed.

→ Sample representativeness is not guaranteed. Regarding the true distribution of the population and of the sample, the researcher has no idea.

## 6) Volunteer Sampling-

- It is a type of non-probability sampling technique, in which participants themselves volunteer to participate in the study and they only approach the researcher to be the part of study sample.
- Researcher publishes an advertisement or informs target population through mass media to participate in the study and interested participants may voluntarily contact researcher to participate in the study.
- E.g. – A nurse researcher is interested to assess the effectiveness of a selected yoga technique on the reduction of blood pressure. She may advertise in newspaper to inform target people to participate in this research to take the scheduled yoga classes and pre and post-assessment of blood pressure.

### ***Merits-***

- Cost-effect sampling technique.
- Needs very limited efforts and time to locate the study participants.

### ***Demerits-***

- Only interested people contact to participate, so there are very chances that sample may not be a representative sample.
- It encounters the systematic error or bias because only people who have the access to particular medium of receiving the researcher's published advertisement may get a chance to participate in the study; the rest will be left unrepresented.

## 7) Genealogy Sampling-

- This is a non-probability sampling technique in which all the members of entire related families are selected rather than selecting the different households in the village or area.
- The genealogy sampling begins with identifying a first participant, who is convinced to participate in the study and then further he/ she is asked to refer to close relatives of his family, who even may be living in other areas of village or area.

### ***Merits-***

- This sampling technique is useful in drawing a representative sample from traditional rural communities, which are socio-culturally and economically homogenous.
- Saves the time and efforts in locating the study subjects because participants are identified through reference from previous participants.

### ***Demerits-***

- This sampling technique encounters problem of systematic errors or bias.
- It lacks the diversity of sample participant characteristics because subjects are selected from a family or related families. Thus, it has limited usefulness.

## **Data Collection-**

### **Introduction-**

- Data are the observable and measurable facts that provide information about the phenomenon under study.
- A systematic collection and analysis of data are most vital to any empirical research.
- It is most essential for a researcher to have appropriate methods of data collation so that planned study variables can be adequately measured and accurate conclusions can be drawn.

### **Concepts of Data Collection-**

- Data are the observable and measurable facts that provide information about phenomenon under study.
- E.g. – To study the quality of perinatal care services provided in the rural area, the first step is to operationally define perinatal care and its different components. Perinatal period for a mother is divided into three phases: antenatal, natal, and postnatal periods. The indicators of the antenatal care are early detection of pregnancy, tetanus toxoid (TT) immunization, identification of high-risk mothers, regular antenatal check-ups, and supply of supplements. Therefore, data collection should aim at identification of observable and measurable facts or variables that would relate to these indicators. The questions that need to be answered are: What are the indicators? How will they be measured or recorded? What is the analysis planned?
- The five ‘Ws’ of data collection are:
  - ✓ What data are to be collected?
  - ✓ From whom data are to be collected?
  - ✓ Who will collect data?
  - ✓ From where the data will be collected?
  - ✓ When is the data to be collected?

**Data Sources-**

→ These are following two types of data sources:

**1) Primary Sources-**

- Primary data are directly collected from the research units, which may be individuals, objects, programmes, or institutions.
- Primary sources provide the first-hand information collected by the researcher directly from the respondents or the situations, which may be collected through interviews, questioning, observation, biochemical measurements, and psychosocial measurement scales.

**2) Secondary Sources-**

- Secondary data are collected from either internal or external secondary sources.
- External sources involve existing materials, such as published or unpublished records.
- Published records may include the journals, magazines, newspapers, government reports, statistical abstracts, and census.

**Methods & Tools (Instruments) of Data Collection-**

→ These are following terms related to methods, techniques, and tools of data collection:

**1) Methods of Data Collection-**

→ The various steps or strategies used for gathering and analyzing data in a research investigation are known as the methods of data collection.

**2) Technique of Data Collection-**

→ The means of gathering data with the use of specific tools used in given methods are known as techniques of data collection.

### 3) Instrument/ Tools for Data Collection-

→ A research instrument is a device used to measure the concept of interest in a research project that a researcher uses to collect data.

#### Types of Methods & Tools of Data Collection-

| Types of Method/Techniques | Tools for Data Collection                                                         |
|----------------------------|-----------------------------------------------------------------------------------|
| 1. Interview               | 1) Interview schedule                                                             |
| 2. Questioning             | 1) Questionnaire<br>2) Attitude scales/Composite scales                           |
| 3. Observation             | 1) Rating scales<br>2) Checklists                                                 |
| 4. Biophysiologic Methods  | 1) In-vivo biophysiologic measurements<br>2) In-vitro biophysiologic measurements |
| 5. Other Methods           | 1) Validity<br>2) Reliability<br>3) Pilot study                                   |

#### 1. Interview-

##### Definition-

→ A method of data collection in which one person (interviewer) asks the questions from another person (respondent) conducted either face-to-face or telephonically.

##### Characteristics of Interview-

- The participants, the interviewer, and the respondent are strangers.
- The relationship between the participants is a transitory one.

- Interview is a mode of obtaining verbal answers to questions put verbally.
- The investigator records information furnished by the respondent in the interview.
- Interview is not a mere casual conversational exchange, but a conversation with a specific purpose.
- The interaction between the interviewer and the respondent need not necessarily be face-to-face because interviews can be conducted over telephone also.

### **Types of Interviews (Schedule)-**

- Interview can be classified in following six broad categories:
  - 1) Structured Interview (Directive Interview)
  - 2) Unstructured Interview (Nondirective Interview)
  - 3) Semi structured Interview
  - 4) In-Depth Interview
  - 5) Focused Group Interview
  - 6) Telephone Interview

### **1) Structured Interview (Directive Interview)-**

- Structured interview is a means of data collection in which the interviewer has an interview schedule in which the questions are listed in the order in which they are to be answered.
- These interviews are also known as standardized interviews.
- In other words, this is an interview made with a detailed, standardized schedule or preplanned.
- Same questions are put to all the respondents and in the same order.

## **2) Unstructured Interview (Nondirective Interview)-**

- These interviews are also known as non-standardized interviews.
- Unstructured interview is a method wherein the questions can be changed to meet the respondent's intelligence, understanding, and beliefs.
- Questions are changed based on how each individual person responds to the questions.
- In this type of interview, the interviewer encourages the respondent to talk freely about given topic with minimum prompting. In this, no preplanned schedule is used.

## **3) Semi-structured Interview-**

- Semi-structured interview is a flexible method that allows new questions to be brought up during the interview, depending upon the situation during the interview.

## **4) In-Depth Interview-**

- This is an intensive and investigative interview conducted and aimed at studying the respondent's opinion and emotions on the basis of interview guide.
- This requires more training and interpersonal skills than structured interviewing.
- This deliberately aims to elicit unconscious as well as extremely personal feelings and emotions.

## **5) Focused Group Interview-**

- Focused group interview is an unstructured group interview technique where 6– 10 members are brought together under the guidance of a trained interviewer to focus on a specific phenomenon.

- Its purpose is to gain insight from the appropriate sample or target population.
- The researcher tries to focus his/ her attention on a particular aspect of a problem.

## **6) Telephone Interview-**

- Telephone interviewing is a nonpersonal method of data collection.
- This method of collecting information consists of contacting respondents on telephone itself.
- It is used widely in industrial surveys, particularly in developed regions.

## **Interviewing Process-**

- The interviewing process includes following steps:
  - 1) Preparation for interview
  - 2) Pre-interview introduction
  - 3) Developing rapport
  - 4) Carrying the interview forward
  - 5) Recording the interview
  - 6) Closing the interview

### **1) Preparation for Interview-**

#### ***a) Fix up appointment-***

- Interviewer should fix up the topic and time of interview with subject ahead of time. It would save time of interviewer.

#### ***b) Appearance & dress-***

- The interviewer should dress in a fashion fairly familiar and with due regards to the sensibilities of people who will be interviewed.

- The interviewer should be pleasant and should communicate a genuine interest.

### ***c) Familiarity with interview schedule-***

- The interview schedule must be studied carefully and the interviewer must practice reading it aloud to be very familiar with all the questions.

## **2) Pre-interview Introduction-**

- Researcher must introduce himself or herself to all respondents respectfully as per their cultural practices.
- Identify the respondent by name.
- Select a setting with as little disturbance as practicable.
- Describe the intention of the interview.
- Mention the confidentiality terms.
- Describe the interview structure.

## **3) Developing Rapport-**

- After primary introduction, the researcher starts developing rapport with respondents.
- This is developed through confidence building and understanding the concerns of the respondents by beginning the conversation on general topics.

## **4) Carrying the Interview Forward-**

- After establishing rapport, the technical task of asking questions from the interview schedule starts.
- While interviewing the respondent, following guidelines should be used:
  - Start the interview.
  - Ask only one question at a time.

- Repeat a question if necessary.
- Try to make sure that the subject understands the questions.
- Listen carefully to the subject's answer.
- Observe the subject's facial expressions, gestures, and tone of voice.
- Allow the subject a sufficient time to answer the questions.
- Do not show signs of surprise, shock, or anger.

### **5) Recording the Interview-**

- It is essential to record responses as they take place.
- Good information can be taken by note-taking.
- A tape recorder or video recording may be used to record the responses of respondent.
- Shorthand may be used to record responses.

### **6) Closing the Interview-**

- After the interview is over, a polite leave of the respondent should be taken, thanking him or her with a friendly smile and saying goodbye.

### ***Advantages of Interview-***

- Interviews are useful to obtain information about people's feelings, perceptions, and opinions.
- Interviews allow more detailed questions to be asked.

### ***Disadvantages of Interview-***

- Interviews are time-consuming and costly affairs.
- Different interviewers may understand and translate interviews in different ways.

## **2. Questioning-**

- Questioning method focusing on the question on the topic.
- The question will have to be complex enough to deserve a thoughtful answer.

### **1) Questionnaire-**

#### **Definition-**

- A questionnaire is a structured instrument consisting of a series of questions prepared by researcher that a research subject is asked to complete, to gather data from individuals about knowledge, attitude, beliefs, and feelings.

#### **Types of Questions-**

- These are following two types:
  - 1) Open format question
  - 2) Closed format question

### **1) Open-Format Questions-**

- Open-ended questions are those questions that provide opportunity to the respondents to express their opinions and answers in their own way.
- Open-format questions have following characteristics:
  - There is no predetermined set of responses.
  - They provide true, insightful, and unexpected suggestions.
- E.g. – State your opinion about the quality of health care services in Punjab?

## 2) Closed-Format Questions-

- These questions offer respondents a number of alternative replies from which the subjects must choose the one that most likely matches the appropriate answer.
- Closed-format questions have following characteristics:
  - They facilitate easy statistical calculation of data.
  - Provide easy preliminary analysis.

### *Types of Closed Format Questions-*

- These are following types of closed format questions:
  - i. Dichotomous questions
  - ii. Multiple choice questions
  - iii. Cafeteria questions
  - iv. Rank order questions
  - v. Contingency questions
  - vi. Rating questions
  - vii. Importance questions
  - viii. Likert questions
  - ix. Bipolar questions
  - x. Matrix questions

#### *i) Dichotomous Questions-*

- These require the respondent to make a choice between two responses, such as Yes/ No or Male/ Female.
- Example: Q. Have you ever been hospitalized?

|        |       |
|--------|-------|
| a) Yes | b) No |
|--------|-------|

**ii) Multiple Choice Questions-**

→ These questions require respondents to make a choice between more than two response alternatives.

→ Example: Q. Which of the following diseases is sexually transmitted?

|                      |                  |             |                 |
|----------------------|------------------|-------------|-----------------|
| a) Diabetes mellitus | b) Ootosclerosis | c) Syphilis | d) Hypertension |
|----------------------|------------------|-------------|-----------------|

**iii) Cafeteria Questions-**

→ These are special type of multiple choice questions that ask respondents to select a response that most closely corresponds to their views.

→ Example: Q. What do you think about hormone replacement therapy?

- a) It is dangerous, should be avoided.
- b) One should be cautious while using it.
- c) I am uncertain about my views.
- d) It is beneficial, should be promoted.

**iv) Rank Order Questions-**

→ These questions ask respondents to rank their responses from most favourable to least favourable.

→ Example: Q. What according to you is most important for your life?  
Rank from most favourable to least favourable.

|          |              |           |           |
|----------|--------------|-----------|-----------|
| a) Money | b) Education | c) Family | d) Health |
|----------|--------------|-----------|-----------|

**v) Contingency Questions-**

→ A question that is asked further only if the respondent gives a particular response to a previous question.

→ Example: Q. Are you stressed?

|       |                                       |
|-------|---------------------------------------|
| a) No | b) Yes, if yes, what are the reasons? |
|-------|---------------------------------------|

**vi) Rating Questions-**

- These questions ask respondents to judge something along an ordered dimension.
- Respondent is generally required to provide rating to a specific element on a scale selecting among the points from poor to good. They may provide a number of choices
- Example: Q. How do you rank the education quality in India?

|      |      |      |           |
|------|------|------|-----------|
| 1    | 2    | 3    | 4         |
| Good | Fair | Poor | Very poor |

**vii) Importance Questions-**

- In this, respondents are asked to rate the importance of a particular issue on a rating scale of 1– 5.
- This helps to know the things/issues that are important to a respondent.
- Example: Q. Exercising every day is..... for the health.

|                     |                |                    |                    |                      |
|---------------------|----------------|--------------------|--------------------|----------------------|
| 1                   | 2              | 3                  | 4                  | 5                    |
| Extremely important | Very important | Somewhat important | Not very important | Not at all important |

**viii) Likert Questions-**

- Likert questions help to know how strongly the respondent agrees with a particular statement.
- These questions help to assess how respondent feels towards a certain issue/services.
- Example: Q. Is this community a good place to raise children?

|                |       |           |          |                   |
|----------------|-------|-----------|----------|-------------------|
| 1              | 2     | 3         | 4        | 5                 |
| Strongly agree | Agree | Uncertain | Disagree | Strongly disagree |

**ix) Bipolar Questions-**

- Bipolar questions are questions that have two extreme answers.
- Respondent has to mark his/ her response between two opposite ends of the scale.
- Example: Q. What is your balance of preference here?

|                               |     |     |     |     |     |                               |
|-------------------------------|-----|-----|-----|-----|-----|-------------------------------|
| [ ] Like<br>going for<br>walk | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] Like<br>watching<br>movie |
|-------------------------------|-----|-----|-----|-----|-----|-------------------------------|

**x) Matrix Questions-**

- Matrix questions include multiple questions, and identical response categories are assigned.
- Questions are placed one under another, forming a matrix. Response categories are placed along the top and a list of questions down the side.
- Example: Q. Please let us know your weekly schedule of the following:

|          | Mon | Tue | Wed | Thurs | Fri | Sat | Sun |
|----------|-----|-----|-----|-------|-----|-----|-----|
| Gym      |     |     |     |       |     |     |     |
| Aerobics |     |     |     |       |     |     |     |
| Eating   |     |     |     |       |     |     |     |
| Drink    |     |     |     |       |     |     |     |

**Advantages of Questionnaires-**

- Questionnaires are cost-effective.
- They are easy to analyze.
- They require less time and energy to administer.

***Disadvantages of Questionnaires-***

- Questionnaires are not suitable for all, for example children, blinds, and illiterates.
- Low response rate.
- Questionnaire sent by mail may be filled by someone other than the intended person.

**2) Attitude Scales/Composite Scales-****Definition-**

- “Composite scales are socio-psychologic measurements directed towards quantifying the qualitative attributes, such as feelings, attitude, self-concept, perceptions, beliefs, etc”.

**Likert Scale-****Definition-**

- Likert scale is a composite measurement scale used to measure attitude, values, and feelings of the people that involve summation of scores on the set of positive and negative declarative statements regarding measuring variables to which respondents are asked to indicate their degree of agreement or disagreement.

**Uses-**

- This scale is used to have quantified measurement of the qualitative attributes of people, such as feelings, values, and attitudes.
- It may also be used to assess the opinions of the respondents about a particular abstract concept.

### Scoring of the Likert Scale-

- Scoring of the Likert scale is done on the basis of type of statement and level of respondents' agreement with statement.
- For a positive statement, respondents get higher score if there is agreement with statement. However, in case of negative statement, respondent gets higher score if there is a disagreement with statement.

| Statement                                                                         | Strongly agree | Agree | Uncertain | Disagree | Strongly disagree |
|-----------------------------------------------------------------------------------|----------------|-------|-----------|----------|-------------------|
| Positive statement<br>(Person with multiple sex partner are at high risk of AIDS) | 5              | 4     | 3         | 2        | 1                 |
| Negative statement (you can get AIDS by sharing utensils)                         | 1              | 2     | 3         | 4        | 5                 |

### Advantages of Likert Scale-

- It is relatively easy to construct this scale.
- Likert scale is considered as more reliable and valid tool to measure the psychosocial variables.

### Disadvantages of Likert Scale-

- In Likert scale, respondents may feel forced to answer the questions against all preplanned items and their categories.
- Feelings of the respondents may not be fully assessed due to researcher's preplanned statements and categories.

### **3. Observations-**

#### **Definition-**

→ Observation is a technique for collecting data or acquiring information through occurrences that can be observed through senses with or without mechanical devices.

#### **Uses of Observation Method-**

→ Observation as a method of data collection is used in following situations:

- ✓ To understand an ongoing process or situation.
- ✓ To gather data on individual behaviors or interaction between peoples.
- ✓ To know about a physical setting.
- ✓ Data collection where other methods are not possible.

#### **Types of Observation-**

→ Observation method of data collection can be classified into following four broad categories:

- 1) Structured observation
- 2) Unstructured observation
- 3) Participant observation
- 4) Non-participant observation

#### **1) Structured Observation-**

→ In structured observation method, researcher prepares a structured or semistructured tool in advance to observe the phenomenon under study.

- Structured observation is guided by the preplanned structured or semistructured instrument, where the researcher observes only specific attributes or behaviour in accordance with planned observation guidelines.
- Structured observation is generally carried out by using following tools:
  - ✓ Checklist
  - ✓ Rating scale
  - ✓ Category system

## **2) Unstructured Observation-**

- Unstructured observation is generally used in qualitative studies, where observation is made with minimally structured or researcher-imposed categories.
- It is used for complete and nonspecific observation of phenomenon, which is very well-known by the researcher.
- The following tools are generally used by the researchers:
  - ✓ Log and field notes
  - ✓ Anecdotes
  - ✓ Field diary
  - ✓ Video recording

## **3) Participant Observation-**

- Historically, field and ethnographical researches have been associated most strongly with participant observation, where observer may live or work in field and actively participate in ongoing activities for an extended period.
- During participatory observation, unstructured tools are used to collect the data, such as logbooks, field notes, field diary, tape, video recording, etc.

#### **4) Nonparticipant Observation-**

- In non-participatory observation, the observer works as an eavesdropper (secretly listening or observing), where an attempt is made to observe people without interacting with them.
- In this technique, the observer is not a participant in the setting but is merely viewing the situation.
- In addition, people are observed without their knowledge that they are being observed.
- This method is routinely used by psychologists to study children or animals.

#### **Planning for Observation-**

##### **1) *Focus determination-***

- The evaluation questions you want to be answered through observation are to be visualized first. Then choose a few focus points for your data collection.
- E.g. – You may need to know how nicely an HIV awareness syllabus is being followed in the classroom.

##### **2) *Data collection design-***

- You need to think about the specific items for which you want to gather data once you have focused your evaluation. Then ascertain the method so as to how you are going to acquire the information that you require.
- The commonly used methods of observation are:
  - Checklists & recording sheets
  - Fields notes
  - Observations guidance list

**3) Site selection-**

- An adequate number of sites need to be selected to ensure that they represent the larger population. This helps in providing overall clear picture of phenomenon, which researcher is interested to observe.

**4) Observer selection-**

- You may prefer to be the only observer or you may add other persons while conducting observations.
- Other professional staff members, stakeholders, interns and graduate students, and volunteers generally act as potential observers.

**5) Observer training-**

- It is important that the observers are well-trained in the data collection process in order to ascertain consistent and high-quality data. The training level will fluctuate on the basis of the data collection complexity and the capabilities of the individual observers.

**6) Appropriate observation timings-**

- A sequence of events generally precedes programmes and processes.
- It is important that you strategize your observations to observe the activity components that will respond to the evaluation questions.

**Advantages of Observation-**

- Collects data so as to when and where an activity or event is happening.
- Does not believe on people's ability or willingness to give information.

**Disadvantages of Observation-**

- Observer's bias.
- Can be time-consuming and expensive as against other data collection techniques.

## 1) Rating Scale-

### Definition-

→ Rating scale refers to a scale with a set of opinion, which describes varying degree of the dimensions of an attitude being observed.

### Types of the Rating Scale-

→ These are following types of rating scale:

- 1) Graphic rating scale
- 2) Descriptive rating scale
- 3) Numerical rating scale
- 4) Comparative rating scale

### 1) Graphic Rating Scale-

→ In this scale, the performance is printed horizontally at various points from lowest to highest.

→ It includes the numerical points on the scale. It is anchored by two extremes presented to respondents for evaluation of a concept or object.

→ E.g. – How much are you satisfied with noise control in your ward?



### 2) Descriptive Rating Scales-

→ This type of rating scales does not use number, but divide the assessment into series of verbal phrases to indicate the level of performance.

→ E.g. – Q. Judge the level of performance of nursing personnel in Medical ICU.

| Nursing personnel<br>in a ward | Level of clinical performance |        |                   |         |
|--------------------------------|-------------------------------|--------|-------------------|---------|
|                                | Very active                   | Active | Moderately active | Passive |
| Ajay                           |                               |        |                   |         |
| Vijay                          |                               |        |                   |         |
| Sanjay                         |                               |        |                   |         |
| Lakshay                        |                               |        |                   |         |

### 3) Numerical Rating Scale-

- It divides the evaluation criteria into a fixed number of points, but defines only numbers at the extremes.
- In these scales, each statement is generally assigned a numerical score ranging from 1 to 10, or even more.
- E.g. – Pain assessment numerical scale.



### 4) Comparative Rating Scale-

- In this type of rating scale, the researcher makes a judgement about an attribute of a person by comparing it with that of a similar another person.
- E.g. – Mr. Ram's decision-making abilities closely resemble those of Mr. Shyam and Mr. Gopal. In this type of rating scale, researcher must have prehand knowledge about the selected attributes of the people with whom the subjects are supposed to be compared.

### Characteristics of Rating Scale-

- Rating scales are value judgements about attributes of one person by another person.
- These scales are most commonly used tools to carry out structured observations.

- Rating scales are generally developed to make quantitative judgements about qualitative attributes.
- Rating scales provide more flexibility to judge the level of performance or presence of attributes among subjects.

### ***Advantages of Rating Scale-***

- Easy to administer and score the measured attributes.
- Rating scales have a wide range of application in nursing research.
- Graphic rating scale is easier to make and requires less time.

### ***Disadvantages of Rating Scale-***

- It is difficult or dangerous to fix up rating about many aspects of an individual.
- Misuse can result in decrease in objectivity.
- There are chances of subjective evaluation; thus the scales may become unscientific and unreliable.

## **2) Checklist-**

### **Definition-**

- A checklist is a simple instrument consisting prepared list of expected items of performance or attributes, which are checked by a researcher for their presence or absence.

### **Characteristics of Checklists-**

- Observe one respondent at one time.
- Clearly specify the characteristics of behaviour to be observed.
- Use only carefully prepared checklist to avoid more complex traits.
- The observer should be trained how to observe, what to observe, and how to record the observed behaviour.

***Advantages of Checklist-***

- Checklists allow inter-individual comparisons.
- It provides a simple method to record observation.
- It is adaptable to subject matter areas.
- It is useful in evaluating learning activities.

***Disadvantages of Checklist-***

- Does not indicate quality of performance, so usefulness of checklist is limited.
- Only a limited component of overall clinical performance can be evaluated.
- Only the presence or absence of an attribute, behaviour, or performance parameter may be assessed.
- It has a limited use in qualitative research studies.

**4. Biophysiologic Methods-****Definition-**

- “Biophysiologic method involves the collection of biophysiologic data from subjects by using the specialized equipment to determine biological and physical status of subjects”.
- E.g. – Blood pressure measurement by using special equipment, such as sphygmomanometer and stethoscope.

**Purposes of Biophysiologic Methods-**

- The purposes of biophysiologic measures are:
  - ✓ To study basic physiologic process.
  - ✓ To study physiologic outcome of nursing care.
  - ✓ To evaluate nursing interventions.
  - ✓ To study correlation of physiologic functioning in patient with health problems.

## Types of Biophysiologic Methods-

→ These are following two types:

- 1) In-vivo biophysiologic methods
- 2) In-vitro biophysiologic methods

### 1) In-Vivo Biophysiologic Method-

→ In in-vivo biophysiologic methods, measurements are directly performed over the organism or study subject by using specialized instruments or equipment.

→ E.g. – Measurement of the blood pressure by using sphygmomanometer, measuring cardiac conduction using ECG machine, measuring temperature using thermometer.

→ In-vivo measuring instrument system involves the following six components:

- ✓ Stimulus
- ✓ Subject
- ✓ Sensory equipment
- ✓ Signal conditioning (amplifiers)
- ✓ Display equipment
- ✓ Recording equipment

### 2) In-Vitro Biophysiologic Method-

→ In in-vitro biophysiologic methods, measurements are carried out outside the organism or study subject by using specialized equipment.

→ E.g. – A blood sample is collected from a patient to assess the blood electrolyte status of the patients.

→ The following measurements can be carried out using in-vitro biophysiologic methods:

- Radiologic measurements
- Biochemical measurements
- Microbiologic measurements
- Cytologic measurements

***Advantages of Biophysiologic Method-***

- Biophysiologic measures are relatively more accurate and errorless.
- Biophysiologic measures are more objective in nature.
- Provide valid measures for targeted variables.

***Disadvantages of Biophysiologic Method-***

- Some of the instruments used for biophysiologic measurements are very expensive.
- The use of biophysiologic method requires significant amount of training, knowledge, and experience.
- The results produced by the biophysiologic measurement instruments may be affected by the environment.

**5. Other Methods-****1) Validity-****Introduction-**

- Validity of an instrument refers to the degree to which an instrument measures what it is supposed to be measuring.
- E.g. – A temperature-measuring instrument is supposed to measure only the temperature.

**Definition-**

- According to Treece and Treece, ‘Validity refers to an instrument or test actually testing what it is supposed to be testing’.
- According to Polit and Hungler, ‘Validity refers to the degree to which an instrument measures what it is supposed to be measuring’.

## **Types of Validity-**

→ Basically validity is classified into following four categories:

- 1) Face validity
- 2) Content validity
- 3) Criterion validity
- 4) Construct validity

### **1) Face validity-**

- Face validity involves an overall look of an instrument regarding its appropriateness to measure a particular attribute or phenomenon.
- E.g. – A Likert scale designed to measure the attitude of the nurses towards the patients admitted with HIV/ AIDS; a researcher may judge the face value of this instrument by its appearance, that is it looks good or not; but it does not provide any guarantee about the appropriateness and completeness of a research instrument with regard to its content, construct, and measurement score.

### **2) Content validity-**

- It is concerned with scope of coverage of the content area to be measured.
- More often it is applied in tests of knowledge measurement.
- It is mostly used in measuring complex psychologic tests of a person.
- It is a case of expert judgment about the content area included in the research instrument to measure a particular phenomenon.

### **3) Criterion validity-**

- This type of validity is a relationship between measurements of the instrument with some other external criteria.
- E.g. – A tool is developed to measure the professionalism among nurses and to assess the criterion validity nurses were separately asked

about the number of research papers they published and number of professional conferences they have attended. Later a correlation coefficient is calculated to assess the criterion validity.

#### **4) Construct validity-**

- A construct is founded in this type of validity, such as a nurse may have designed an instrument to measure the concept of pain in amputated patients.
- The pain pattern may be due to anxiety; hence the results may be misleading.
- Construct validity is a key criterion for assessing the quality of a study, and construct validity has most often been addressed in terms of measurement issues.

#### **2) Reliability-**

##### **Introduction-**

- The quality and adequacy of quantities data can only be assessed by establishing the reliability of an instrument.
- Reliability is the degree of consistency with which the attributes or variables are measured by an instrument.
- E.g. – A blood pressure-measuring instrument gave a reading of 120 mmHg systolic blood pressure; after some time when blood pressure is again measured for the same subject, it gave a reading of 160 mmHg systolic blood pressure. In this situation, this instrument is not considered reliable. However, if a research instrument yields similar or close to similar results on repeated administration of instrument, it is considered a highly reliable research instrument.

##### **Definition-**

- Reliability is the degree of consistency and accuracy with which an instrument measures the attribute for which it is designed to measure.

→ ‘Reliability is defined as the ability of an instrument to create reproducible results’. Therefore, reliability is concerned with consistency of the measurement tools.

## **Measuring of Reliability-**

→ The main aspects of the reliability considered important in quantitative research include:

- 1) Stability
- 2) Internal consistency
- 3) Equivalence

### **1) Stability-**

- The stability aspect of reliability means research instrument provides same results when used consecutively for two or more times.
- Stability is estimated to make sure that research instrument is consistent in providing similar results with repeated administration.
- It is also known as reliability of Test– Retest function.
- To measure test– retest reliability, the test is given twice at two different points in time.
- It is used to measure the consistency of a test across time. This type of reliability supposes that there will be no change in the quality or construct being assessed.
- Test– retest reliability is suitably best used for things that are strong over time (e.g. intelligence). Reliability will be, generally, higher when little time has passed between tests.

### **2) Internal consistency-**

- It is also called homogeneity.
- Internal consistency ensures that all the subparts of a research instrument measure the same characteristics.

- E.g. – A patient's satisfaction measurement scale developed to measure his or her satisfaction with nursing care must include all the subparts related to the measurement of satisfaction with nursing care only.

### **3) Equivalence-**

- This aspect of reliability is estimated when a researcher is testing the reliability of a tool, which is used by two different observers to observe a single phenomenon simultaneously and independently or two presumably parallel instruments are administered to an individual at about the same time.
- This is also known as interrater or interobserver reliability.
- E.g. – A rating scale is developed to assess cleanliness of the bone marrow transplantation unit; this rating scale may be administered to observe the cleanliness of the bone marrow transplantation unit by two different observers simultaneously but independently.

### **3) Pilot Study-**

- In nursing science research, the term pilot study is utilized in two different ways.
- It refers to the so-called feasibility studies that are trial run or small-scale versions, which are done in preparation for the major study. However, a pilot study can also be the 'trying out' or pretesting of a specific research instrument.
- A pilot study is referred to a small-scale preliminary tryout of the method to be used in an actually large study, which acquaints the researcher with problems that can be corrected in proportion for the large research study or is done to provide the researcher with an opportunity to try out the procedure, methods, and tools of data collection.

**Definition-**

- Pilot study is a smaller version of a study carried out before the actual investigation is done. Researchers use information gathered in pilot studies to refine or modify the research methodology for a study and to develop large-scale studies.
- Pilot study is a small-scale study conducted to test the plan and method of a research study.

**Purposes of Pilot Study-**

- Assessing study feasibility and practicability of research study.
- Assessing the availability of study subjects.
- Establishing validity and reliability of the research tools.
- Ensuring the appropriateness of methods and procedures of data collection.
- Understanding the study variables and other confounding variables.
- Estimating the actual time and potential problems researcher may encounter during the actual large research study.
- Refining the study methodology.
- Planning for the data analysis and interpretation of the final larger research.

**Problems & Issues of Pilot Study-**

- It should be recognized that the pilot studies may also have a number of problems, issues, and limitations.
- Following are some of the significant problems or issues of pilot study:
  - ✓ It may be a possibility that the researcher may make inaccurate predictions or assumptions on the basis of pilot data, specifically when the data of pilot study is used for estimating the sample size.
  - ✓ Contamination is another concern. This may arise in two ways:

- a) Where pilot participants are included in the main study, but new data are collected from these people.
- b) Where data from the pilot study are included in the main results.
- ✓ The main study is generally not conducted in the same area where pilot study was conducted to avoid the risk of contamination.

## **Procedure of Data Collection-**

- Data collection is a long and tedious process, which is generally done by researchers themselves in small size studies; however, in large size studies, data collectors are used by the investigators for data collection.
- The data collection procedure involves following steps:

### **1) Operationally Defining the Data to Be Collected-**

- It is essential for researcher to identify and clearly define the data to be collected in the study.
- Researcher must identify the nature of data to be collected whether they are quantitative or qualitative in nature.

### **2) Selecting Appropriate Methods of Data Collection-**

- After clearly identifying and defining the data to be collected; the next step of data collection is to select appropriate methods of data collection.
- There are different methods of data collection available such as self-reporting methods (questionnaire, composite scales), interviews, and observations and so on.
- Researcher has to select most suitable methods of data collection based on the nature of data to be collected.

### **3) Selecting or Developing Data Collection Instrument-**

- Once researcher makes a choice about most appropriate method of data collection, the next step is to identify an existing tool for data collection, which can be used for present study.
- In case, the existing tool does not exactly suit the present data collection needs, it can be modified based on the needs of the present study.

### **4) Try-out of Data Collection Instrument & Procedure-**

- Researcher either chose a data collection instrument, modify an existing tool or develop a new data collection tool and decide the data collection methodology, which need to be tried-out to identify practical difficulties may be faced in data collection procedure and modifications are made accordingly in the data collection instrument, as well as data collection methodology.

### **5) Developing Data Collection Supportive Forms & Procedure-**

- Once the researcher has a valid and reliable data collection instrument ready, he or she also prepares the supportive forms needed for the data collection such as informed consent, administrative permission letters, list, contact details of participants.

### **6) Training of Data Collectors-**

- Investigators involve data collectors in most of the large studies.
- These data collectors need different levels of training based on their prior experience of data collection.
- Training usually covers both general procedures (such as how to conduct a data collection interview) and the specific to particular study

(such as how to administer particular specialized data collection instruments).

### **7) Inform the Local Authority for Data Collection & Obtain Due Permissions-**

→ Before actual data collection begins, it is the responsibility of the investigator to inform the local authority in command and obtain their permission for data collection, so that their support and cooperation can be obtained.

### **8) Data Collection & Data Management-**

- The final step in the procedure of data collection is data collection and data management.
- This is the most crucial step of data collection procedure, where consent is obtained and data are collected carefully and if needed changes are made in data collection methodologies based on ongoing needs of data collection procedure.

## **Introduction to Statistics**

### **Introduction-**

- The root of the word statistics comes from the Italian word *statista* (meaning ‘statesman’) and the German word *statistik* (meaning ‘political state’).
- It was used for the first time by Gottfried Achenwall, a professor in Marlborough, in 1749 to refer to the subject matter as a whole.
- The word statistics conveys a variety of meanings to people in different walks of life.
- To some, it could be an imposing form of mathematics containing tables, charts and figures, which one can commonly find in newspapers, journals, books, reports, speeches, classroom lectures, etc.

### **Definition-**

- Statistics can be defined as numerical data involving variability and treatment of such data.
- According to Croxton and Cowden, ‘Statistics is defined as collection, presentation, analysis, and interpretation of numerical data’.

### **Uses & Applications of Statistics-**

#### **Uses-**

- It presents facts in a definite form.
- It facilitates comparisons.
- It simplifies the masses of figures.
- It helps in formulating and testing hypothesis.
- It helps in prediction.

#### **Applications-**

- Applications of statistics in health sciences can be conceptualized into one of the following categories:
  - Defining normal and not normal in context of various aspects related to health and illness.

- Establishing the accuracy of diagnostic procedures.
- Planning of experiments and analysis of results.
- Collection, analysis, and dissemination of various population health statistics.

## **Types of Data/Scales of Measurement-**

→ There are four levels of measurements:

- 1) Nominal level measurement
- 2) Ordinal level measurement
- 3) Interval level measurement
- 4) Ratio level measurement

### **1) Nominal-Level Measurement-**

→ The nominal-level measurement is described as follows:

- ✓ It is the lowest of the four levels of measurements.
- ✓ It involves the assignment of number to represent categories or classes of things.
- ✓ It consists of categories that are not more or less than each other but are different from one another in some way. In other words, they have no quantitative values.
- ✓ Its categories are mutually exclusive and exhaustive categories.
- ✓ E.g. –
  - Gender: Male, Female
  - Habitat: Urban, Rural, Slums

### **2) Ordinal-Level Measurement-**

→ The ordinal-level measurement is described as follows:

- ✓ It ranks objects based on their relative standing on a specific attribute.

- ✓ The order of ranking is imposed on categories.
- ✓ The numbers must preserve some level of order but do not need to be precise.
- ✓ It reflects only magnitude and does not equal intervals or an absolute zero point.
- ✓ E.g. –
  - Health status: Poor, Fair, Good, Excellent
  - Income status: Low income, Middle income, Upper income

### **3) Interval-Level Measurement-**

→ The interval-level measurement is described as follows:

- ✓ In this level, there is specification of ranking of objects on an attribute of the distance between those objects.
- ✓ There is, more or less, equal numerical distance between intervals.
- ✓ There is no absolute zero because there is no zero point or absence of concept.
- ✓ E.g. – Temperature: 40 – 50<sup>0</sup>C

### **4) Ratio-Level Measurement-**

→ The ratio-level measurement is described as follows:

- ✓ It is the highest level for measurement.
- ✓ It has all the three attributes: magnitude, equal intervals, and absolute zero point.
- ✓ It represents continuous values.
- ✓ E.g. – Biophysical parameters: Weight, Height, Volume

### **Classification of Statistics-**

→ Statistics is broadly classified into two categories are:

1. Descriptive statistics
2. Inferential statistics

## **1. Descriptive Statistics-**

- Descriptive statistics deals with the enumeration, organization, and graphical representation of data.
- These are following types:
  - 1) Measures of data condensation
  - 2) Measures of central tendency
  - 3) Measures of dispersion
  - 4) Measures of relationship (Correlation coefficient)

### **1) Measures of Data Condensation-**

- These are following two subcategories:
  - a) Frequency distribution
  - b) Graphical presentation of data

#### **a) Frequency Distribution-**

- Quantitative data are generally condensed and frequency distribution is presented through tables, charts, graphs, and diagrams.

### **Tabulation-**

- A table presents data in a concise, systematic manner from masses of statistical data.
- Tabulation means a systematic presentation of information contained in the data in rows and columns in accordance with some common features and characteristics.
- Rows are horizontal and columns are vertical arrangements.

### ***General Principles of Tabulation-***

- These are following principles:
  - A table should be precise, understandable, and self-explanatory.
  - Every table should have a title, which is placed at the top of the table.

- The title must describe the content clearly and precisely.
- Items should be arranged alphabetically or according to size, importance, or causal relationship to facilitate comparison.
- Rows and columns to be compared with one another should be brought together.
- The unit of measurement must be clearly stated.

### ***Parts of a Table-***

→ These are following parts of a table:

- ✓ **Table number:** It should be placed at the top of the table.
- ✓ **Title:** Every table must have a suitable title. A title should be brief and concise and describe the contents of the table. The content of the table should be self-explanatory.
- ✓ **Head notes:** Head note is given just below the title in a prominent type usually enclosed in brackets for further description of the contents of table.
- ✓ **Captions and stubs:** Captions are the headings designated for vertical columns and stubs are the headings for horizontal rows.
- ✓ **Body of table:** Arrangement of the data according to description given in the form of captions and stubs form the body of the table.
- ✓ **Footnotes:** When some characteristics or items of the table are not adequately explained, footnotes are used to explain those items.
- ✓ **Source note:** Source note is used when secondary data is used to mention the source from which the data for the table or the table itself is retrieved.

***Types of Tables-***

→ Basically tables are following four types:

**1) Frequency Distribution Table-**

- These tables present the frequency and percentage distribution of the information collected, where an attribute is grouped into number of classes, which may vary between three and eight.
- Too many or too few groups or classes may fail to reveal the salient features of the data.

**2) Contingency Table-**

- Tables that report the frequency distribution of two nominal variables simultaneously and that include the totals are known as contingency tables.
- The categories considered should be mutually exclusive as well as exhaustive.
- Contingency tables are also known as cross tables, which present the frequency distribution of two or more variables to establish the relationship or association among them.

**3) Multiple-Response Table-**

- When classification of the cases is done into categories that are neither exclusive nor exhaustive, then it is called a multiple-response table.
- For example, a patient can have two or more complaints, but only the major ones may be listed. In such cases, the sum total of frequencies would exceed the total number of subjects and may lead to confusion.

**4) Miscellaneous Table-**

- These tables are used to present data other than frequency or percentage distributions, such as mean, median, mode, range, standard deviation.

**b) Graphical Presentation of Data-**

→ The main reasons for using the diagrammatic and graphic representation of data are as follows:

- ✓ It is the most convenient and appealing way in which statistical results may be presented.
- ✓ It gives an overall view of entire data.
- ✓ It is visually more attractive than other ways of representing data.
- ✓ It is easier to understand and memorize data through graphical representation.

***Types of Diagrams & Graphs-***

→ These are following types of diagrams & graphs:

**1) Bar Diagram-**

- It is a convenient graphical device that is particularly useful for displaying nominal or ordinal data.
- It is an easy method adopted for visual comparison of the magnitude of different frequencies.
- Length of the bars drawn vertically or horizontally indicates the frequency of a character.
- The bar charts are called vertical bar charts (or column charts), if the bars are placed vertically.
- When the bars are placed horizontally, it is called horizontal bar charts.

**2) Pie Diagram/Sector Diagram-**

- It is another useful pictorial device for presenting discrete data of qualitative characteristics, such as age groups, genders, and occupational groups in a population.
- The total area of the circle represents the entire data under consideration.
- Researcher must remember that only percentage data must be used to prepare pie diagrams.

**3) Histogram-**

- It is the most commonly used graphical representation of grouped frequency distribution.
- Variable characters of the different groups are indicated on the horizontal line (x-axis) and frequencies (number of observation) are indicated on the vertical line (y-axis).
- Frequency of each group forms a column or rectangle. Such a diagram is called a histogram.

**4) Frequency Polygon-**

- It is the curve obtained by joining the middle top points of the rectangles in a histogram by straight lines.
- It gives a polygon that is figure with many angles.
- In this, the two end points of the line drawn are joined to the horizontal axis at the midpoint of the empty class-intervals at both ends of the frequency distribution.
- Frequency polygons are simple and sketch an outline of data pattern more clearly than histograms.

**5) Line Graphs-**

- In this, variables in the frequency polygon are depicted by a line.
- It is mostly used where data is collected over a long period of time.
- On x-axis, values of independent variables are taken and values of dependent variables are taken on y-axis.
- Vertical axis may not start from zero, but at some point, from where frequency starts.

**6) Cumulative Frequency Curve or Ogive-**

- This graph represents the data of a cumulative frequency distribution.
- For drawing an ogive, an ordinary frequency distribution table is converted into cumulative frequency table.

- The cumulative frequencies are then plotted corresponding to the upper limits of the classes.
- The points corresponding to cumulative frequency at each upper limit of the classes are joined by a free-hand curve. The diagram made is called ogive.

### **7) Scattered or Dotted Diagrams-**

- It is a graphic presentation that shows the nature of correlation between two variable characters  $x$  and  $y$  on the similar features or characteristics.
- It is also called correlation diagram.

### **8) Pictograms or Picture Diagram-**

- This method is used to impress the frequency of the occurrence of events to common people, such as attacks, deaths, number of operations, admissions, accidents, and discharges in a population.

### **9) Map diagram or Spot Map-**

- These maps are prepared to show geographical distribution of frequencies of characteristics.

## **2. Measures of Central Tendency-**

- A researcher collects the values of any variable under study, like Hb% from a large population, and the observation values cluster around a central value. This feature of concentration of the value around a central value is known as central tendency.

### **Definition-**

- According to Ya-Chou, “The average is sometimes described as a typical value in the sense that it is sometimes employed to represent the entire individual in a series of a variable”.

**Objectives of Central Value-**

- ✓ To get a single value that describes the characteristics of the entire group.
- ✓ To facilitate comparisons.

**Requisites of Good Averaging-**

- ✓ Easy to understand
- ✓ Simple to compute
- ✓ Based on all the items
- ✓ Capable of further algebraic treatment
- ✓ Sampling stabilities

**Measures of Central Tendency-**

→ The important measures of central tendency are as follows:

- 1) Mean
- 2) Median
- 3) Mode

**1) Mean-**

→ The arithmetic mean is the quantity obtained by summing two or more numbers and then dividing by the total number of numbers.

→ It is also known as average or average value.

$$\bar{X} = \frac{\text{Sum of the values } (\Sigma x)}{\text{Number of the values } (n)}$$

**Question-** The haemoglobin levels of the 10 women are given here, that is 12.5, 13, 10, 11.5, 11, 14, 9, 7.5, 10, and 12.

**Solution-**

→ The mean haemoglobin of this sample of women will be calculated as follows:

$$\begin{aligned}\text{Formula } \bar{X} &= \frac{\text{Sum of the values } (\Sigma x)}{\text{Number of the values } (n)} \\ &= \frac{12.5 + 13 + 10 + 11.5 + 11 + 14 + 9 + 7.5 + 10 + 12}{10} = \frac{110.5}{10} \\ \bar{X} &= 11.05\end{aligned}$$

→ The mean haemoglobin of the study sample is 11.05.

**Condition: I- Calculating arithmetic mean from discrete frequency table-**

→ In discrete frequency table, the mean is calculated using the following formula:

$$\bar{X} = \frac{\Sigma xf}{\Sigma f}$$

➤ Where  $x$  = corresponding value variable,  $f$  = frequency.

**Question-** The following data gives the age of 100 adolescent girls. Find the mean age.

| Age in years ( $x$ ) | No. of students ( $f$ ) |
|----------------------|-------------------------|
| 16                   | 35                      |
| 17                   | 31                      |
| 18                   | 20                      |
| 19                   | 14                      |

**Solution-**

| Age in years (x) | No. of students ( f ) | xf                 |
|------------------|-----------------------|--------------------|
| 16               | 35                    | 560                |
| 17               | 31                    | 527                |
| 18               | 20                    | 360                |
| 19               | 14                    | 266                |
|                  | $\Sigma f = 100$      | $\Sigma fx = 1713$ |

$$\bar{X} = \frac{\Sigma xf}{\Sigma f} = \frac{1713}{100} = 17.13$$

→ Mean age of the adolescents in this group is 17.13 years.

**Condition: II- Calculating arithmetic mean from continuous frequency table-**

→ In case of continuous frequency table, mean can be calculated by the following formula:

$$\bar{X} = \frac{\Sigma xf}{\Sigma f}$$

- Where x = Midpoint of class interval, f = Corresponding frequency.
- Midpoint of the class interval is calculated by the following formula:

$$\text{Midpoint} = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

**Question-** Calculate the mean age of the following group of people.

| Class interval of age | No. of people ( $f$ ) |
|-----------------------|-----------------------|
| 15 – 20               | 15                    |
| 20 – 25               | 20                    |
| 25 – 30               | 40                    |
| 30 – 35               | 60                    |

**Solution-**

| Class interval of age | No. of people ( $f$ ) | Midpoint (x) | $xf$                 |
|-----------------------|-----------------------|--------------|----------------------|
| 15 – 20               | 15                    | 17.5         | 262.5                |
| 20 – 25               | 20                    | 22.5         | 450                  |
| 25 – 30               | 40                    | 27.5         | 1100                 |
| 30 – 35               | 60                    | 32.5         | 1950                 |
|                       | $\Sigma f = 135$      |              | $\Sigma xf = 3762.5$ |

➤ Midpoint (x) =  $15+20/2 = 17.5$  (calculate complete this method)

$$\bar{X} = \frac{\Sigma xf}{\Sigma f} = \frac{3762.5}{135} = 27.87$$

→ Mean age of the people in this group will be 27.87 years.

**Merits-**

→ It is a simple average to understand and easiest to compute.

→ It is affected by the value of each item in a series.

→ It is defined by a fixed mathematical formula with the result that irrespective of whoever computes the average gets the same answer.

→ It can be treated algebraically.

**Demerits-**

- Very small and very large items usually affect the value of average.
- In the distribution with open-ended classes, values of mean cannot be computed without making assumptions.
- It is not always ‘a good measure of central tendency’.

**2) Median-**

- Median of a set of values is the middle-most value when the data is arranged in ascending order of magnitude.
- The middle value will divide the number of observations in the data into two equal parts.
- The median is denoted by **M**.
- It is also called positional average.

**Condition: I- Data with Odd set of numbers:**

- In case of discrete data with odd set of numbers, median is calculated using following formula:

$$\text{Formula} = M = \left( \frac{n+1}{2} \right)^{\text{th}} \text{ observation or item}$$

**Question-** The following data gives the weight of the seven persons in pounds: 158, 167, 143, 169, 172, 146, and 151. The median of the data given is calculated.

**Solution-**

- First arrange the data in ascending or descending order of magnitude:  
143, 146, 151, 158, 167, 169, 172.

$$M = \left( \frac{n+1}{2} \right); \quad M = \frac{7+1}{2} = \frac{8}{2} = 4\text{th observation or item}$$

→ The fourth observation in above given data is the median. Therefore, in this example, the fourth value, 158, will be the median.

**Condition: II- Calculating median for discrete data with Even set of numbers:**

→ In case of discrete data with even set of numbers, the median is calculated by using following steps as given in example.

**Question-** Find the median for the following data set: 102, 56, 34, 99, 89, 101, 10, 54.

**Solution-**

→ Place the data in ascending order (smallest to highest): 10, 34, 54, 56, 89, 99, 101, 102.

→ Find two numbers in the middle (where there are an equal number of data points above and below the two middle numbers).

→ Add the two middle numbers ( $56 + 89 = 145$ ) and then divide by 2, to get the average:  $145/2 = 72.5$ .

→ Thus the median will be 72.5 for above given odd set of data.

**Condition: III- Calculation of median from the discrete frequency table:**

**Question-** Calculate the median for the following frequency table data.

|                             |     |     |     |     |     |     |
|-----------------------------|-----|-----|-----|-----|-----|-----|
| Income/day (x)              | 100 | 150 | 200 | 250 | 300 | 350 |
| Number of households ( f )  | 05  | 19  | 03  | 11  | 06  | 09  |
| Cumulative frequency ( cf ) | 05  | 24  | 27  | 38  | 44  | 53  |

**Solution-**

- In case of the discrete frequency table, one has to calculate cumulative frequency first.
- Then consider the final cumulative frequency as  $n$ , and calculate the median by the following formula.

$$M = \left( \frac{n+1}{2} \right); \quad M = \frac{53+1}{2} = \frac{54}{2} = 27\text{th observation or item}$$

- The 27th observation in array, which means 27th cumulative frequency and x value in front of 27th cumulative frequency is 200. Therefore, median value will be 200.

**Condition: IV- Calculation of median from the continuous frequency table:**

**Question-** Calculate the median for the following frequency table data.

| Income/day (x)              | 100-150 | 150-200 | 200-250 | 250-300 | 300-350 | 350-400 |
|-----------------------------|---------|---------|---------|---------|---------|---------|
| Number of households ( f )  | 05      | 19      | 03      | 11      | 06      | 09      |
| Cumulative frequency ( cf ) | 05      | 24      | 27      | 38      | 44      | 53      |

**Solution-**

- In case of continuous frequency table, median can be calculated by using following formula:

$$M = l + \frac{(N/2 - m)w}{f}$$

- Where  $l$  = the lowest limit = 200
- $w$  = the width = 50
- $f$  = the frequency of the median class = 03
- $m$  = the cumulative frequency of a class just before the median class = 24
- $N = \Sigma f = 53$

$$\begin{aligned} M &= 200 + \frac{(26.5 - 24)50}{03} = 200 + \frac{2.5 \times 50}{03} = 200 + \frac{125}{03} \\ &= 200 + 41.67 = 241.7 \end{aligned}$$

→ The median is 241.7.

### ***Merits-***

- It is useful in case of open-ended and unequal classes.
- Extreme values do not affect the median.
- Most appropriate average in dealing with qualitative data.

### ***Demerits-***

- For calculating the median it is necessary to arrange the data.
- Since it is the positional averages, the value is not determined by all the observations.
- It is not capable of further algebraic treatment.

## **3) Mode-**

- Mode is most frequently occurring number in a data set.
- In other words, it is the value which has the highest frequency in the data.
- It may also be regarded as the most typical of a series of values.
- It is denoted by **Z**.

**Question-** Calculate the mode in given table.

| Income/day (x)             | 100-150 | 150-200 | 200-250 | 250-300 | 300-350 | 350-400 |
|----------------------------|---------|---------|---------|---------|---------|---------|
| Number of households ( f ) | 5       | 19      | 3       | 11      | 6       | 9       |

**Solution-**

$$Z = l + \frac{(f - f_1)w}{2f - f_1 - f_2}$$

- Where  $l$  = lowest limit of the modal class
- $f$  = frequency of the modal class
- $w$  = width of the modal class
- $f_1$  = frequency of class just before the modal class
- $f_2$  = frequency of class just after the modal class.

$$Z = l + \frac{(f - f_1)w}{2f - f_1 - f_2} \text{ where } l = 150, f = 19, w = 50, f_1 = 5, f_2 = 3,$$

$$Z = 150 + \frac{(19 - 5)50}{2 \times 19 - 5 - 3} = 150 + \frac{14 \times 50}{38 - 8}$$

$$Z = 150 + \frac{(700)}{30} = 150 + 23.33 = 173.33$$

→ The mode for the above tabulated data is 173.33.

**Merits-**

- It is not affected by extreme values.
- It can be used to describe the qualitative phenomenon.
- Values of mode can be determined graphically.

**Demerits-**

- The value of mode cannot always be determined.
- It is not capable of algebraic manipulation.
- It is not based on each value.

**3. Measures of Dispersion-**

- The observation deviating from the central value is different in different set of values of character.
- In some distributions, the difference may be less whereas in some others, it may be more.
- This property of deviation of the values from the average is called variation or dispersion.

**Standard Deviation-**

- It is the most prominently used measure of dispersion.
- It is denoted by 'SD' or ' $\sigma$ '.

**Definition-**

- Standard deviation is the positive square root of mean of the squared deviations of values from the arithmetic mean.

**Condition: I- Calculation of the SD from discrete data:**

**Question-** Blood cholesterol levels of 10 persons are 240, 260, 290, 245, 255, 288, 272, 263, 277, and 257. Calculate the standard deviation of the given discrete data.

**Solution-**

| $x$             | $(x - \bar{x})$       | $(x - \bar{x})^2$               |
|-----------------|-----------------------|---------------------------------|
| 240             | $240 - 264.7 = -24.7$ | 610.09                          |
| 245             | $245 - 264.7 = -19.7$ | 388.09                          |
| 255             | $255 - 264.7 = -9.7$  | 94.09                           |
| 257             | $257 - 264.7 = -7.7$  | 59.29                           |
| 260             | $260 - 264.7 = -4.7$  | 22.09                           |
| 263             | $263 - 264.7 = -1.7$  | 2.89                            |
| 272             | $272 - 264.7 = 7.3$   | 53.29                           |
| 277             | $277 - 264.7 = 12.3$  | 151.29                          |
| 288             | $288 - 264.7 = 23.3$  | 542.89                          |
| 290             | $290 - 264.7 = 25.3$  | 640.09                          |
| $\sum x = 2647$ | $\bar{x} = 264.7$     | $\sum (x - \bar{x})^2 = 2564.1$ |

$$SD = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \quad \text{Where; } \sum (x - \bar{x})^2 = 2564.1 \text{ and } n (\text{number of observation}) = 10$$

$$SD = \sqrt{\frac{2564.1}{10}} = \sqrt{256.41}$$

$$SD = 16.01$$

**Condition: II- Calculation of the SD from continuous frequency data:**

**Question-** The following data depicts the weight of the adolescent girls. Calculate the standard deviation for the following continuous frequency data.

| Weight          | 60-64 | 64-68 | 68-71 | 71-74 |
|-----------------|-------|-------|-------|-------|
| No. of subjects | 10    | 9     | 7     | 2     |

**Solution-**

| Weight (kg) | $f$             | Midpoint (x) | $xf$               | $(x - \bar{x})$   | $(x - \bar{x})^2$ | $(x - \bar{x})^2 \cdot f$                 |
|-------------|-----------------|--------------|--------------------|-------------------|-------------------|-------------------------------------------|
| 60-64       | 10              | 62           | 620                | -4.14             | 17.14             | 171.40                                    |
| 64-68       | 9               | 66           | 594                | 00                | 00                | 00                                        |
| 68-72       | 7               | 70           | 490                | 3.86              | 14.90             | 104.3                                     |
| 72-76       | 2               | 74           | 148                | 7.86              | 61.78             | 123.56                                    |
|             | $\Sigma f = 28$ |              | $\Sigma xf = 1852$ | $\bar{x} = 66.14$ |                   | $\Sigma (x - \bar{x})^2 \cdot f = 399.26$ |

$$\bar{x} = \frac{\Sigma xf}{\Sigma f} = \frac{1852}{28} = 66.14$$

$$SD = \sqrt{\frac{\Sigma (x - \bar{x})^2 \cdot f}{n}} \quad \text{Where, } \Sigma (x - \bar{x})^2 \cdot f = 399.26$$

$$n(\Sigma f) = 28$$

$$SD = \sqrt{\frac{399.26}{28}} = \sqrt{14.26}$$

$$SD = 3.78$$

**4. Correlation Coefficient-**

- Sometimes two continuous characters are measured in a series or in a similar pattern such as weight and cholesterol, weight and height, IQ in brothers and sisters.
- This relationship or association between two quantitatively measured or continuous variables is called correlation.
- The extent or degree of relationship between two sets of figures is measured in terms of another parameter called correlation coefficient. It is denoted by **r**.

## Types of Correlation Coefficient-

→ These are following types:

### 1) Perfect Positive Correlation ( $r = +1$ ):

→ X is directly proportional to Y.

→ E.g. – Designation and salary, height and weight.

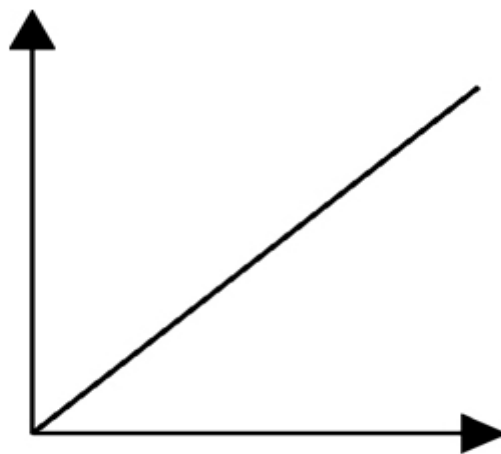


Fig. – Perfect Positive Correlation

### 2) Perfect Negative Correlation ( $r = -1$ ):

→ X and Y are inversely proportional to each other.

→ E.g. – Leniency of teacher and discipline among students.

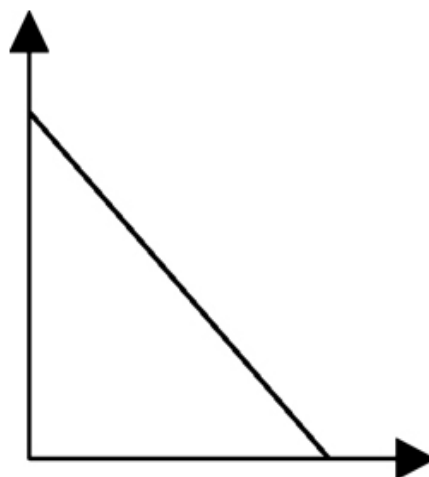


Fig. – Perfect negative correlation

**3) Moderately Positive Correlation ( $0 < r < 1$ ):**

→ E.g. – The correlation between infant mortality rate (IMR) and overcrowding, and correlation between the plasma value and circulating albumin in grams.

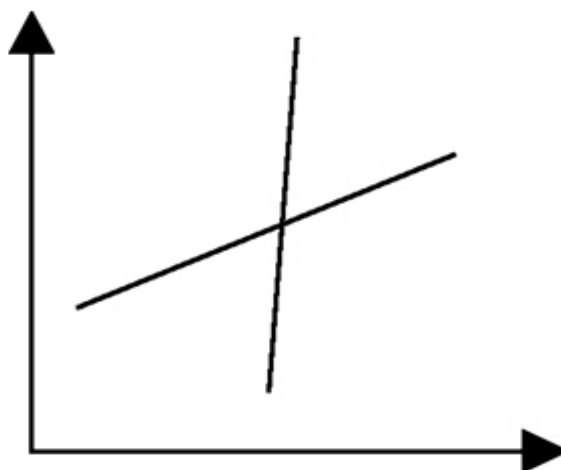


Fig. – Moderately positive correlation

**4) Moderately Negative Correlation ( $-1 > r > 0$ ):**

→ E.g. – Age and vital capacity in adults, IMR, and income.

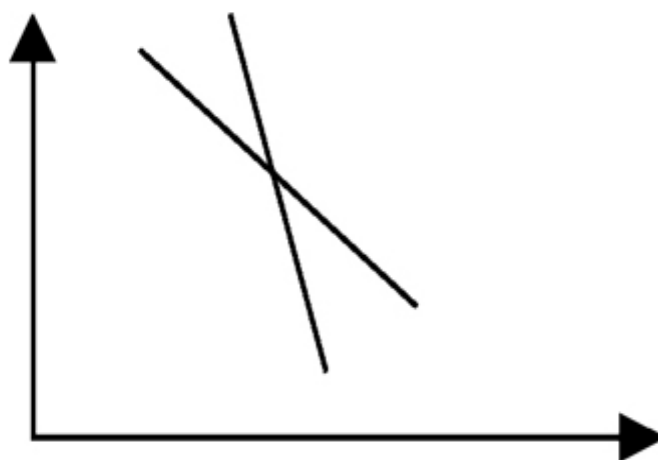


Fig. – Moderately negative correlation

**5) Absolutely No Correlation ( $r = 0$ ):**

→ E.g. – Habit of smoking and housing facility of person.

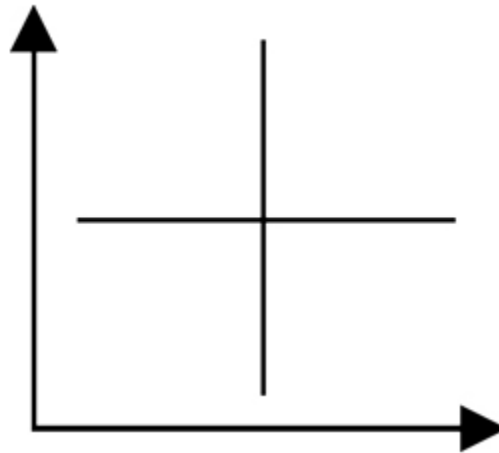


Fig. – Absolutely no correlation

**Method of Computing the Correlation-****Karl Pearson's Correlation Coefficient-**

- It is used to measure the degree of linear relationship between two variables.
- It is also called product moment correlation coefficient.
- It is denoted by  $r$ .
- It can be computed using the following formula:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

**Question-** Compute the correlation coefficient from following data.

|                     |     |     |     |     |
|---------------------|-----|-----|-----|-----|
| Weight (kg)         | 60  | 70  | 80  | 90  |
| Cholesterol (mg/dl) | 120 | 130 | 140 | 150 |

**Solution-**

| $x$            | $y$             | $(x - \bar{x})$ | $(y - \bar{y})$ | $(x - \bar{x})(y - \bar{y})$           | $(x - \bar{x})^2$           | $(y - \bar{y})^2$           |
|----------------|-----------------|-----------------|-----------------|----------------------------------------|-----------------------------|-----------------------------|
| 60             | 120             | -15             | -15             | 225                                    | 225                         | 225                         |
| 70             | 130             | -5              | -5              | 25                                     | 25                          | 25                          |
| 80             | 140             | 5               | 5               | 25                                     | 25                          | 25                          |
| 90             | 150             | 15              | 15              | 225                                    | 225                         | 225                         |
| $\bar{x} = 75$ | $\bar{y} = 135$ |                 |                 | $\sum(x - \bar{x})(y - \bar{y}) = 500$ | $\sum(x - \bar{x})^2 = 500$ | $\sum(y - \bar{y})^2 = 500$ |

$$r = \frac{500}{\sqrt{500 \times 500}} = \frac{500}{\sqrt{250,000}} = \frac{500}{500} = 1$$

→ This means that there was increase in cholesterol level with increase in weight of the patients.

**Merits-**

→ It summarizes in a figure not only the degree of correlation but also the direction.

**Demerits-**

→ The correlation coefficient always assumes linear relationship regardless of the fact that assumption is correct or not.

→ Great care must be exercised in interpreting the values of coefficient.

→ The value of the coefficient is unduly affected by the extreme items.

**Normal Probability Curve (Z Score)-**

- If large values are collected for any character and a frequency table is prepared with small class interval, the frequency curve of this data will give a bell-shaped symmetrical curve, which is known as Gaussian or normal curve.
- The shape of this curve depends on the mean and SD of data.
- If the standard deviation (variation) is very high, the width of the curve is also more.
- If we move both sides from the midpoint ( $x = \text{mean}$ ), the height of the curve decreases.
- If the mean and standard deviation are 0 and 1, respectively, then normal curve is called standard normal curve.
- If the total area under the curve is considered as unity, the normal curve is called normal probability curve.

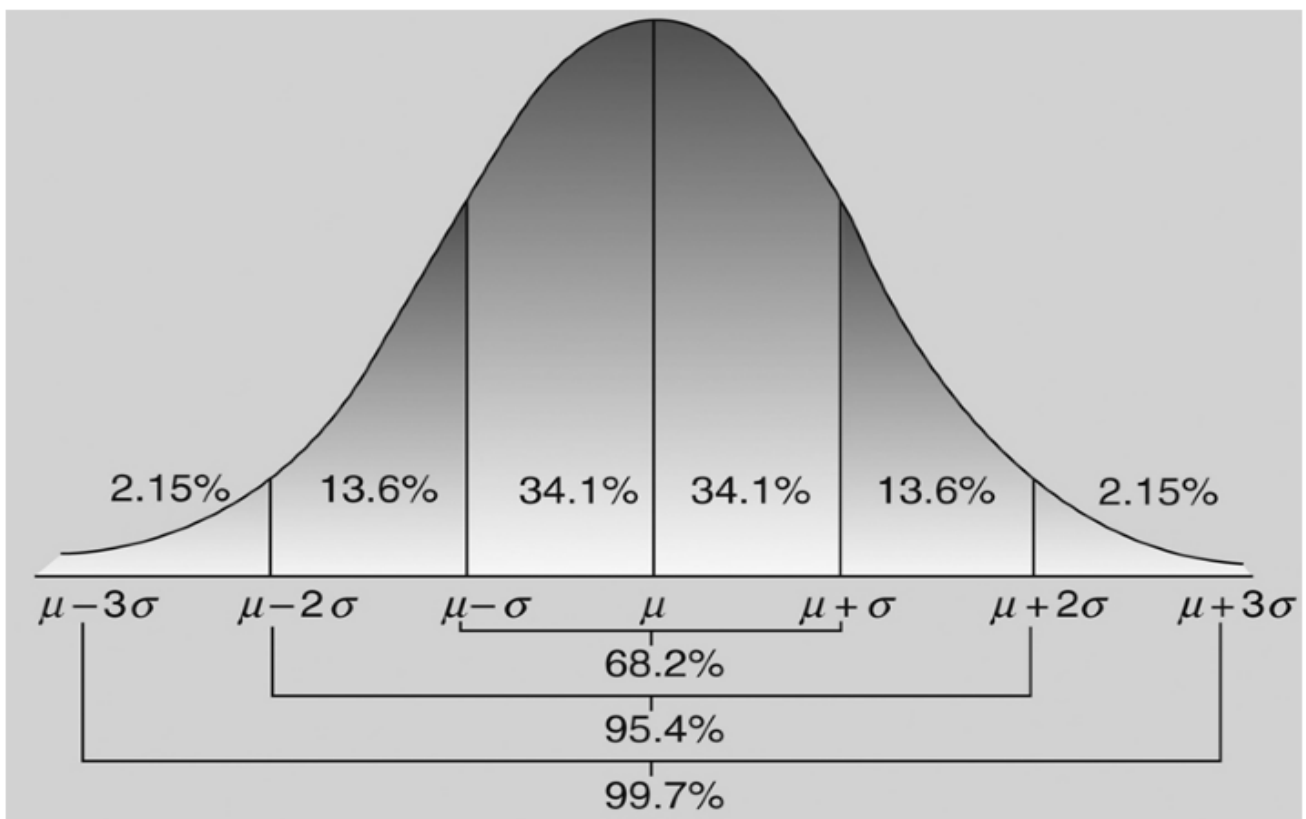


Fig. – Normal probability curve

→ Formula is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

➤ In this formula:

- $\mu$  = Mean distribution (it also may be median/mode)
- $\sigma$  = Standard deviation, and its variance is therefore  $\sigma^2$

### Properties of Normal Probability Curve-

- It is a normal bell-shaped curve.
- Normal probability curve is continuous-type probability curve.
- The curve is symmetrical and asymptotic (i.e. it touches at infinity).
- All the measures of central tendency are equal and stable on the highest peak axis, that is mean = median = mode.
- The total area under the curve is equal to unity.
- The quartiles Q1 and Q3 are equidistant from the mean  $\mu$  and  $Q1 = \mu - 0.67\sigma$ ,  $Q3 = \mu + 0.67\sigma$  approximately.
- The normal curve has two parameters, that is mean ( $\mu$ ) and standard deviation ( $\sigma$ ).
- On the basis of mean ( $\mu$ ) and standard deviation ( $\sigma$ ), the area of normal curve is distributed as:
  - $\mu \pm 1\sigma = 68.27\%$
  - $\mu \pm 2\sigma = 95.45\%$
  - $\mu \pm 3\sigma = 99.73\%$
  - $\mu \pm 0.67\sigma = 50\%$
  - $\mu \pm 1.96\sigma = 95\%$
  - $\mu \pm 2.58\sigma = 99\%$
- Normal curve is mesokurtic.

**Importance of Normal Probability Curve-**

- Normal distribution plays an important role in the sampling theory.
- Data obtained from biological measurements approximately follow normal distribution.
- Binomial and Poisson distribution can be approximated to normal distribution.
- For a large sample, any statistics (i.e. sample mean, sample standard deviation) approximately follow normal distribution, and as such it can be studied with the help of a normal curve.
- Normal curve is used to find confidence limits of the population parameters.

**Inferential Statistics-  
Introduction-**

- Inferential statistics is concerned with populations and uses sample data to make an inference about the population or to test the hypotheses considered at the beginning of the research study.
- Inferential statistics helps the researcher to determine whether the difference found between two or more groups, such as an experimental and a control group, is a real difference or only a chance difference that occurred because an unrepresentative sample was chosen from the population.

**Parametric Tests-**

- In this parametric tests included:
  - 1) t-test
  - 2) ANOVA

**1) t – test-**

- It is applied to find the significant difference between two means.
- This test can be applied when following criteria are fulfilled:
  - Randomly selected homogeneous sample.
  - Quantitative data measures on interval or ratio scale.
  - Variability normally distributed.

**Types of t-Tests-**

- The t-test is further divided in two types:

**1) Unpaired t-test-**

- It is also known as independent sample t-test.
- It is applied when we obtain data from subjects of two independent separate groups of people or samples drawn from two different populations.
- E.g. – When you are comparing post-interventional mean systolic blood pressure of male and female participants.
- Formula:

$$\text{Standard error (SE)} = \sqrt{\frac{(\text{SD}_1)^2}{n_1} + \frac{(\text{SD}_2)^2}{n_2}}$$

$$t\text{-Test} = \frac{\text{Observed difference}(\bar{X}_1 - \bar{X}_2)}{\text{SE}}$$

## ➤ Where:

- SD1: Standard deviation of the first sample
- SD2: Standard deviation of the second sample
- n1: Number of subjects in first sample
- n2: Number of subjects in the second sample

➤ Observed difference =  $\bar{X}_1 - \bar{X}_2$ .

- Where  $\bar{X}_1$  is mean of the first sample and  $\bar{X}_2$  is the mean of the second sample

**2) Paired t-test-**

→ It is applied on paired data of independent observations made on same sample before and after the intervention.

→ E.g. – Comparing mean oxygen saturation among same group of patients before and after suctioning with a new device.

→ Formula:

$$t = \frac{\sum d}{\sqrt{\frac{n(\sum d^2) - (\sum d)^2}{n-1}}}$$

➤ Calculate d (the difference between the first and the second paired observations)

➤ Compute the sum of the calculated difference ( $\sum d$ )

**2) Analysis of Variance (ANOVA) Test-**

→ When a researcher wants to compare the difference between more than two samples means, t-test will not be useful and an alternative inferential statistical will be needed.

→ This need can be fulfilled by test known as analysis of variance (ANOVA) test. Therefore, it is clear that ANOVA is used to compare

more than two sample means drawn from corresponding normal population.

→ E.g. – A researcher wants to examine the difference of blood pressure in three different groups, that is executives, managers and clerks; here t-test cannot be applied to examine difference in the mean blood pressure scores because there are more than two groups; therefore, ANOVA will be used in this case.

→ Formula:

$$\sum X^2 - \frac{(\sum X)^2}{N}$$

### **Nonparametric Tests-**

→ In this nonparametric test included:

### **Chi-Square Test-**

→ This is nonparametric test used to find out the association between two events in binomial or multinomial samples.

→ It is represented by a symbol ' $\chi^2$ ' and is used to find out association between two discrete attributes.

→ E.g. – This test can be used if one wants to find the association between smoking in pregnancy and low-birth-weight babies, blood pressure and renal diseases, and obesity and coronary diseases, etc.

### **Prerequisites of Chi-square Test-**

→ There are four essential requirements to apply  $\chi^2$  test, which are as follows:

- Preferably random sample but not necessarily.
- Qualitative data measured on nominal or ordinal scale (frequency data, not the means).

- Sample size should be more than 30.
- Lowest expected frequency not less than 5.

→ Formula:

$$E = \frac{\text{Column or vertical total} \times \text{Row or horizontal total}}{\text{Sample total}}$$

**Note-**

### 1) Type-I and Type-II Errors-

- Type-I error occurs when null hypothesis is rejected, where it should have been accepted. It is also called alpha error.
- Type-II error occurs when null hypothesis is accepted, where it should actually have been rejected. It is also known as beta error.
- These errors generally occur due to unrepresentative sample drawn from the population.

|                | <b>Decision</b>                |                                |
|----------------|--------------------------------|--------------------------------|
| <b>Reality</b> | <b>Accept <math>H_0</math></b> | <b>Reject <math>H_0</math></b> |
| $H_0$ (True)   | Correct decision               | Type – I error                 |
| $H_0$ (False)  | Type – II error                | Correct decision               |

Table- Showing Explanation about Type-I & Type-II Errors

### 2) Tests of Significance-

- There are several parametric (t-test, ANOVA) and nonparametric tests (chi-square test) available to establish the statistical significance.

**Statistical Packages & Applications-**

- Traditionally, in the field of healthcare biostatistics, data has been handled manually; it involves laborious work, needing lots of paperwork.
- The results often become available too late and contain different types of errors.
- The advent of computers and their popularity in almost every sphere of life have significantly changed the way biostatistics is now applied to health sciences.
- Computer hardware and appropriate software have optimized the various processes involved in the application of biostatistics by making the data analysis and storage easier, faster, and errorless.
- Several computer packages are available to be used for the statistical analysis; among them commonly used packages are Microsoft Excel, SPSS, SAS, Minitab, Stata, Systat, NCSS, etc.

**1) Microsoft Excel-**

- Microsoft Excel is a very popular and useful spreadsheet program that can be used for data entry and analysis.
- It has the capacity to generate random numbers and can be used for the computation of many standard statistical applications like computation of mean, range, standard deviation, etc. It can also be used for relatively sophisticated work like computing regression and ANOVA.

**2) SPSS-**

- Originally known as Statistical Package for Social Sciences, SPSS was developed in 1960 at Stanford University to help solve problems in the social sciences.
- SPSS now stands for Statistical Product and Services Solutions and is among the most comprehensive and popular statistical packages.
- To know more about SPSS, visit the website [www.spss.com](http://www.spss.com).

**3) SAS-**

- The Statistical Analysis System (SAS) is very comprehensive software developed by North Carolina State University.
- This software is divided into many modules and its licensing is flexible, based upon the need for functions.
- This system contains a very large variety of statistical methods and is the software of choice for many major businesses, including the entire pharmaceutical industry.
- To learn more visit their website [www.asa.com](http://www.asa.com).

**4) Minitab-**

- This is another statistical package designed to facilitate the teaching of statistical methods by using computer.
- The Minitab is also very user-friendly product with well-designed documentation facilities being widely used in educational institutions.
- To learn more about Minitab visit their website [www.minitab.com](http://www.minitab.com).

## **Analysis of Data**

### **Introduction-**

- Analysis and interpretation of data is the most important phase of the research process, which involves the computation of the certain measures along with searching for patterns of relationship that exists among data groups.
- Data collection is followed by the analysis and interpretation of data, where collected data are analyzed and interpreted in accordance with study objectives.
- Analysis and interpretation of data includes compilation, editing, coding, classification, and presentation of data.

### **Purposes of Analysis of Data-**

- The main purposes for analysis of data are as follows:
  - To make the raw data meaningful.
  - To test null hypotheses.
  - To test the statistical significance of data or related data.
  - To draw inferences and make generalizations.
  - To estimate parameters.

### **Definition of Data Analysis-**

- Analysis is the process of organizing and synthesizing the data so as to answer research questions and test hypothesis.
- Analysis is referred to as a method of organizing data in such a way that research questions can be answered and hypothesis can be tested.

### **Compilation-**

- Compilation process includes gathering together all the collected data in a manner that a process of analysis can be initiated.

- While compiling the data, care is to be taken to arrange all the data in an order so that editing and coding process can be implemented with ease.

### **Interpretation Data-**

- Interpretation of data refers to the critical examination of the analyzed study results to draw inferences and conclusions.
- Interpretation of the research findings of a study involves a search for their meaning in relation to the research problem, objectives, conceptual framework, and hypotheses.
- This is an activity of critical thinking, which is done carefully through brainstorming to infer the condensed and statistically computed data, so that research questions can be answered and hypothesis can be tested.
- Interpretation is one of the most essential tasks in a research process to frame the recommendations of a research study, so that empirical evidences generated through research process can be used during implementation phase of the research study.
- Interpretation is a very subjective activity; it is not guarded with the scientific methods and procedures. Therefore, it is liable to bias and errors such as publication bias or reporting bias (selective reporting of results).
- To present this adverse consequence, researcher must be very careful and critical while examining the study results to draw the interpretations.

## Interpretation Process-

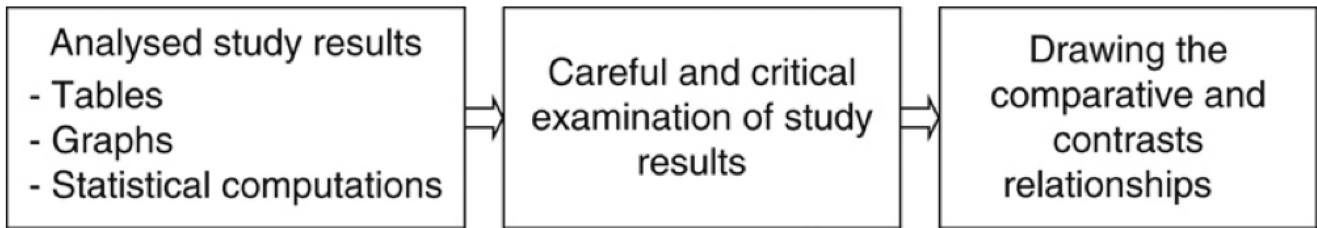


Fig. – Interpretation Process

## Strategies for Effective Interpretations-

- Interpretations must be made in light of research problem, objectives, conceptual framework, hypotheses, and assumptions.
- Critical examination of each element of study results before framing the interpretations.
- Careful consideration and recognition of the limitations of the research study so that inappropriate interpretation can be avoided.
- Interpretations must be based on the study results only, so that chances of misinterpretations or over-interpretations of the unstudied facts can be avoided.
- Each part, aspect, and segment of the analyzed result must receive close attention, so that misrepresentation can be avoided.

## **Communication & Utilization of Research**

### **Introduction-**

- A research project cannot be considered complete until its results are effectively communicated to its users and consumers. Therefore, communication of the research findings is one of the essential and final steps of the research process.
- Communication of the research findings is carried out through the dissemination of the empirical research evidences generated through research study.

### **Criteria/Finding for Communication of Research-**

- In order to share or report the research findings, researcher has to follow some criteria, which are as follows:
  - 1) Selecting proper channel for communicating.
  - 2) Knowing the consumers.
  - 3) Developing an effective plan for writing a research report.
  - 4) Careful selection of a journal for publication of a report.
  - 5) Careful review of the author's manuscript guidelines.

### **Writing a Research Report-**

- A research report is the ultimate outcome of the research process.
- A research process cannot be considered complete until its report has been written and disseminated.
- Writing a report highlights the research project and helps in the dissemination of the research findings.
- Dissemination of research findings serves scientific, professional, and public functions.

## Characteristics of a Good Research Report-

→ A good research report has following characteristics:

- ✓ A research report must have characteristics of conciseness, clarity, honesty, completeness, and accuracy.
- ✓ A research report must be long enough to cover the subject content and short enough to maintain interest among its users and consumers.
- ✓ It must be written and presented logically so that different pieces of the research process fit effectively.
- ✓ Abstract terminology and technical jargons must be avoided in the research report.
- ✓ Presentation of the research report must be lucid and visually attractive, so that it can be interesting to its users.
- ✓ Research report must reflect its originality.

### *Note:*

## Tips for Writing a Good Research Report-

- Use short, simple sentences.
- Write in active voice.
- Use headings and subheadings to guide the readers.
- Use bold type in headings to attract attention.
- Use tables, graphs to break up print pages.
- Use left and right margin justification.
- Make sure there are good transitions between sections.
- Follow directions as stated in the guidelines for writing report.

## Format of Research Report-

- The format of the research report varies from institution to institution and university to university.
- Quantitative research reports typically follow a conventional format referred to as the **IMRAD format**.
- It involves the organization of the study material in four sections:
  - Introduction, **M**ethods, **R**esults, **A**nd **D**iscussion.

### 1) Introduction-

- It is often written in a funnel-shaped structure.
- The researcher begins broadly to establish a framework to understand the study and narrows it down to the specifics of what he or she intended to learn.
- Introduction generally includes:
  - ✓ Background of the study
  - ✓ Need of the study
  - ✓ Research problem
  - ✓ Objectives
  - ✓ Hypothesis/assumptions
  - ✓ Operational definitions
  - ✓ Conceptual framework
  - ✓ Literature review

### 2) Methods-

- The methods section is often subdivided into several parts, which include stages, such as:
  - Research design
  - Research setting
  - Target population
  - Sampling technique and sample size
  - Development and description of data collection tools

- Validity and reliability of research tools
- Methods and procedures of data collection
- Pilot study and tool tryout
- Feasibility of the study

### **3) Results-**

- It is the heart of a research report.
- If both descriptive and inferential statistics have been used, then results start with descriptive statistics to present sample characteristics.
- Later analyzed data are presented in accordance with study objectives using descriptive and inferential statistics.

### **4) Discussion-**

- A typical discussion section addresses the following questions:
  - What were the main findings?
  - What do the findings mean?
  - How do the results compare with prior knowledge on the topic?

### **5) Conclusion-**

- Conclusion includes the paragraphs that summarize what was done and found from present study as precise as possible.

### **6) Other aspects of the report-**

#### ***a) Title:***

- It should include the variables under study, study population, and place of the study.
- The title should be clear and concise, ideally not more than about 15 words.

#### ***b) Abstract:***

- It is an unstructured paragraph of 100– 200 words or in a structured form with subheads such as Background and Objectives, Methods, Results and Conclusion.

**c) Keywords:**

- Substance, methodology, and theoretical terms can be used as keywords; these keywords are used in the index to help others locate the study.

**d) References:**

- The reference style should be selected after reviewing the manuscript or report.
- Usually, Vancouver reference style is used in health sciences literature.

**Steps in Writing Report-**

- The steps involved in writing a report include:

**1) Logical analysis of the subject matter-**

- It is the first step which is primarily concerned with the development of a subject.

**2) Preparation of the final outline-**

- It is the next step in writing the research report. 'Outlines are the framework upon which works are constructed.
- They are an aid to the logical organization of the material and a reminder of the points to be stressed in report'.

**3) Preparation of the rough draft-**

- This step is of utmost importance for the researchers as it involves actually writing down what they have done in the context of their research studies.
- The researchers write down the procedures adopted by them in collecting the material for their study along with various limitations, the techniques of analysis adopted by them, the broad findings and generalizations, and the various suggestions they want to offer regarding the problems encountered.

**4) Rewriting and polishing the rough draft-**

- This step happens to be the most difficult part of all formal writing.
- The careful revision makes the difference between a mediocre and a good piece of writing.
- While rewriting and polishing, one should check the report for weaknesses in logical development or presentation.
- The researcher should check the mechanics of writing— grammar, spelling, and usage.

**5) Preparation of final bibliography-**

- Next in the order comes the task of the preparation of the final bibliography.
- It is a list of books, journal articles, papers presented, etc. that have contributed or are in some way pertinent to the research which has been done.

**6) Writing the final draft-**

- This constitutes the last step.
- The final draft should be written in a concise and objective style and in simple language, avoiding vague expressions, such as ‘it seems’, ‘there may be’, etc.
- While writing the final draft, researcher must avoid abstract terminology.

**Utilization of Research Findings-****Introduction-**

- From the beginning, nursing has been recognized as both an art and a science.
- Florence Nightingale's book, Notes on Nursing, provided the information necessary for those who nursed others back to health.

- At the time of its first publication, nursing schools did not exist, there was no distinct body of nursing science, nor were there recognizable nursing practices.
- Today, 140 years later, the need for research-based practice is still widely accepted.
- But unfortunately, many current nursing interventions are based on tradition rather than science.
- Some traditional procedures seem very logical and helpful when considered at face value, but a review of the scientific literature reveals that intervention is not only ineffective, but also potentially harmful.

### **Barriers in Utilization of Nursing Research-**

- These are following barriers:
  - ✓ Nurses lack time to actively participate in conducting and implementing research.
  - ✓ They lack the resources to read current research findings.
  - ✓ Nurses do not understand the importance of research.
  - ✓ Research is a minute and difficult component of undergraduate nursing programmes.
  - ✓ Health care system in our own scenario pays little attention to research on nursing issues.
  - ✓ Lack of access to research literature for the bedside clinical nurses.
  - ✓ Nurses failing to read research articles.
  - ✓ Published research may have few clinical applications, and staff nurses may not read nursing research journals.

## **Strategies to Facilitate Utilization of Nursing Research-**

→ Strategies for each of these areas are presented below:

### **1) Nurse Educators-**

- Nurse educators are crucial role models in providing a foundation for research-based practice.
- They must have the ability to foster an appreciation for research and make it interesting for the nursing students.
- Nursing students must see that change occurs, but only if the need for change is identified.
- It allows reflecting the skills they need to use after graduation. Research-based clinical practices that will solve problems, while making work procedures more efficient and, therefore, less costly, are good reasons for change.

### **2) Nurse Researchers-**

- Dissemination methods used by researchers in publications and presentations often inhibit rather than support research-based practice.
- The clinical nurse often finds the language and the results difficult to interpret and apply to the patient.
- For patients to benefit from research-based practice, researchers must make an elaborate effort to include information that specifically addresses practical clinical applications of their findings.

### **3) Nurse Administrators-**

- Establish a research-friendly culture.
- Encourage clinicians to question traditions.
- Reward risk-taking and innovation.
- Require research basis for practice changes.
- Incorporate research role in job descriptions.
- Provide research resources, such as medical and nursing literature, Internet access, consultants, etc.

**4) Nurse Clinicians-**

- Nurses should be encouraged to update their research skills.
- This can be accomplished by seeking advanced nursing degrees, or the organization can work to provide education regarding research utilization.
- Institutions can send key nurses to conferences, which provide practice updates, or bring in researchers to share their findings with nurses, as well as other members of the interdisciplinary team.
- This can increase acceptance by physicians and other health care providers of nursing research findings.
- Support research conduct in the clinical setting.
- Take the risks to make changes and improve practice.

**Developing & Presenting a Research Proposal-  
Introduction-**

- A research proposal is an outline of the research idea, which helps in communicating this idea to faculty or the funding agency, so that an approved project can be implemented.
- Development of the research proposal starts when the research problem is sufficiently specified to begin work and the researcher is satisfied that the problem is feasible.

**Meaning of Research Proposal-**

- A research proposal is a written document specifying what the investigator proposes to study, and is, therefore, written before the project has commenced.
- Proposals serve to communicate the research problem, its significance, and planned procedure for solving the problem to the interested party. That party may be a funding agency, a faculty advisor, or institutional officers depending upon the circumstances.

## **Purposes of Proposal-**

- It serves as a blueprint and guiding path for the researcher to carry out the research project.
- Research proposal helps the researcher to communicate to the supervisor, faculty, department, and funding agency what he or she is going to study, so that the desired suggestions and support to carry out the research project can be obtained.
- Proposal also helps the department to make a right decision about allotment of guide for the candidate pursuing a particular project.
- It also gives an opportunity to receive feedback from supervisor and others in the academic community as well as possible funders.
- Research proposal also serves as a contract between researcher, guide, and university.
- Research proposal also can be used to seek ethical approval from the institutional, regional, or national-level research ethical committee.

## **Developing the Proposal-**

- Development of the research proposal involves:
  - Choosing a topic.
  - Narrowing down and focusing on the topic.
  - Formulating research objectives or questions and ideas for analysis.
  - Outlining the key literature in the topic area.
  - Deciding on research methodology, research design, and methods.
  - Proposing an approach to data analysis.
  - Proposing a format, for example how many chapters and suggested chapter headings.
  - Developing a timeline.
  - Developing a budget and resources required for the project.
  - Organizing references and bibliography.

**Format of Research Proposal-**

- Different funding agencies or colleges may have different guidelines/format requirements.
- Where this is the case, the principal investigator should prepare the proposal in the required format, provided that the format used gives all or most of the details as contained in the guidelines of the particular institute.
- A research proposal usually consists of the following elements:

**1) Title-**

- The research project title should demarcate the following:
  - Who or/what will be researched
  - Where
  - When
  - How

**2) About Investigators-**

- Full names, qualifications, academic titles of all the investigators, including trainees (assistant lecturers or postgraduate students), and their institutional/ departmental affiliation.

**3) Institution under who's Umbrella the Research Project will be Conducted-**

- For example:
  - World Health Organization
  - College of Nursing, All India Institute of Medical Sciences, Rishikesh
  - Indian Council of Medical Research

**4) Background Information & Introduction-**

- A review of the relevant literature. It should be most recent. (Majority being in the past 5–10 years at most.)
- Locally available information published or unpublished. It may include clinical or laboratory observations.

**5) Objectives of the Study-****a) Broad objectives:**

- The main issues that are being looked at/ for are laid down in the broad objectives.

**b) Specific objectives:**

- These must be measurable, either qualitatively or quantitatively, and form a guide to the research methodology, data analysis, and presentation of results.

**6) Formulating Hypotheses-**

- The following aspects should be kept in mind when formulating a hypothesis:
  - Hypotheses can only be formulated after the researcher has gained enough knowledge regarding the nature, extent, and intensity of the problem.
  - Hypotheses should figure throughout the research process in order to give structure to the research.
  - Hypotheses are tentative statements/ solutions or explanations of the formulated problem.
  - Care should be taken not to oversimplify and generalize the formulation of hypotheses.

**7) Methodology-**

→ In this included:

- The type of research study.
- Location of study/study setting/study period.
- Study population.
- Sample size.
- Data collection.
- Data management & analysis.
- Result presentation.

**8) Budgetary Estimates-**

→ Each line item should be quantified in monetary terms.

→ The investigator should indicate the amount to be asked for and for what the institution under which the research project will be conducted will contribute.

→ Each item should be quantified, if possible.

→ If it is not possible (e.g. premises or space), it should just be mentioned as an institutional contribution.

→ Subtotals for each group should be indicated and then the grand total.

**Computer in Nursing Research-**

→ Use of computer in research is so extensive that it is difficult to conceive today a scientific research project without a computer.

→ In-depth and extensive studies cannot be carried out without the use of computer, particularly those involving complex computations and data analysis.

→ These are following uses of computer in nursing research:

- To Research problem, purpose, main question, or hypothesis.
- To literature review.
- To conceptual framework.
- To Research design.
- To sampling.
- To research instruments & data collection.
- To legal & ethical considerations,
- To discussion of finding, recommendations & implementation.
- To budgetary information.
- To references & bibliography.